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Article

Smart Product-Service System for Parking Furniture—Sale of Storage Space in Parking Places

Mariusz Salwin [1,*] and Tomasz Chmielewski [2]

[1] Faculty of Mechanical and Industrial Engineering, Institute of Organization of Production Systems, Warsaw University of Technology, 85 Narbutta Street, 02-524 Warsaw, Poland
[2] Faculty of Mechanical and Industrial Engineering, Institute of Manufacturing Technology, Warsaw University of Technology, 85 Narbutta Street, 02-524 Warsaw, Poland; tomasz.chmielewski@pw.edu.pl
* Correspondence: mariusz.salwin@onet.pl

Abstract: Growing competition, changing customer needs, and growing environmental protection requirements mean companies are forced to change their approach to business. Traditional product sales are being replaced by systemic solutions focused on meeting specific customer requirements while reducing negative impacts on the environment. One such solution is the Product-Service System (PSS). This allows manufacturers to offer their products' functionalities and features through related services. By extending the life of products, promoting the reuse of materials, and reducing the amount of waste, the implementation of PSS strongly supports sustainable development. The paper focuses on a new product group—garage boxes (GB). It discusses a new PSS business model that responds to the needs of people living in blocks of flats with no tenant storage lockers or rooms in the basement. The new business model sells the function (storing various possessions) and eliminates problems faced by tenants due to the lack of sufficient storage space. It provides customers with high-quality GB for as long as they need them. Customers can pick and choose equipment with additional services depending on their needs. The idea of the model is the outcome of a nationwide study carried out in Poland on a group of 500 residents of blocks of flats and consultations with manufacturers, homeowner associations, wholesale and retail traders, and the financial sector. The study provided us with information and data that provided a comprehensive picture of the problem of the absence of storage lockers or rooms for residents and the needs connected with GB. The results of the conducted research indicate that the developed business model responds to the diverse requirements of residents and supports sustainable solutions. It is an alternative to the lack of a storage unit assigned to each apartment. The business model developed in the paper is highly innovative and comprehensive. This makes it an attractive solution for residents of apartment blocks, and its implementation can significantly reduce the environmental impact.

Keywords: product-service system; product-service system design; parking furniture; storage; business model; sustainable development

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1. Introduction

Over recent years, manufacturing has evolved towards models intended to meet the sophisticated needs and requirements of clients [1–4]. This marks a truly breakthrough transition from mass consumption to highly personalised solutions [5–7]. The evolution goes hand in hand with a reconsideration of the manufacturer's business model and the offerings that he makes to his clients [8–10]. What we observe is a shift from manufacturers selling products to a consortium that delivers system solutions covering products, services, functions, and opportunities [11–13].

On the other hand, the whole world is currently facing challenges such as growing production, unsustainable consumption practices, reduced consumption of resources, and environmental protection [14–17]. Subject-matter literature suggests there is huge potential for changing the manufacturer's business model [18–20]. Using new business models to target customers with new offers that are alternative to traditional sales will help counterbalance the aforementioned challenges with more productive manufacturing practices and sustainable consumer choices [21–23].

Product-Service System (PSS) is a business model that satisfies the requirements of the evolution of industrial production and the above-listed challenges faced all over the world [24–26]. Such a revolution of traditional business models appears to be increasingly more interesting to researchers, businesses, and consumers [27–29]. A PSS enables the reduction of the number of products in the market and the increase in their availability [30–33]. This extends the useful life of products and reduces the consumption of resources in production [23,34]. This is an approach in which, instead of product ownership, a manufacturer sells a service, access, functionality, productivity or an effect [32–35]. In the design stage, attention is paid to the best possible meeting of consumer needs, durability and product regeneration possibilities that allow multiple uses of a product [36–38]. PSS is a special case of servitisation linked with the performance economy [39–41]. By minimising the consumption of raw materials and waste, extending the useful life of a product, promoting its sharing, and reducing pollution and energy losses, the system is seen as a way to a circular economy (CE) [29,42,43]. PSS can be defined as a business model that helps suppliers provide customers with comprehensive, environment-friendly solutions [25,44]. There are various categories of proposed PSS on offer, which differ in terms of how a product is used, the dominant revenue mechanism, the outcome of the contract, product ownership, or the degree of integration between a product and services [45–48]. PSS categories that are most often discussed in the literature include product-oriented PSS, use-oriented PSS, and result-oriented PSS [13,49].

The park and street furniture industry focuses on the production of furniture, systems and fittings for car parks or garage areas. They are usually resistant to corrosion, have an unlimited lifetime, and are recyclable. They are mainly used by people living in blocks of flats who have their own parking or garage spaces to store their belongings. The rapid development of this industry is connected with savings made in modern construction. The latter, combined with construction and utility requirements, make developers replace storage rooms with parking spaces or additional flats. The lack of a storage locker or room assigned to each flat is a serious inconvenience for people using the flat for a longer period of time, as they have to store many things that do not fit in the flat. This is why they decided to buy and install a garage box (GB). Fire safety regulations and regulations imposed by homeowner associations make it difficult, and often impossible, to obtain permission to install a garage box. On the other hand, installation without permission, which often occurs, is in breach of the law and places the responsibility for the resulting hazardous situation and associated costs on the GB owner. This means there is a serious problem that can be solved by developing a PSS for garage boxes [50].

This article focuses on parking furniture such as garage boxes (GB). They are used to store all sorts of items. Suggested locations for their installation are parking spaces in underground garages or other closed garages. Garage boxes differ significantly from other furniture in their construction and durability. They are simple in construction, use steel and plastic parts and have a relatively high value for the average customer. In addition, each garage box can be described by its capacity (in litres), load capacity (in kilograms) and overall height (in metres). Misuse (overloading), damage and defects that occur during use can lead to a loss of their mechanical properties. Dangerous situations such as attempted break-ins or vandalism can cause permanent damage to major garage box components. All this can cause accidents, which can result in damage to your vehicle. It is, therefore, essential that a garage box is inspected and regularly maintained and the items stored in it are additionally protected.

Offering garage boxes to customers requires specialised solutions, which is why this paper sets out to develop a new PSS model for garage boxes. The business model developed is the result of collaboration with an enterprise. It relies on data and information obtained from a nationwide questionnaire-based survey of residents of blocks of flats with their own parking spaces, statistical analyses and consultation panels with representatives of homeowner associations, housing developers and people involved in issuing approvals and permits for the use of garage boxes.

Compared to traditional models related to services or products alone, the described PSS business model is more complex and extensive. The customers, and at the same time, the users of the presented PSS model, are people who suffer from a shortage of storage space in residential buildings and have their own parking space. The financial analyses carried out (financial scenarios and evaluation of economic efficiency) were based on data on garage box production costs, sales price and costs associated with its use. The data were obtained from manufacturers of analysed products and the residents of blocks of flats who use them.

The article is structured as follows: the first part is an introduction. Part two outlines the research methodology. The third part presents the most important features of garage boxes. The next part discusses the results of the survey and statistical analysis. The fifth part proposes a new PSS business model for garage boxes. The next part is the financial analysis, and the last part is a discussion and conclusions.

2. Literature Review

PSS design is crucial for environmental protection, business competitiveness and creating tailor-made solutions for customers. With this in mind, researchers and industry have started to focus on PSS design methods and PSS development in different industries. At this stage, a systematic literature review was conducted (Table 1). It was a two-stage review of PSS design methods and practical cases of PSS operating in the industry.

Table 1. Guidelines for systematic literature review.

	Product-Service System Design	**Product-Service System in Industrial Practice**
Analysis period	2001–2020	
Information sources	ProQues, Springler Link, Science Direct, Taylor and Francis Online, EBSCOhost, Scopus, Emerald, Insight, Web of Science, Ingenta, Dimensions, Wilma, IEEE Xplore Digital Library and Google Scholar	
Keywords	"Product-Service System in design" or synonyms	"Product-Service System in industry" or synonyms
Result	60 PSS design methods	150 works describing PSS functioning in industry

Figure 1 shows 60 PSS design methods, which were analysed in relation to industries in which they have been applied. Most of the analysed methods have been developed for mechanical engineering (14), production sector (13), and domestic appliances and consumer electronics (11). Noteworthy, 10 of the analysed methods target several industries, while 12 do not target any specific industry. We can see that there are many studies which propose diverse approaches to the development of a new PSS. They consider different customer needs and problems. The fact is that there are no design methods that would target solutions connected with additional storage space for tenants of blocks of flats [51].

Table 2 shows examples of PSS models used in industrial practice and the companies that use them. Companies that offer PSS know the market in which they operate very well. With PSS, they foster their market position and expand the scope of their business operations. Their activities are supported by a lot of analyses and well thought-over. Manufacturers know their products very well and are constantly developing them together with the range of services linked to these products. Most of the analysed PSS focus on solving specific problems related to the products or arising during their use or even before their purchase. This is a new way of delivering added value to customers. The products used in these models belong to the premium class in their segment and are characterised by high quality, price and long life cycle. All the models analysed aim to protect the environment. They assume zero waste production through maximum reuse of materials in a closed loop. In some PSS, different customers can use products several times. This reduces the number of products and packaging produced, thus reducing waste. In addition, they contribute to the spread of sustainable lifestyles and environmentally friendly values among their customers.

Among the analysed PSS design methods and industrial use cases for PSS, no solutions were found for additional storage space in blocks of flats or for parking furniture that could fulfil such a function.

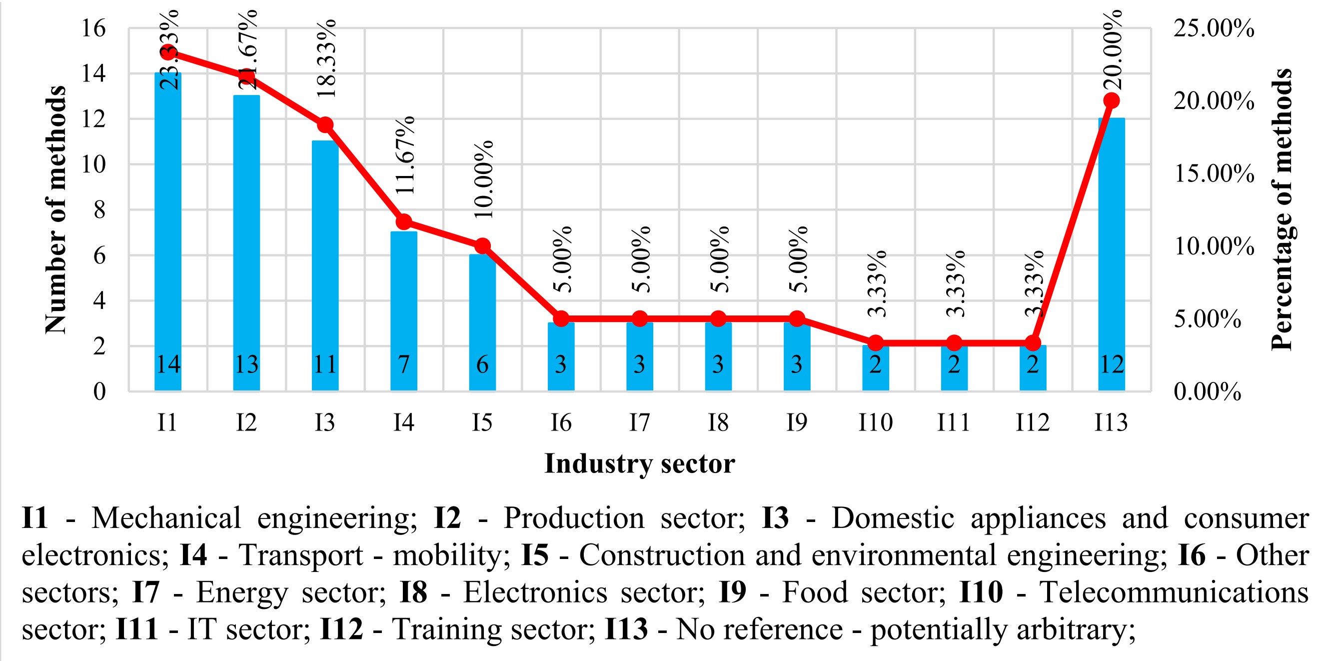


Figure 1. PSS design methods—classification by sectors.

Table 2. PSS in industrial practice—examples.

	Company	Country	Product	PSS Type	PSS Description
1.	Alup	Germany	Air compressor	Result-oriented PSS	Rental services or sales of compressed air per m^3
2.	Be park	Belgium	Parking space	Use-oriented PSS	"Parksharing", i.e., making available parking spaces in certain locations, e.g., at hotels, supermarkets or in other not sufficiently used car parks. An automated management system allows clients to find, reach, and book a parking space and pay for it.
3.	Beneens and Zonen byba	Belgium	Shop-fitting	Product-oriented PSS	'Shop fitting as a service' model offers four components in one package: a design, construction, maintenance, and financing store fittings.
4.	Bosch	Germany	Electrical appliances	Use-oriented PSS	Customer pays a monthly subscription which covers the rental of an electrical appliances (e.g., an oven, espresso maker), maintenance, repairs, and necessary materials (cleaning detergent, filters, etc.). Contracts are signed for 5 years. A customer can enjoy top-notch electrical appliances without having to invest in product purchase.
5.	Desso	Netherlands	Carpets	Product-oriented PSS	The company sells carpets that are not harmful to the health of people and to the environment and enables customers returning 'used up' carpets. Next, by recycling, they are re-used in the production of new carpets.

Table 2. *Cont.*

	Company	Country	Product	PSS Type	PSS Description
6.	Dimdom	France	Toys	Use-oriented PSS	Clients may choose toys that their children like from a catalogue. Then, they buy a subscription whose fee depends on how many high quality toys are ordered. Toys are delivered to clients and used by their children. Toys that are not used or those that children got bored with are replaced or sent back. Subscription period is not precisely identified and depends on child's likes and interests.
7.	L'art en loc	France	Picture	Use-oriented PSS	Short-term or long-term lease of original works of art. By renown artists to private individuals or business people for the purpose of interior decoration. Depending on the period for which a contract is signed the offer includes also the change of works selection.
8.	Philips Lighting	Netherlands	Lighting systems	Result-oriented PSS	Selling a promised level of luminance in a building, according to a Pay per Lux concept.

3. Research Methodology

The aim of this paper is to design a PSS business model for garage boxes. The purpose of the developed PSS is to eliminate the problem of storing items that are not used on a daily basis and that have to be stored in the flat. This will be an alternative to not having a storage room allocated to each flat.

The following research question has been formulated in this article:

- "What opportunities does a PSS offer to manufacturers of parking furniture, residents of blocks of flats (clients), and homeowner associations?"

In order to answer the research question formulated in the paper and accomplish its goal, we adopted a 7-stage research methodology:

1. Identification of the research problem. Savings of property developers and construction regulations (with respect to the number of parking spaces in a block of flats), as well as utility-related requirements, make them replace storage rooms for tenants with parking spaces or more flats. That leads to the absence of storage rooms for all tenants. Residents of blocks of flats suffer from the lack of storage space where they can store their possessions. Therefore, they buy and install garage boxes (GB) in their parking spaces. This, in turn, is against binding regulations adopted by homeowner associations. Hence, a comprehensive solution seems necessary to provide tenants with their own safe and secure storage space in the form of PSS for garage boxes.
2. Analysis of PSS-related literature. This stage of research focuses on a systematic literature review. At this stage, PSS design methods were examined together with cases of industrial PSS.
3. Description and analysis of parking furniture, such as garage boxes. The goal of this stage is to discuss the features and application of garage boxes as a space for storing items. At this stage, we also listed seven factors that explain why garage boxes are worth considering in the context of PSS.
4. A questionnaire-based survey of tenants of blocks of flats and statistical analyses. At this stage, we investigated 500 residents of blocks of flats situated in urban areas who have parking spaces that they either own or rent. A questionnaire containing closed and open questions was used as a research tool. Next, statistical analysis was carried out using the single-factor analysis of variance ANOVA and post hoc test–Fisher's Least Significant Difference test. Additional bits of equipment and services were statistically tested for traits of tenants of blocks of flats (e.g., sex, age, size of a city they live in, etc.).

5. Construction of a PSS business model. Relying on data taken from the analysis of literature, data for the industry, questionnaire-based studies, and statistical analysis, a draft PSS business model was designed for garage boxes.
6. Consultations with experts. Works at this stage focused on improving the PSS business model drafted at the previous stage. The initial framework was consulted with representatives of the manufacturers, homeowner associations, wholesalers and retailers, finance, and residents of blocks of flats. Their main role was to analyse and diagnose aspects and elements that could be improved or added on. Next, the model was updated with their suggestions. At this stage, the final PSS business model for garage boxes was worked out.
7. Financial analysis of the proposed solution. At this stage, a financial analysis was conducted for the developed PSS model by comparing it to a traditional model. Analyses were carried out for both cases based on formulas that were especially developed. At the next stage of analysis, the following methods of assessment of the economic effectiveness of investment projects were deployed: Net Present Value (NPV), Internal Rate of Return (IRR), and Modified Internal Rate of Return (MIRR). All the above-mentioned analyses were conducted using data concerning costs of production, purchase/sales prices, and cost of maintenance of garage boxes.

4. Parking Furniture—Garage Box Case

4.1. What Is a Garage Box

A garage box is a modern and functional metal parking furniture (Figure 2). It can be used as additional storage space in underground car parks or other parking spaces. It is made of a thick galvanised steel sheet (sheet thickness from 1 to 3 mm). It has height-adjustable legs made of 6 mm thick steel. The legs support the entire structure and are universally mounted without anchoring to many types of surfaces. The entire structure is powder painted, thanks to which the coating is not only exceptionally smooth but also durable. The specialist powder paint used for the coating does not contain any harmful or toxic substances. The side and rear walls are additionally reinforced. Bolted sheet metal parts are used in the construction of garage boxes, which makes assembly very quick. The garage box is stable and resistant to loads—its load-bearing capacity is 500 kg, and its capacity is approximately 1900 L. It is a non-flammable construction. The product meets stringent health and safety requirements and has fire protection certificates and opinions. It allows for very safe storage of various types of items. The garage box is equipped with an adjustable shelf and a special handle, brackets and rails, which significantly facilitate keeping things in order, even with a large number of items. It closes with a solid metal hatch door with an anti-burglar lock. It has an actuator which holds the hatch when it is open. This cabinet effectively protects valuable contents against unauthorised access. The described product is addressed mainly to parking spaces in underground garages. However, it has a much wider application. It can also be used in single-family houses, utility rooms, mechanical and vulcanisation workshops, and other places. The product does not interfere with parking the car, as it is placed on legs, which makes it possible to adjust its height above the bonnet of any car. This is a great advantage, thanks to which parking can be performed as usual and is safe.

Figure 2. Photos of a garage box installed in a manufactured parking space.

4.2. Why Garage Boxex?

The motivation to focus on garage boxes rests on the following premises:

1. An analysis of the literature devoted to PSS does not give any practical examples or design methods for parking furniture, including garage boxes.
2. Modern developer buildings tend to be less functional than those built 20 or 30 years ago. An example of this is the fact that in today's modern residential buildings, no storage room is assigned to each flat. This is the result of savings made by housing developers. Each square meter of their investments is very valuable, so developers prefer to allocate space for an additional flat or garage space. At present, tenant storage rooms are a bonus available to the largest flats. In practice, this means that

in a building with 100 flats, the number of tenant storage rooms is approximately 15. The lack of storage units causes problems with storing things which are not used on a daily basis and have to be stored in the flat.

3. In order to meet the construction and use requirements, investors are obliged to provide a certain number of parking spaces for users. Therefore, for new buildings, a designated parking space is already a standard which has to be paid for. This means that a parking space is available to the vast majority of residents. A consequence of the shortage of storage space in residential buildings is the desire to adapt any available space. Thus, a garage box set up in a parking space in an underground car park can be a solution to the problem of storing many things when there is no room in the flat.
4. For long-term users of a dwelling, setting up a garage box can be a solution to storage problems. Garage boxes installed in parking spaces in underground garages seem to be a safe place for storage. From the point of view of housekeeping regulations imposed by homeowner associations and fire safety regulations, obtaining permission to install a garage box can prove cumbersome and, in some cases, impossible. Installation without permission is a significant breach of the law and places the responsibility for the resulting dangerous situation, property damage, compensation problems and costs on the owner. In addition, it can all create a major risk to the health and life of users for which the person installing the garage box without permission is responsible.
5. Another argument in favour of garage boxes is their high regenerative value at the end of their useful life. These are products recently introduced in the market whose main characteristics are functionality, quality and longevity. Garage boxes are durable and, with proper maintenance, can be used for many years. Components that will wear out faster include the handles and locks, the actuators that hold the hatch door, the hatch and the components responsible for adjusting the legs (Figure 3). The framework, sheet metal walls, and leg bases last much longer. There is, therefore, a discrepancy in the life span of the individual garage box components. Observations indicate that damaged garage boxes are not repaired but end up dumped in landfills. This leads to low material efficiency, environmental pollution and economic losses for the users.
6. Currently, garage box manufacturers offer their customers the traditional option of a cash purchase. The purchase involves a large investment, which is not always affordable for people living in blocks of flats. All garage boxes offered by a given manufacturer are identical in terms of dimensions and equipment. Besides, the standard garage box equipment is very poor, while obtaining additional equipment is usually very difficult and expensive. Additional services include only delivery and installation for a fee. This means that the initiatives taken by manufacturers focus on the design, production and sale of garage boxes. There is an apparent problem with the supply of spare parts. Customer needs and various issues arising during use are not taken into account. It is, therefore, interesting to investigate what garage box equipment and services are needed by current and potential users. All this will be helpful in developing and proposing a new offer (based on PSS) that meets customer requirements and solves the installation permit issues.
7. Survey results show that 351 (70.20%) respondents noticed a lack of storage rooms for tenants in their blocks of flats (Table 3). Storing unnecessary things in a flat (316 residents (63.2%)) and lack of tenant storage rooms (260 residents (52%)) are the most frequently mentioned reasons for being interested in purchasing a garage box (Table 4). As many as 317 (63.40%) respondents are interested in a garage box in order to create their own storage space. On the other hand, 441 (88.20%) believe that garage boxes should be a standard feature of parking spaces or garages in blocks of flats that do not have storage rooms assigned to each flat (Table 3). The research also shows that 284 (56.80%) respondents are interested in PSS for garage boxes in order to store items that do not fit in the flats (Table 3).

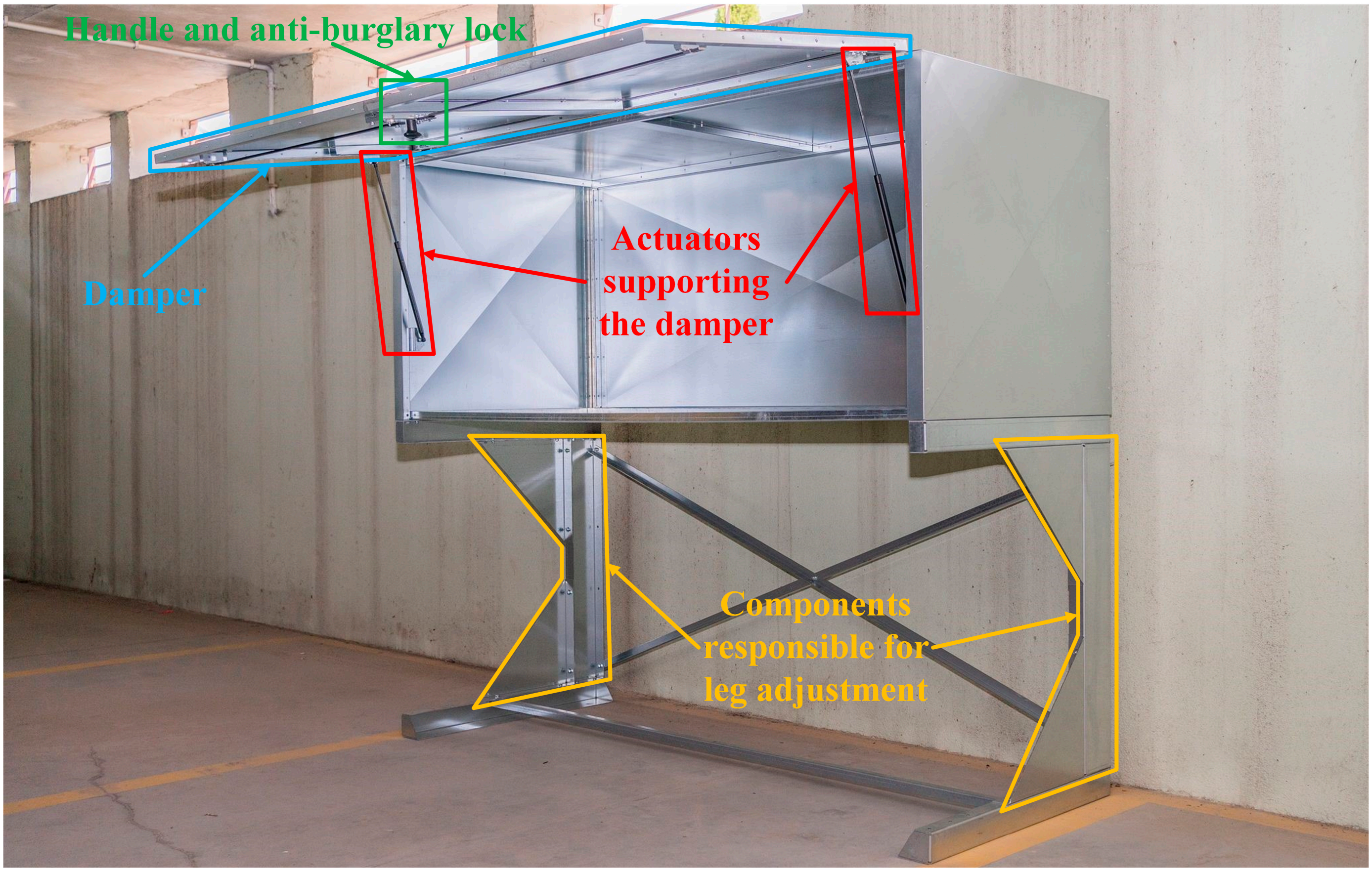


Figure 3. Parts of a garage box that need to be replaced during modernisation.

The above arguments show that there is a need to develop a new business model that would integrate the manufacturer, the users (residents of blocks of flats with their own parking spaces) and the homeowner associations; a model that would provide residents of blocks of flats with their own parking spaces with the storage space they need and eliminate problems involved in obtaining permission for installation. With this model, customers will get a garage box equipped to suit their needs and the services associated with it. All this is available at an affordable subscription fee for each customer. With this model, the manufacturer will be able to make more profit, which will be based on the use of the garage box storage space, the addition of additional equipment and the services provided. The customer will receive support for installation, spare parts replacement, dismantling and take-back. The agreement between the homeowner association and the manufacturer will eliminate problems with obtaining permission for the installation, situations such as installation without permission and the consequences associated with this. It will also eliminate the scrapping of garage boxes by the customer, as the manufacturer will remanufacture and re-rent them to the same or new customers.

Table 3. Questionnaire-based research—the sense of lack of storage rooms for tenants in blocks of flats; tenants interested in Garage Boxes (GB) as a solution that provides them with self-organised storage space; interest in PSS for garage boxes amongst respondents; Garage Boxes (GB) as a standard equipment of parking space in blocks of flats.

No.	Questionnaire-Based Research	Yes		No	
		Number of Surveyed	Percentage of Surveyed	Number of Surveyed	Percentage of Surveyed
1.	The feeling of a lack of storage in block of flats	351	70.20%	149	29.80%
2.	Interest in GB for self-organization of storage space	317	63.40%	183	36.60%
3.	Interest in PSS for GB among the residents of block of flats	284	56.80%	216	43.20%
4.	GB as a standard equipment for a garage blocks	441	88.20%	59	11.80%

Table 4. Reasons of interest in Garage Boxes (GB).

	Questionnaire-Based Research	Yes		No	
		Number of Surveyed	Percentage of Surveyed	Number of Surveyed	Percentage of Surveyed
1.	No space in the apartment	316	63.20%	184	36.80%
2.	There are no storage rooms	260	52.00%	240	48.00%
3.	Storage of unnecessary things in the apartment	234	46.80%	266	53.20%
4.	Some things cannot be stored in the apartment	255	51.00%	245	49.00%
5.	The ability to safely store your belongings	130	26.00%	370	74.00%
6.	Low price compared to extra storage room	121	24.20%	379	75.80%

5. Customers of Garage Boxes in Poland

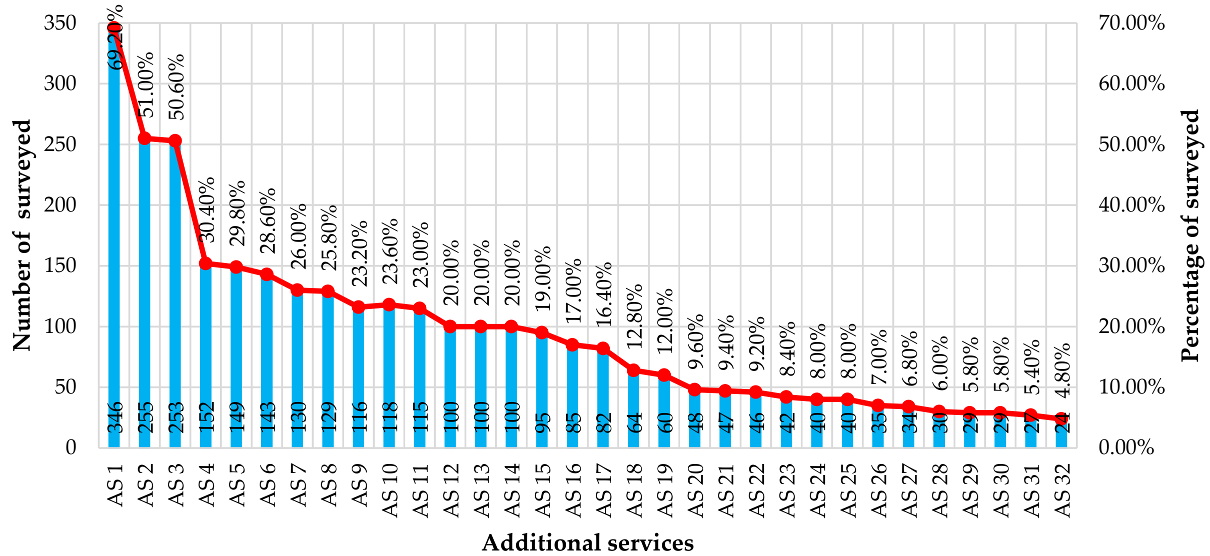
5.1. Survey of Potential Customers

A survey of residents of blocks of flats who owned or rented parking spaces was carried out. Respondents were city dwellers from all over Poland of different ages, marital statuses, and educational backgrounds. The research tool used in the study was a survey questionnaire containing closed and open questions. The output of the research was 500 questionnaires filled out with data suitable for further analysis. The aim of the study was:

- to find out if residents of blocks of flats need new solutions in the area of storing that would replace storage rooms for tenants,
- to identify what extra services linked with garage boxes [AS] are necessary for users (Figure 4),
- to find out what additional equipment [AE] should be installed in garage boxes (Figure 5),
- to gather opinions on PSS for garage boxes and learn about the scale of interest in such a solution (Tables 3 and 4),

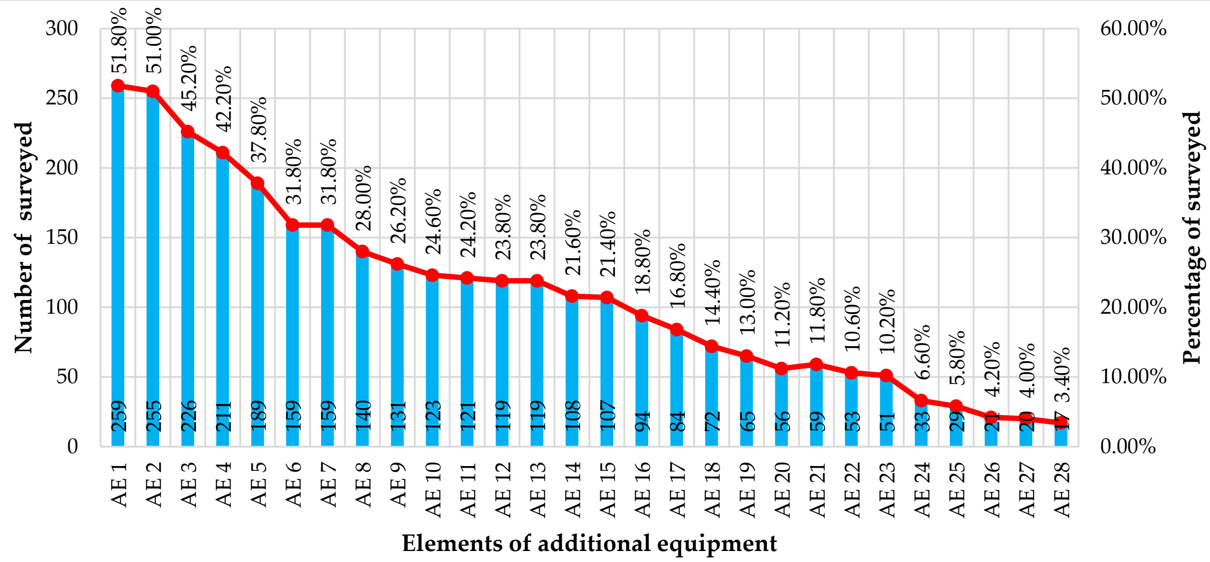
The aim of the research was to understand the current situation of residents of blocks of flats when it comes to storing their belongings and to collect information that will be used in the construction of a new PSS business model. The answers will help to choose the elements of the garage box equipment and related services that will create a PSS addressed to the residents of blocks of flats with a parking space.

Results of studies suggest that residents of blocks of flats are interested in PSS for garage boxes. The reason is the insufficient availability of storage rooms for tenants living in blocks of flats. The results of the studies are summarised in the table below (Table 5).



AS 1 - Transport to the customer; **AS 2** - Assembling and installation at the customer's site; **AS 3** - Dimensioning of the parking space and custom-made furniture; **AS 4** - Powder coating in a selected color from the RAL palette; **AS 5** - Alarm; **AS 6** - Installation of additional shelves; **AS 7** - Transport and submission of GB after moving; **AS 8** - Possibility to extend the lower part; **AS 9** - Insurance; **AS 10** - Supply of spare parts; **AS 11** - Installation of additional partitions; **AS 12** - Mobile application; **AS 13** - GB monitoring; **AS 14** - Maintenance; **AS 15**- Protection; **AS 16** - Ventilation; **AS 17** - Cleaning; **AS 18** - Monitoring of the presence of hazardous substances (gas, CO_2 and others); **AS 19**- Parking assistant; **AS 20** - Advice on the maximum use of GB capacity; **AS 21** - Advice on the maximum use of GB storage space; **AS 22** - Waste disposal; **AS 23** - Training in operation and use; **AS 24** - Fill indicator; **AS 25** - Heating; **AS 26**- Training and advice on parking a car in a parking space from GB; **AS 27** - Disinfection; **AS 28** - Loading indicators; **AS 29** - Air conditioning; **AS 30** - Polishing; **AS 31** - Location service; **AS 32** - Glazing;

Figure 4. Additional services linked with a garage box [AS] that are of interest to customers.



AE 1 - LED lighting – inside; **AE 2** - Actuators supporting the damper; **AE 3** - Height adjustable shelf; **AE 4** - Additional anti-burglary locks; **AE 5** - Adjustable bicycle holder; **AE 6** - Motor and remote control for automatic opening of the hatch; **AE 7** - Three-bolt lock (two side bolts and middle one); **AE 8** - Sensors, detectors, cameras and LCD display; **AE 9** - Divider with adjustable width; **AE 10** - Hangers, e.g. for skis; **AE 11** - Elements for screwing the GB to the wall; **AE 12** - Adjustable Tire Holder / Suspended Tire Rack; **AE 13** - Thicker sheet; **AE 14** - Elements for screwing the GB to the floor; **AE 15** - LED lighting – outside; **AE 16** - Toolbox; **AE 17** - Parking threshold; **AE 18** - Lugs enabling expansion of the lower part of the GB; **AE 19** - Glare; **AE 20** - Elements enabling GB installation without screwing; **AE 21** - Additional supports for long elements; **AE 22** - Ramp threshold; **AE 23** - Ladder; **AE 24** - Owner's address / Information card; **AE 25** - Work boots and overalls; **AE 26** - Tightening belts; **AE 27** - Adjustable beam; **AE 28** - Mirrors;

Figure 5. Additional components of garage box equipment [AE] which are interesting to clients.

Table 5. Summary study results.

Study Results
Additional services and equipment make a vital component of the offer
464 (92.80%) residents of blocks of flats believe that their needs, wishes, and requirements should be considered when a PSS is designed
478 (95.60%) residents of blocks of flats believe that functionalities and utility features of GB are relevant
Major features that are critical for residents of blocks of flats when making GB choices include: GB capacity (384 (76.80%)), storage space (282 (56.40%)), solid structure (277 (55.40%)), capacity load (194 (38.80%)), adjustable height (182 (36.40%)), thickness of steel sheet (thicker than 1.5 mm/2 mm) (164 (32.80%)), expansion potential (96 (19.20%)), lower case (95 (19.00%)), modern design (59 (11.80%))
When purchasing or renting a GB, residents of blocks of flats consider the following certificates of the product: - 205 (41.00%)—certificate of operation issued by a duly authorised fire-fighting expert; - 150 (30.00%)—Product Certificate issued by State Institute of Construction Technology (ITB)—Fire Research Department; - 145 (29.00%)—Load attestation for a garage locker; - 57 (11.40%)—Decision of 4 October 1996 establishing the list of products belonging to Class A 'No contribution to fire' provided for in the Decision 94/611/EC implementing Art. 20 of Council Directive 89/106/EEC on construction products; - 41 (8.20%)—Decision of 2 December 2010 establishing the classes of reaction-to-fire performance for certain construction products as regards steel sheets with polyester coating and with plastisol coating; To 166 (33.20%) residents of blocks of flats none of the above documents is relevant and they can buy or rent a GB without them
280 (56%) residents of blocks of flats are interested in GB rental if liability for damage that might occur when using a GB falls upon the manufacturer
421 (84.20%) residents of blocks of flats would like to return a GB when they do not need it anymore
420 (84%) residents of blocks of flats are interested in renting a GB after a major repair
455 (91%) residents of blocks of flats want to be able to inform the manufacturer about defects and damage to a GB resulting from its wear and tear
Currently, one of the major problems linked with the GB is the fact that homeowner associations do not issue permits for installing GBs which is why as many as 258 (51.60%) residents of blocks of flats believe that the administration would not permit them to install GBs
A monthly subscription model is a preferred payment option for the PSS
Residents of blocks of flats are interested in new technologies being used in GB

5.2. Statistical Analysis

5.2.1. One-Way ANOVA

In order to analyse the relationship between the level of interest in the researched services [AS] and GB equipment elements [AE] and the characteristics of the inhabitants of blocks with parking spaces, a statistical study was carried out using the one-way analysis of variance (ANOVA). The services and additional elements of GB equipment were tested according to the following characteristics of the residents of the blocks:

- Sex [A] (female [A1], male [A2]);
- Marital status [B] (in a relationship [B1], single [B2]);
- Having children [C] (yes [C1], no [C2]);
- Age [D] (18–25 years old [D1], 26–35 years old [D2], 36–45 years old [D3], 46–55 years old [D4], >55 years old [D5]);
- Level of education [E] (Ph.D. or higher [E1], higher [E2], secondary [E3], vocational [E4]);
- Level of monthly earnings [F] ($\leq$250 Euro [F1], 251–500 Euro [F2], 501–1000 Euro [F3], 1001–1500 Euro [F4], 1501–2000 Euro [F5], 2001–2500 Euro [F6], 2501–4000 Euro [F7], >4000 Euro [F8]);
- Place of residence [G] (city of $\leq$ 20,000 inhabitants [G1], city of 20.001–100,000 inhabitants [G2], city of 100,001–500,000 inhabitants [G3], city > 500,000 inhabitants [G4]);
- The following hypotheses were formulated for the statistical test used in the study:
- Null hypothesis: The interest in a given service or element of GB equipment is the same regardless of the feature that characterises the residents of the blocks;
- Alternative hypothesis: The interest in a given service or element of GB equipment depends on a given feature that characterises the residents of blocks;

The aforementioned statistical method is based on checking the equality of means characterising several populations that have one common (classifying) factor. In the described method, the factor that classifies individual populations is the willingness to use a given service or a specific item of GB equipment. The tested populations are created based on the preferences of the residents of the blocks.

The significance level for which statistical tests were performed is 0.05. In the one-way ANOVA analysis, the Fisher statistic was used, which consists of comparing the mean square of deviations within groups with the mean square of deviations between individual groups.

If the null hypothesis is rejected, it is necessary to check which of the analysed features characterising the residents of blocks are responsible for it. In such a situation, it is necessary to study the differences between the values in individual populations more carefully. For this purpose, post hoc tests are used.

The table below (Table 6) presents all cases for which the value of the Fischer-Snedecor Statistics is in the critical region, which is equivalent to rejecting the null hypothesis. In these situations, a relationship was identified between the feature that characterises the residents of the blocks and the interest in a given service or element of GB equipment. Based on the average values determining the probability of having a service or an item of equipment, it was distinguished which of them are important for specific features characterising the residents of blocks. The results of all statistical analyses are presented in Appendix A.1. Tables A1–A3.

Table 6. One-way ANOVA—results [the list of service designations can be found in Figures 9 and 10].

Test Factor	[AS]	Fisher Statistics	*p*-Value	[AS] Selected by:	Mean	Standard Deviation	[AE]	Fisher Statistics	*p*-Value	[AE] Selected by:	Mean	Standard Deviation
[A]	AS 4	12.87	0.000	[A1]	0.386	0.488	AE 5	6.027	0.014	[A2]	0.425	0.495
	AS 6	5.845	0.016	[A1]	0.341	0.475	AE 7	9.779	0.002	[A1]	0.391	0.489
	AS 13	5.105	0.024	[A1]	0.245	0.431	AE 12	5.098	0.024	[A1]	0.286	0.453
	AS 19	4.462	0.035	[A1]	0.155	0.362	AE 15	6.924	0.009	[A1]	0.268	0.444
	AS 25	6.426	0.012	[A2]	0.107	0.31	AE 16	4.683	0.031	[A2]	0.221	0.416
	AS 26	5.139	0.024	[A2]	0.093	0.291	AE 19	7.853	0.005	[A1]	0.177	0.383
	AS 30	4.958	0.026	[A2]	0.079	0.27						
[B]	AS 4	5.959	0.015	[B1]	0.358	0.48	AE 1	4.526	0.034	[B1]	0.569	0.496
	AS 9	4.377	0.037	[B2]	4.377	0.037	AE 10	5.211	0.023	[B1]	0.293	0.456
	AS 12	8.076	0.005	[B1]	0.254	0.436	AE 12	5.619	0.018	[B2]	0.28	0.45
	AS 14	15.632	0.000	[B2]	0.265	0.442	AE 13	4.264	0.039	[B1]	0.28	0.28
	AS 26	4.837	0.028	[B2]	0.093	0.291	AE 16	5.722	0.017	[B1]	0.233	0.233
	AS 28	6.895	0.009	[B2]	0.086	0.281						
[C]	AS 4	4.374	0.037	[C1]	0.351	0.478	AE 16	4.709	0.03	[C2]	0.276	0.448
	AS 14	6.846	0.009	[C2]	0.243	0.429	AE 16	5.543	0.019	[C1]	0.263	0.441
[D]	AS 2	3.727	0.005	[D1]	0.517	0.501	AE 2	2.805	0.025	[D2]	0.609	0.49
				[D2]	0.641	0.482				[D5]	0.612	0.492
	AS 4	6.588	0.000	[D4]	0.491	0.505	AE 6	3.268	0.012	[D4]	0.528	0.504
				[D5]	0.429	0.5	AE 14	2.78	0.026	[D5]	0.347	0.481
	AS 9	5.829	0.000	[D1]	0.357	0.481	AE 17	2.97	0.019	[D4]	0.208	0.409
	AS 12	3.115	0.015	[D4]	0.321	0.471				[D5]	0.265	0.446
	AS 14	4.439	0.002	[D1]	0.273	0.447	AE 20	2.602	0.035	[D2]	0.18	0.385
				[D2]	0.321	0.471	AE 21	3.17	0.014	[D2]	0.141	0.349
	AS 18	2.467	0.044	[D2]	0.195	0.398				[D3]	0.189	0.393
				[D5]	0.163	0.373	AE 22	2.766	0.027	[D2]	0.172	0.379
	AS 20	2.704	0.03	[D4]	0.208	0.409				[D3]	0.11	0.314
							AE 24	3.824	0.005	[D4]	0.113	0.32
										[D5]	0.113	0.32
[E]	AS 2	2.976	0.031	[E1]	0.645	0.486	AE 5	3.125	0.026	[E1]	0.484	0.508
				[E2]	0.645	0.486				[E2]	0.408	0.492
				[E3]	0.475	0.501	AE 14	8.656	0.000	[E1]	0.484	0.508
	AS 8	2.718	0.044	[E3]	0.248	0.434	AE 19	2.886	0.035	[E4]	0.212	0.412
	AS 18	3.42	0.017	[E1]	0.29	0.461	AE 24	6.135	0.000	[E4]	0.182	0.389
				[E2]	0.29	0.461	AE 25	5.293	0.001	[E3]	0.099	0.3
	AS 19	4.029	0.008	[E4]	0.227	0.422				[E4]	0.121	0.329
	AS 24	2.77	0.041	[E1]	0.161	0.374	AE 27	4.048	0.007	[E2]	0.069	0.253
				[E2]	0.161	0.374						
	AS 29	3.888	0.009	[E2]	0.092	0.289						
[F]	AS 6	2.261	0.028	[F7]	0.444	0.511	AE 5	2.193	0.034	[F6]	0.578	0.499
				[F8]	0.563	0.512				[F8]	0.563	0.512
	AS 7	2.305	0.026	[F1]	0.395	0.495	AE 7	3.802	0.000	[F5]	0.613	0.495
				[F2]	0.302	0.463				[F8]	0.5	0.516
				[F6]	0.333	0.477	AE 15	2.704	0.009	[F2]	0.333	0.475
	AS 8	2.188	0.034	[F2]	0.302	0.463				[F8]	0.375	0.5
				[F4]	0.315	0.467	AE 18	2.917	0.005	[F2]	0.27	0.324
				[F6]	0.311	0.468				[F6]	0.244	0.435
	AS 12	2.623	0.011	[F8]	0.5	0.516				[F8]	0.25	0.447
	AS 15	2.48	0.016	[F2]	0.349	0.481	AE 25	2.05	0.047	[F1]	0.163	0.374
	AS 27	2.657	0.01	[F8]	0.25	0.447						
[G]	AS 2	3.117	0.026	[G1]	0.6	0.492	AE 6	3.986	0.008	[G3]	0.435	0.498
				[G3]	0.467	0.502	AE 14	3.134	0.025	[G4]	0.281	0.45
				[G4]	0.531	0.5	AE 19	2.787	0.04	[G3]	0.196	0.399
	AS 10	3.488	0.016	[G1]	0.322	0.469	AE 22	4.067	0.007	[G3]	0.185	0.39
	AS 11	3.962	0.008	[G3]	0.272	0.447	AE 24	2.816	0.039	[G3]	0.12	0.326
				[G4]	0.291	0.455						
	AS 14	2.989	0.031	[G1]	0.296	0.458						
	AS 21	3.455	0.016	[G1]	0.139	0.348						
				[G4]	0.117	0.323						
	AS 22	2.733	0.043	[G1]	0.148	0.356						
				[G4]	0.097	0.297						
	AS 32	2.821	0.038	[G1]	0.096	0.295						

The use of one-way analysis of variance (ANOVA) shows that the features characterising the inhabitants of blocks which have the greatest correlation with interest are as follows:

- Services: sex (7), age (7), place of residence (7), marital status (6), level of education (6), level of monthly earnings (6), having children (2).
- GB equipment items: age (8), sex (6), education level (6), level of monthly earnings (6), marital status (5), place of residence (5), and having children (2).

In both cases, having children by the residents of blocks shows less dependence on the examined services and elements of GB equipment.

5.2.2. Test Post Hoc—Test Least Significant Difference Fisher

The previously performed ANOVA allowed for the demonstration of differences in the analysed groups. Therefore, in the next step, we tested the populations that caused the null hypothesis to be rejected in the ANOVA. Further tests were carried out to identify which of the k means differed and which were equal. Fisher's LSD (Least Significant Difference) was used to identify this relationship. This is the oldest test of multiple comparisons of individual pairs of means.

The following hypotheses were adopted for the Fisher LSD test checking the occurrence of dependencies between individual averages for the factors from a given pair:

- Null hypothesis: the pair averages are equal.
- Alternative hypothesis: the pair averages are not equal.

The significance level of the tests performed was 0.05. This statistic compares all pairs of the independent variable with a student's *t*-test and adjusts the significance level for multiple comparisons.

The Fisher's LSD test did not show the relationship only between the marital status of the block's inhabitants and the elements of GB equipment, which are AE 10, AE 13, and AE 16. In the remaining cases, there were dependencies, as presented in Table 7. The results of all statistical analyses are presented in Appendix A.1, Table A4.

Table 7. Fisher's LSD test results—significant relationships between the pairs under consideration [the list of service designations can be found in Figures 9 and 10].

Test Factor	[AS]	Pair					[AE]	Pair				
		Feature 1	Mean	Feature 2	Mean	*p*-Value		Feature 1	Mean	Feature 2	Mean	*p*-Value
[A]	AS 4	[A1]	0.239	[A2]	0.386	0.000	AE 5	[A2]	0.318	[A1]	0.425	0.014
	AS 6	[A1]	0.243	[A2]	0.341	0.016	AE 7	[A1]	0.261	[A2]	0.391	0.002
	AS 13	[A1]	0.164	[A2]	0.245	0.024	AE 12	[A1]	0.2	[A2]	0.286	0.024
	AS 19	[A1]	0.093	[A2]	0.155	0.035	AE 15	[A1]	0.171	[A2]	0.268	0.009
	AS 25	[A2]	0.045	[A1]	0.107	0.011	AE 16	[A2]	0.145	[A1]	0.221	0.031
	AS 26	[A2]	0.041	[A1]	0.093	0.024	AE 19	[A1]	0.093	[A2]	0.177	0.005
	AS 30	[A2]	0.032	[A1]	0.079	0.026						
[B]	AS 4	[B1]	0.358	[B2]	0.257	0.015	AE 1	[B1]	0.569	[B2]	0.474	0.033
	AS 9	[B2]	0.269	[B1]	0.19	0.037	AE 10			No dependency		
	AS 12	[B1]	0.254	[B2]	0.153	0.005	AE 12	[B2]		[B1]	0.193	0.03
	AS 14	[B2]	0.265	[B1]	0.125	0.000	AE 13			No dependency		
	AS 26	[B1]	0.043	[B2]	0.093	0.028	AE 16			No dependency		
	AS 28	[B2]	0.086	[B1]	0.03	0.009						
[C]	AS 4	[C1]	0.351	[C2]	0.265	0.037	AE 12	[C2]	0.276	[C1]	0.193	0.03
	AS 14	[C2]	0.243	[C1]	0.149	0.009	AE 14	[C1]	0.263	[C2]	0.176	0.019
[D]	AS 2	[D2]	0.641	[D1]	0.517	0.041	AE 2	[D2]	0.609	[D1]	0.448	0.008
		[D2]	0.641	[D3]	0.449	0.002		[D5]	0.612	[D1]	0.448	0.046
		[D2]	0.641	[D4]	0.415	0.005		[D2]	0.609	[D3]	0.465	0.02
		[D2]	0.641	[D5]	0.408	0.005	AE 6	[D4]	0.528	[D1]	0.301	0.002
	AS 4	[D2]	0.305	[D1]	0.168	0.013		[D4]	0.528	[D2]	0.266	0.001
		[D4]	0.491	[D1]	0.168	0.000		[D4]	0.528	[D3]	0.315	0.005
		[D3]	0.331	[D1]	0.168	0.003		[D4]	0.528	[D5]	0.286	0.008
		[D5]	0.429	[D1]	0.168	0.001	AE 14	[D5]	0.347	[D1]	0.196	0.026
		[D4]	0.491	[D2]	0.305	0.012		[D5]	0.347	[D2]	0.172	0.011
		[D4]	0.491	[D3]	0.331	0.03		[D5]	0.347	[D4]	0.132	0.043

Table 7. *Cont.*

Test Factor	[AS]	Pair					[AE]	Pair				
		Feature 1	**Mean**	**Feature 2**	**Mean**	***p*-Value**		**Feature 1**	**Mean**	**Feature 2**	**Mean**	***p*-Value**
[D]	AS 9	[D1]	0.357	[D2]	0.219	0.006	AE 17	[D3]	0.268	[D4]	0.132	0.008
		[D1]	0.357	[D4]	0.189	0.012		[D1]	0.189	[D3]	0.079	0.015
		[D1]	0.357	[D3]	0.189	0.001		[D3]	0.079	[D2]	0.063	0.03
		[D1]	0.357	[D5]	0.061	0.000		[D4]	0.208	[D3]	0.079	0.034
		[D2]	0.219	[D5]	0.061	0.024		[D5]	0.265	[D3]	0.079	0.003
	AS 12	[D3]	0.244	[D1]	0.119	0.01	AE 20	[D2]	0.18	[D3]	0.079	0.01
		[D4]	0.321	[D1]	0.119	0.002		[D4]	0.208	[D2]	0.18	0.006
	AS 14	[D1]	0.273	[D3]	0.087	0.000	AE 21	[D2]	0.141	[D1]	0.063	0.047
		[D2]	0.25	[D3]	0.087	0.001		[D3]	0.189	[D1]	0.063	0.001
	AS 18	[D2]	0.195	[D1]	0.112	0.04		[D3]	0.189	[D4]	0.075	0.031
		[D2]	0.195	[D3]	0.094	0.016		[D3]	0.189	[D5]	0.082	0.047
		[D2]	0.195	[D4]	0.057	0.011	AE 22	[D2]	0.172	[D1]	0.084	0.019
	AS 20	[D4]	0.208	[D1]	0.07	0.004		[D2]	0.172	[D5]	0.02	0.003
		[D4]	0.208	[D2]	0.063	0.003	AE 24	[D4]	0.113	[D2]	0.016	0.015
		[D4]	0.208	[D3]	0.11	0.043		[D5]	0.163	[D1]	0.056	0.009
								[D5]	0.163	[D2]	0.016	0.000
								[D5]	0.163	[D3]	0.071	0.026
[E]	AS 2	[E1]	0.645	[E4]	0.379	0.014	AE 5	[E1]	0.484	[E3]	0.277	0.031
		[E2]	0.546	[E4]	0.379	0.015		[E2]	0.277	[E3]	0.277	0.009
	AS 8	[E2]	0.302	[E4]	0.152	0.013		[E3]	0.424	[E3]	0.277	0.041
	AS 18	[E1]	0.29	[E2]	0.137	0.016	AE 14	[E2]	0.302	[E4]	0.152	0.013
		[E1]	0.29	[E3]	0.085	0.002	AE 19	[E2]	0.302	[E3]	0.071	0.045
		[E1]	0.29	[E4]	0.106	0.011		[E3]	0.212	[E3]	0.071	0.005
	AS 19	[E4]	0.227	[E1]	0.032	0.006	AE 24	[E4]	0.182	[E1]	0.000	0.001
		[E4]	0.227	[E2]	0.126	0.023		[E4]	0.182	[E2]	0.182	0.000
		[E4]	0.227	[E3]	0.078	0.002		[E4]	0.182	[E3]	0.057	0.001
	AS 24	[E4]	0.227	[E2]	0.099	0.024	AE 25	[E4]	0.121	[E2]	0.023	0.002
		[E4]	0.227	[E1]	0.161	0.019		[E3]	0.099	[E2]	0.023	0.002
	AS 29	[E2]	0.092	[E3]	0.021	0.004	AE 27	[E2]	0.069	[E3]	0.007	0.003
		[E2]	0.092	[E4]	0.015	0.017		[E2]	0.069	[E4]	0.015	0.046
[F]	AS 6	[F8]	0.563	[F1]	0.256	0.02	AE 5	[F6]	0.578	[F4]	0.407	0.047
		[F8]	0.563	[F2]	0.302	0.038		[F6]	0.578	[F3]	0.33	0.002
		[F8]	0.563	[F3]	0.256	0.009		[F6]	0.578	[F1]	0.326	0.014
		[F8]	0.563	[F5]	0.29	0.049		[F6]	0.578	[F2]	0.286	0.002
		[F8]	0.563	[F6]	0.133	0.001		[F8]	0.563	[F2]	0.286	0.041
		[F7]	0.444	[F6]	0.133	0.013	AE 7	[F5]	0.613	[F6]	0.178	0.000
		[F4]	0.333	[F6]	0.133	0.012		[F5]	0.613	[F4]	0.278	0.000
	AS 7	[F1]	0.395	[F7]	0.056	0.006		[F5]	0.613	[F3]	0.273	0.000
		[F6]	0.333	[F7]	0.056	0.022		[F5]	0.613	[F2]	0.413	0.047
		[F2]	0.302	[F7]	0.056	0.035		[F5]	0.613	[F1]	0.279	0.002
		[F3]	0.284	[F7]	0.056	0.034		[F8]	0.5	[F6]	0.178	0.016
		[F1]	0.395	[F4]	0.167	0.004		[F7]	0.444	[F6]	0.178	0.037
		[F3]	0.284	[F4]	0.167	0.028		[F2]	0.413	[F6]	0.178	0.009
		[F6]	0.333	[F4]	0.167	0.031		[F2]	0.413	[F3]	0.273	0.038
	AS 8	[F2]	0.397	[F1]	0.186	0.015	AE 15	[F8]	0.375	[F4]	0.13	0.024
		[F2]	0.397	[F3]	0.188	0.001		[F8]	0.375	[F7]	0.000	0.007
		[F4]	0.315	[F3]	0.188	0.017		[F3]	0.233	[F7]	0.000	0.021
	AS 12	[F8]	0.5	[F1]	0.093	0.000		[F2]	0.333	[F7]	0.000	0.002
		[F8]	0.5	[F2]	0.397	0.048		[F1]	0.256	[F7]	0.000	0.025
		[F8]	0.5	[F3]	0.153	0.001		[F3]	0.233	[F4]	0.13	0.038
		[F8]	0.5	[F4]	0.194	0.004		[F2]	0.333	[F4]	0.13	0.002
		[F8]	0.5	[F5]	0.258	0.048	AE 18	[F8]	0.25	[F7]	0.000	0.036
		[F8]	0.5	[F7]	0.222	0.042		[F6]	0.244	[F7]	0.000	0.012
		[F6]	0.289	[F3]	0.153	0.041		[F2]	0.27	[F7]	0.000	0.004
		[F6]	0.289	[F1]	0.093	0.021		[F6]	0.244	[F4]	0.093	0.014
	AS 15	[F2]	0.349	[F7]	0.111	0.022		[F6]	0.244	[F3]	0.125	0.04
		[F2]	0.349	[F6]	0.156	0.011		[F2]	0.27	[F5]	0.097	0.023
		[F2]	0.349	[F5]	0.129	0.01		[F2]	0.27	[F4]	0.093	0.001
		[F2]	0.349	[F4]	0.111	0.000		[F2]	0.27	[F3]	0.125	0.005
		[F2]	0.349	[F3]	0.21	0.015		[F2]	0.27	[F1]	0.116	0.026
		[F2]	0.349	[F1]	0.186	0.034	AE 25	[F1]	0.163	[F8]	0.000	0.017
		[F3]	0.21	[F4]	0.111	0.037		[F1]	0.163	[F7]	0.000	0.013
	AS 27	[F8]	0.25	[F7]	0.000	0.004		[F1]	0.163	[F6]	0.022	0.005
		[F8]	0.25	[F6]	0.000	0.001		[F1]	0.163	[F5]	0.000	0.003
		[F8]	0.25	[F5]	0.097	0.046		[F1]	0.163	[F4]	0.056	0.011
		[F8]	0.25	[F4]	0.056	0.004		[F1]	0.163	[F3]	0.068	0.017
		[F8]	0.25	[F3]	0.091	0.015		[F1]	0.163	[F2]	0.048	0.013
		[F8]	0.25	[F2]	0.016	0.001						
		[F8]	0.25	[F1]	0.093	0.032						
		[F3]	0.091	[F6]	0.000	0.029						
		[F3]	0.091	[F2]	0.016	0.041						

Table 7. *Cont.*

Test Factor	[AS]	Pair					[AE]	Pair				
		Feature 1	**Mean**	**Feature 2**	**Mean**	***p*-Value**		**Feature 1**	**Mean**	**Feature 2**	**Mean**	***p*-Value**
[G]	AS 2	[G1]	0.6	[G2]	0.402	0.004	AE 6	[G3]	0.435	[G4]	0.24	0.001
		[G4]	0.531	[G2]	0.402	0.038	AE 14	[G4]	0.281	[G1]	0.139	0.003
	AS 10	[G1]	0.322	[G2]	0.134	0.001	AE 19	[G3]	0.196	[G4]	0.082	0.007
		[G4]	0.24	[G2]	0.134	0.044	AE 22	[G2]	0.031	[G1]	0.096	0.037
	AS 11	[G4]	0.291	[G2]	0.165	0.016		[G4]	0.112	[G2]	0.031	0.032
		[G4]	0.291	[G1]	0.148	0.004		[G3]	0.185	[G2]	0.031	0.001
		[G3]	0.272	[G1]	0.148	0.034	AE 24	[G3]	0.12	[G1]	0.043	0.028
	AS 14	[G1]	0.296	[G2]	0.186	0.045		[G3]	0.12	[G4]	0.041	0.012
		[G1]	0.296	[G3]	0.152	0.01						
		[G1]	0.296	[G4]	0.173	0.009						
	AS 21	[G4]	0.117	[G2]	0.031	0.017						
		[G1]	0.139	[G3]	0.054	0.037						
		[G1]	0.139	[G2]	0.031	0.007						

6. Proposed Business Model for Garage Boxes—Problem Solution

The examination of the needs and requirements of the surveyed population has led to the drafting of a New Concept Canvas (Table 8). That allowed us to put in order the acquired data based on which we have developed the PSS business model in question.

Table 8. New Concept Canvas.

Issue	Solution	Competition
Lack of storage rooms for residents of recently constructed blocks of flats	Product-Service System business model for garage boxes—a garage box rental system with diverse type of equipment for residents of blocks of flats and terraced houses who own a parking space	Currently in Poland garage boxes are manufactured by 5 firms—small family businesses
Many items are stored in flats	Making garage boxes available to residents of blocks of flats and terraced houses who own a parking space in cycles adjusted to their needs	Garage boxes are sold through traditional channels
Conflicts with homeowner association/ car park owner when a garage box is installed at a parking space without prior permission	Garage box equipment options respond to concrete customer requirements	Assembly and installation are additional services
Installation without permission constitutes a serious infringement of law while the burden of liability for dangerous situations falls on a GB owner	Transportation, measurement, assembly, installation, and transfer are all standard services	Poor equipment (shelf, bicycle rack, ski rack, threshold) available for extra fee
Difficulties in obtaining installation permits	Extra service packages that help using garage boxes and protect items kept in them	No such solution available for garage boxes
Fire-fighting regulations and attestations	Assistance in utilising storage space in the most effective way	
Key customer needs	**Value proposition**	**What makes our solution better than that offered by competitors?**
A safe place to store items	A customer receives a garage box (a new one or in a perfect condition) where he can store his possessions	No such solutions are available in the market
Storing items which: are too big to be stored in a flat, may cause inconvenience if stored in a flat, cannot be stored in a flat	A customer receives a garage box with a specific equipment option, additional services and a mobile application	An opportunity for residents of single-bedroom flats, small flats and terraced houses to get a cheap and safe place where they can store items that are too big to be stored in their flats
Low investment cost	Problems with homeowner associations/car park owner are eliminated	Stored items are permanently protected and insured
No conflict with homeowner association	Affordable subscription fee	No installation without permission
No need to repair or provide maintenance to a GB	Safe storage of items	
	More space in a flat	

Table 8. *Cont.*

Issue	Solution	Competition
Customer segment	**Technical features**	**Necessary technological knowledge and *know-how***
Residents of blocks of flats who own a parking space	Garage box made of 1 mm or 3 mm thick tinplate	Knowledge about quick installation, assembly, transport of a garage box and its maintenance
Residents of terraced houses	Made exclusively of screwed components which enables its extension, adding on new elements or assembly and re-installation	Knowledge about how to use new technologies in a given solution
People who own a garage or a parking space	Capacity/load capacity 500 kg	Giving advice in the area of servicing and achieving the highest utility index
Other (car repair shops, tyre vulcanising shops, carsharing businesses)	No need to anchor GBs in the floor	Risk: repairs
Property developers who want to install garage boxes at their car parks	A mobile application monitors a garage box and how much of its capacity has been used, sensors and cameras, LCD display	
	Lamps and a parking system	
	Attestation	

The main objectives of the model being developed include finding a way to extend the useful life, avoiding scrapping through remanufacturing and reusing garage boxes on the secondary market, and eliminating the problems with homeowner associations and the consequences caused by installing a garage box without permission.

The company (manufacturer) will make garage boxes available in ten-year cycles. Along with the product, the customer will receive additional equipment and services (Table 9). Everything comes in packages so that the customer can choose what he needs. Agreements between the homeowner association and the company will eliminate the installation problems that have arisen to date. From now on, the manufacturer will become an operator responsible for all garage boxes installed in a given underground car park. This will bring standardisation and order to the car parks. This solution will improve the utilisation potential of garage boxes. The solution will allow for the monitoring of product parts so that they can be replaced quickly. At the end of life, the manufacturer will take back the garage box, inspect it, and modernise it. Components that will wear out faster will be regenerated or replaced with new components—depending on their condition. After all these steps are completed, a garage box will be made available again to the same or another resident in the block who has a parking space. This will reduce the environmental impact of the model.

A survey of residents of blocks of flats who have their own parking spaces made it possible to divide garage box amenities and related services into baskets (packages). The baskets that bring together equipment components and services were ranked in ascending order from basic to the most comprehensive. This division, as well as the whole new business model (Figures 6–8), will enable the manufacturer to reach out to the customers with the solution they want.

The study carried out on a sample of people living in blocks of flats who own parking spaces has revealed their interest in the solution proposed in the model. Out of 500 respondents, as many as 284 (56.80%) are interested in a PSS model for garage boxes as a solution, offering them storage space where they can store items for which they have no space in the flat. That only confirms the big market potential for the newly developed solution.

The key elements of the new model include a delivery, installation, dismantling and reverse logistics system, waste management, a spare parts delivery system and agreements with homeowner associations.

The delivery, installation, dismantling and reverse logistics system are key components of the system under development. The manufacturer delivers a GB and installs it at a customer's parking space. He is also responsible for dismantling the garage box at the end of the contract period or at the termination of the contract. The aim of these activities is to eliminate additional damage, quickly remanufacture, and reuse the product by making it available to a new or the same user. As garage boxes are bolted products, transporting them to the customer and returning them for reconditioning will not be problematic. In both cases, it is possible to transport several dozen products at a time, which makes the process economically viable. Installation and dismantling times may be an issue, especially if the customer has opted for additional equipment versions. The key action will be to reduce these times, which will be significantly influenced by the selection of an optimum number of employees, the arrangement of individual elements in the delivered parcel and the developed schemes of assembly and disassembly activities, as well as the place where these activities are carried out. The developed system will use reusable packaging to protect the environment.

The spare parts supply system will focus on the supply of all parts that will need to be replaced. Above all, this concerns components that wear out more quickly. These include handles and locks, actuators supporting the hatch, and the hatch or components responsible for leg adjustment. The relationship between the manufacturer and the customer will play an important role here. A precise and fast flow of information is necessary to replace a defective component. Monitoring the garage box, sensors, and a mobile application through which the customer can report the need for replacement will be crucial for this solution.

Waste management means all waste management activities from the moment a garage box is manufactured through its installation, use, disassembly and recovery. It includes the segregation and recovery or disposal of all materials used. It aims to segregate all parts into those that can be reused without remanufacturing, those that need remanufacturing and those that can be returned to stakeholders for reprocessing. This element of the developed PSS is critical due to the fact that the process of making steel products is environmentally burdensome. The proposed system provides energy savings compared to the production of steel from virgin raw materials. Steel components that are not suitable for reconditioning are sent to cooperating steelworks and processed there. The recycled materials are then sent to the manufacturer to be turned into new garage boxes. It is worth noting that recycled steel is also cheaper. This solution will be an important link in the recycling chain, which is dictated by the fact that the product involved is practically made of steel components.

Table 9. Service packages linked with a garage box and equipment components in a new PSS business model.

No.	Package	Services	Equipment Components	Manufacturer's Operations
1.	Elementary	Measuring a parking space, transport to customer, assembly and installation at customer's facilities, transport and assembly in another location	Actuators to hold the hatch, burglar-proof locking, elements for installation (on the wall, on the floor or without screwing), slots for extending the lower GB	Being in touch and establishing cooperation with homeowner associations and developers; Collaboration and permanent contact with a resident of a block of flats, homeowner association as long as a GB is used Data collection and GB monitoring; Data analysis and generating new solutions tailored to the needs of a surveyed group; Responding to changes in customer and market preferences; Components of circular economy and environmental protection; Development of a mobile application and a cloud platform; Instruction manual to ensure correct use of a GB;
2.	Spare parts	Delivery of spare parts, installing additional partitions and shelves, maintenance, cleaning, glittering, polishing	Partitions adjustable in width, height adjustable shelves, supports for long items, reflectors, information card	
3.	Personalization	GBs made to measure. Powder coating in any colour from the RAL colour chart, training, advice, possibility to extend the lower part, waste disposal, insurance, ventilation, and disinfection	Additional burglar-proof locks, interior and exterior lighting, handles, hangers, racks, beams, clamping straps, thicker sheet metal, fan, toolbox, ladder, boots and work overalls	
4.	Smart Garage Box	Localisation service, mobile application, monitoring GB and the presence of dangerous substances, parking assist, indicators of: occupancy and load, protection, alarm, heating, air-conditioning	Motor and remote control for automatic hatch opening, sensors and detectors, parking and overrun threshold, cameras, LCD display, intelligent LED lighting (inside and outside), heater, air conditioner	
5.	Storing possessions of a resident of a block of flats	Storage and protection of all items of a resident of a block of flats		

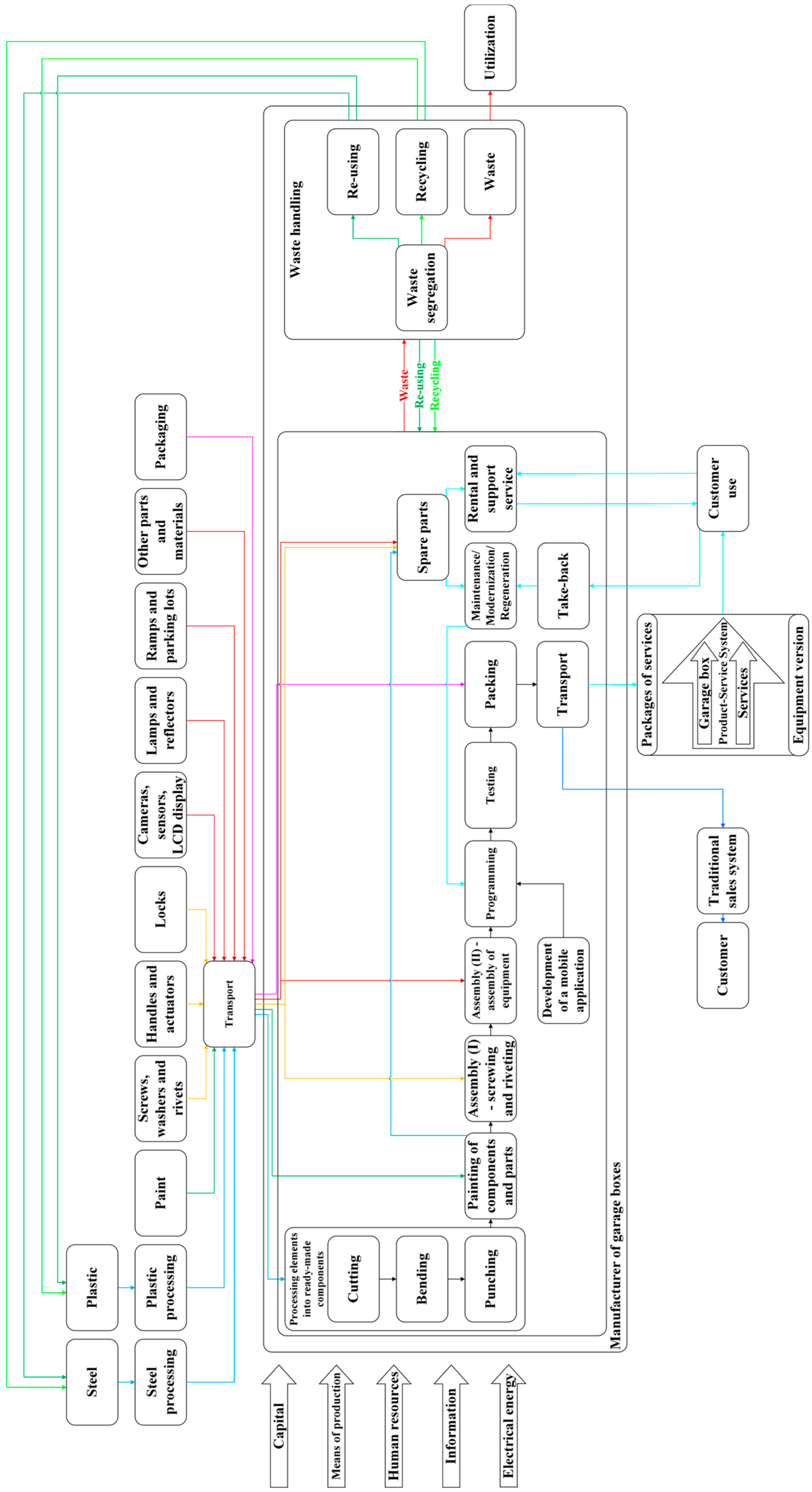


Figure 6. PSS model for garage boxes.

Key Partners	Key Activities	Value Proposition	Customer Relationships	Customer Segments
Banks Insurance firms Steel mills Plastics processing plants Component suppliers (screws, locks, actuators) Suppliers of electronic components, Cloud computing suppliers Media suppliers Producers of paints Machine manufacturers Housing associations Car parks owners Developers	Design, assembly, transport and installation of a garage box Maintenance, repairs and reconditioning Putting secondhand garage boxes back on the market Monitoring and protection Contracts and guidelines that meet the requirements of housing associations	Safe storage of customer possessions Elimination of conflicts with housing association/car park owner Additional equipment components and services A well thought out solution made to measure	These solutions call for coordinated effort to understand customer needs. A manufacturer interacts with the customer throughout the entire lifetime of the solution	Residents of blocks of flats or terraced houses who own parking spaces. Developers wishing to equip underground car parks with garage boxes Other potential customers (car repair shops, tyre vulcanising, training centres for drivers, carsharing companies)
	Key resources		**Channels**	
	Garage box with equipment adjusted to customer needs. Human resources (designers, technicians, machine operators, assemblers, and servicemen) Machines (metal bending presses, CNC punch presses, powder paint shop, and other) Mobile application Training and consulting		Direct marketing Website Online store A network of intermediaries Construction material stores Wholesalers Fairs	

Cost structure	Revenue streams
Investment in garage box rental Cost of attestation and installation permission Costs of use include depreciation and maintenance, monitoring and insurance Reconditioning after the expiry of the rent period Application development and maintaining Cloud Platform Capital investment in people in a plant with specialist knowledge and servicing teams	Revenue from subscription fee that includes usage fee and service fee Revenue maximisation through selling integrated solutions. Additional revenue means additional equipment and services for clients Revenue includes contractual penalties Final fee if a customer wants to buy out a garage box

Figure 7. Canvas Business Model for a new solution.

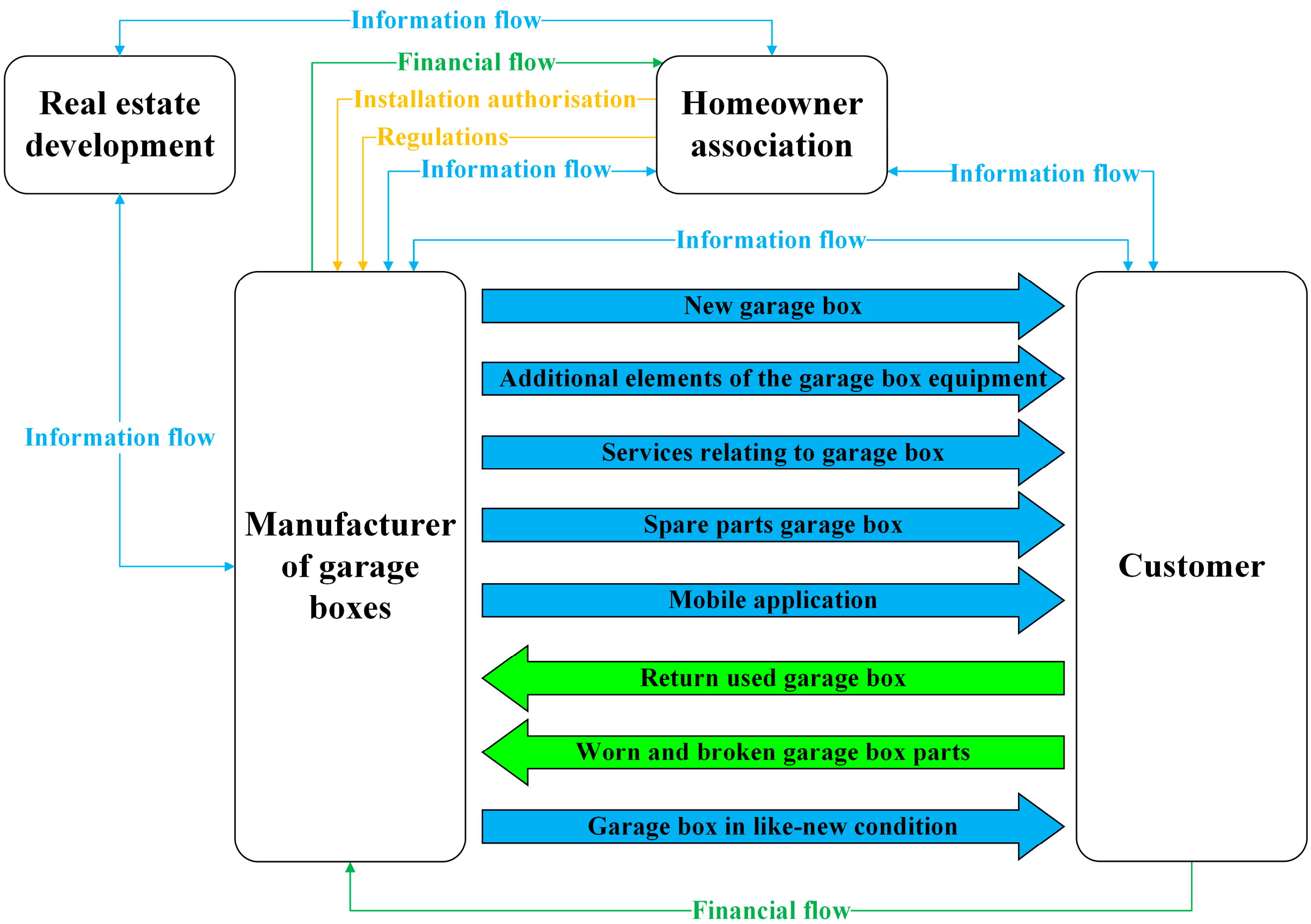


Figure 8. PSS for garage boxes—flows.

Contracts between manufacturers and homeowner associations will lay down guidelines for maintenance (supervision, on-call maintenance and inspections) and use of garage boxes in parking spaces. They will also specify the scope of work for design and installation. In addition, they will enumerate functions that the PSS for garage boxes must perform in order to be viewed as useful by all potential users. Agreements will provide details of access to the monitoring portal and the data that will be supplied to the administration from the installed garage box, as well as the systems with which it will interact in the car park. Thanks to the agreements, the manufacturer will be able to offer high-quality products, reaching a wider range of customers who will be able to use the garage boxes legally and without consequences.

Another important element is the use of new technologies in the PSS solution in question. Their goal is to eliminate any potential risks and meet the stringent requirements of homeowner associations by offering permanent contact with the installed garage box. A display integrated into the garage box, Wi-Fi or SIM card connectivity, mobile application, sensors and cameras as well as the use of Cloud Platform are the pillars of this solution (Figure 9). Management and control of the garage box is possible via the app. The devices connect via the Internet to a smartphone, on which a dedicated application is installed. Through the app, the customers will be able to report their needs, damages and dangerous situations. The garage box will recognise who is using it. Built-in entertainment modules and Internet access will make a GB much more attractive to use and eliminate the possibility of dangerous situations occurring. This will bring innovation, modernity and security to garages and underground car parks. It will help manage stored items that do not fit in the flat. By using cameras, lights and a mobile app, the user will be able to check the current contents of the garage box and the surroundings of their parking space on their smartphone anytime, anywhere. Being able to see the images from the cameras fixed in the garage box on the screen of one's phone also allows one to talk to a member of the household who is currently in the box. In addition, if necessary, the hatch can be opened remotely. In this way, belongings, garage/parking spaces, and cars are monitored, guarded, and protected around the clock. In addition to a real-time view, the built-in cameras also allow taking pictures every time the garage box is closed. These pictures will be displayed on the customer's Smartphone a few seconds later. Thanks to external and internal sensors, the garage box automatically informs us of the distance when the car is being parked. It also informs the user if it is safe to use the GB when the car is parked. Motion detectors register movements and send a message if any unusual activity is detected (for example, if a stranger stays too long near the garage box and observes it). The person watching the garage box will be visible to the owner and the homeowner association, and it is also possible to establish voice contact with him in order to find out what he is doing at the moment. Motion detectors will switch on the lighting only in certain situations, e.g., when stacking things, parking the car, or performing an unusual activity. The sensors also inform the user about the air temperature (inside and outside), concentration of hazardous substances, and other parameters required by the manufacturer and indicated by the homeowner's association. The solution allows the create a list of stored items and categorises them in terms of value and danger. By adding things to the garage box, the customer also adds them to the list on the LCD screen or via his phone. The list will be visible to all household members, so no one will forget what is stored. In addition, the LCD screen is a multimedia board that will allow the display of photos and make notes about stored items. The base that supports customers in using a garage box is a relevant component. Displaying videos and training on how to best use and arrange the stored items will eliminate incorrect use of storage space and the risk of potential damage.

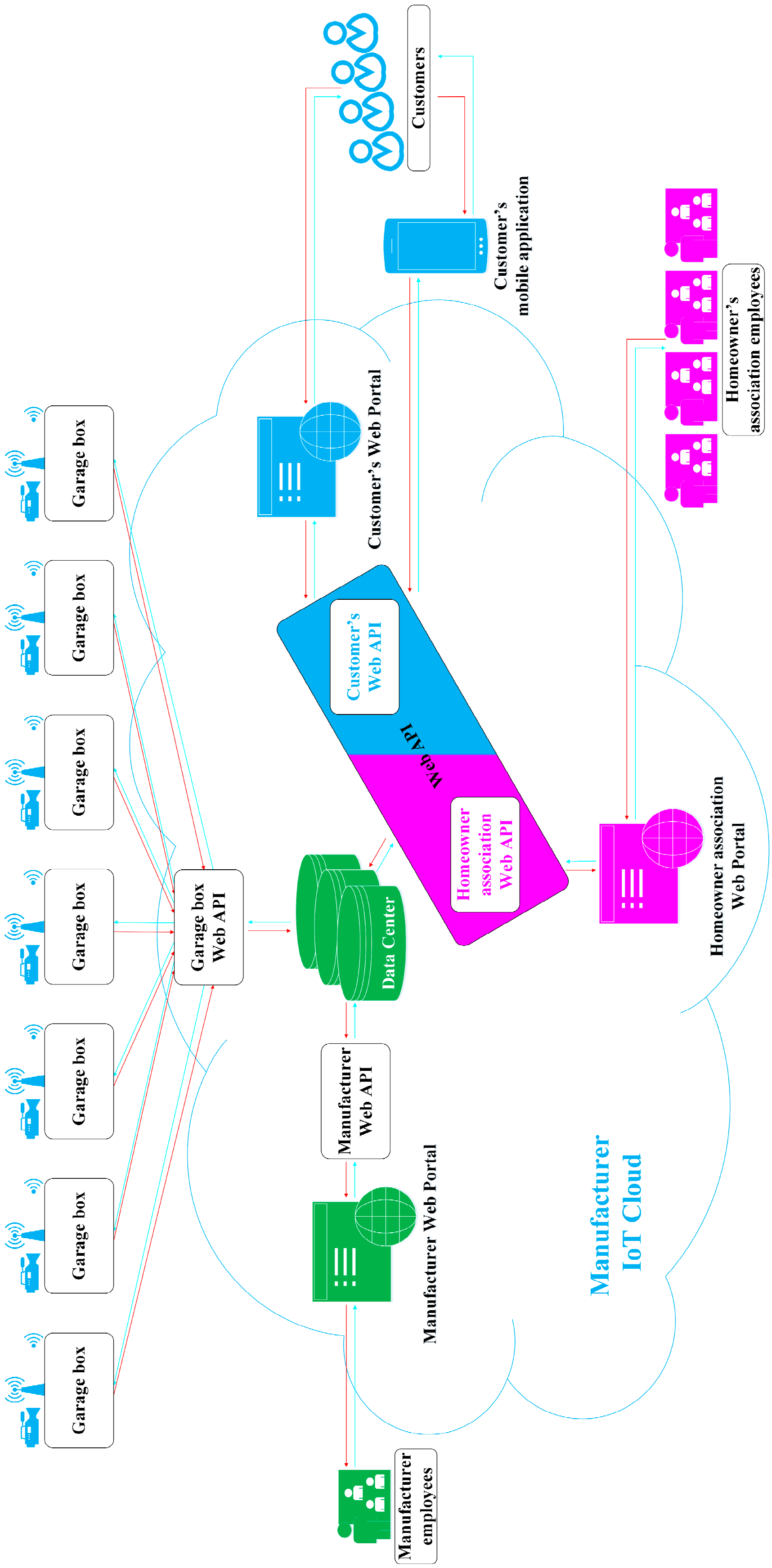


Figure 9. System operation flow.

On top of the previously discussed contracts, the use of new technology will make it possible to obtain permission for the installation. Apart from cameras, a system of remotely controlled detectors for moisture, smoke, carbon dioxide, gas, and carbon monoxide will be installed on each GB in the car park or garage. These detectors help prevent flooding, fire, and even poisoning caused by carbon monoxide or other substances. In the event of any hazard, everyone (homeowner association, users, and the manufacturer), as well as the relevant services and systems that are connected to garage boxes in the car park, is immediately informed. Thanks to this solution, the homeowner association can immediately take preventive measures proportional to received notifications. Additionally, air quality sensors are installed to monitor the concentration of particulate matter inside the car park. Whenever the allowable concentration of PM is exceeded, the system will notify all those concerned, at the same time suggesting to the homeowner association or automatically activating ventilation, air purifiers and humidifiers. In the event of heavy rainfall, the system sends out information about any flooding of the underground car parks. Sensors installed on the garage box monitor what is going on in the car park or garage. Access to data from monitoring, protection, marking of stored items, alarm, and notification in dangerous situations enables eliminating the homeowner associations' concerns regarding the installation of a garage box.

All data regarding the use, conditions in which a product is used and dangerous situations are passed on to the manufacturer. The manufacturer knows how the garage box is used and when the best time for maintenance is. He gets data that can be used in the design and development of new products. The manufacturer receives information in real-time and can be in touch with the user and the homeowner associations.

Providing services to residents of blocks of flats who own a parking space or a garage will be one of the key aspects of the newly developed solution. The company in question can create a special department responsible for this, which will involve hiring new staff and training them. Another solution is trying to seek external assistance, i.e., finding a suitable partner. This may prove difficult due to the initial stage of development of the garage boxes market.

7. Financial Analysis: New Concept

Relying on the information obtained in the research, a financial analysis of the proposed solution was conducted. A market price of EUR 900 for the garage box was used, together with the manufacturing cost of EUR 400, a value after 10 and 20 years of use, and the cost of spare parts, painting and repairs. The analysis takes into account the 20-year lifespan of the garage box—each user will use the product for 10 years. This period of time, suggested also by respondents taking part in the study, will allow for the continuous regeneration of the product.

Two cases were examined (Figures 10 and 11):

- Traditional sales/purchase format—a manufacturer sells a garage box to a customer for cash payment. The ownership is transferred to the customer when the purchase takes place. The user bears all costs connected with the maintenance, additional equipment, and administration.
- PSS for garage boxes—a manufacturer makes a garage box available to a customer for a period of 10 years. Ownership remains on the part of the manufacturer, who is responsible for the maintenance of the product, servicing, additional equipment, and administrative fees. In addition, the manufacturer pays for the depreciation, major repairs or remanufacturing of the product after its useful life has been completed. The user pays a monthly subscription fee for the garage box.

$$\begin{aligned} PS\ fee = {} & Production\ cost - (Usage\ fee + Equipment\ version\ fee + Service\ fee) \\ & -(Depreciation + Service\ delivery\ cost + Administration\ fee) \end{aligned}$$

$$Usage\ fee = \frac{Initial\ value\ of\ the\ garage\ box - Final\ value\ of\ the\ garage\ box}{Life\ cycle\ period}$$

$$Equipment\ version\ fee = (Equipment\ version\ cost + Margin) \cdot Equipment\ version\ level$$

$$\begin{aligned} Service\ fee &= Percentage\ of\ warehouse\ space\ utilization \cdot Period\ of\ time \\ &+ Percentage\ utilization\ of\ capacity \cdot Period\ of\ time \end{aligned}$$

$$Traditional\ system = Garage\ box\ sale\ price + Additional\ equipment\ price + Service\ fees$$

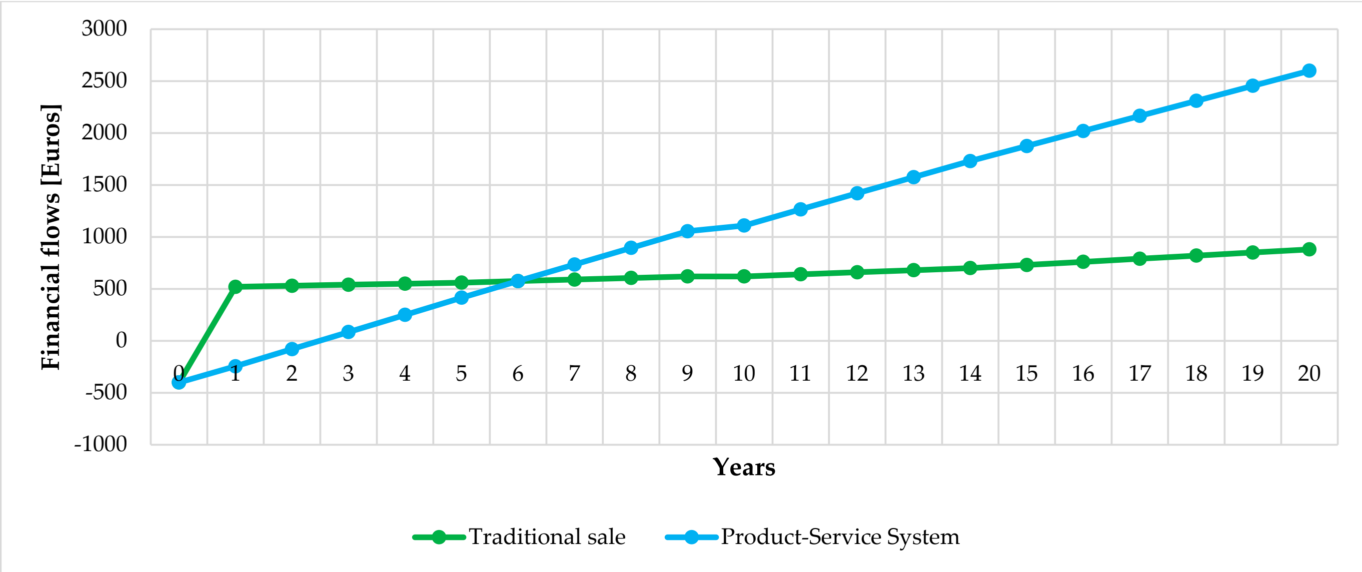


Figure 10. Manufacturer perspective: annual financial flows for traditional sales and a Product-Service System—option for a 20-year garage box life cycle.

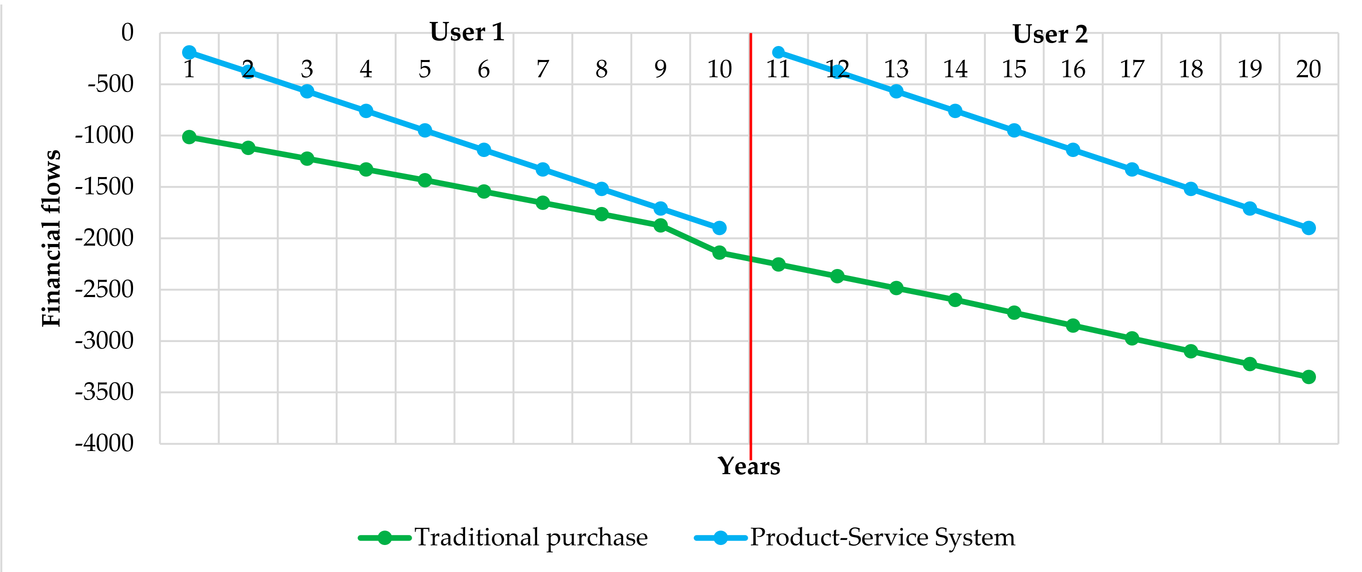


Figure 11. Customer perspective: annual fees.

When examining these two cases, one can realise that in the traditional sales option, a manufacturer receives the entire sum for a garage box on the day when a customer buys it. If a garage box is purchased for 20 years, the manufacturer reports profit in the first year. In subsequent years, the manufacturer may expect profit from selling spare parts and additional equipment components. In the proposed PSS, a manufacturer makes a garage box available and every year receives profits from a monthly subscription fee paid by the user. By monitoring how a product is used and in what condition it is, the manufacturer reduces costs. After 6 years, a PSS becomes more profitable for the manufacturer than traditional sales. Over 20 years of using a garage box, the manufacturer has made 139% more than in traditional sales format.

$$PPS\ fee = (Usage\ fee + Equipment\ version\ fee + Service\ fee)$$

$$\begin{aligned} &Traditional\ system \\ &\quad = Garage\ box\ sale\ price + Additional\ equipment\ price + Service\ fees \\ &\quad + Depreciation + Administration\ fee \end{aligned}$$

The proposed solution is more profitable from the user's perspective. The user does not bear the cost of purchasing a garage box with additional equipment components, nor does he pay for repairs or administrative charges. Depreciation costs are also eliminated as the garage box is owned by the manufacturer.

The next stage of financial analysis uses the Net Present Value (NPV) (Figures 12 and 13). The indicator is used in capital budgeting to examine the profitability of a planned project or investment. The analysis was conducted for two cases: traditional sales/purchase and a PSS for garage boxes. For calculations, we adopted a 5% reference interest rate.

$$NPV = \sum_{t=1}^{n} \frac{CF_t}{(1+r)^t} - I_0$$

Analysing pure cash flows without discounting for traditional sales shows that the manufacturer will receive a return of EUR 1280 (20 years) against an investment of EUR 400, hence a surplus of EUR 880. In a PSS for garage boxes, the return is € 3000 (20 years) against the same investment, hence a surplus of € 2600. However, when assessing the investment, it must be borne in mind that the company could invest this EUR 400 in an alternative venture where it would receive a return of 5%. This is, therefore, the opportunity cost of the capital invested and the reason for discounting future cash inflows at this rate.

After discounting, the proceeds in both cases under consideration exceed the value of the investment, which means that the NPV is positive. Comparing the two cases, after 7 years, the NPV of PSS is already higher than that of the traditional option. In both examples, the present value of the investment is EUR 400. However, the present value of the proceeds (after 20 years) is EUR 640.08 for traditional sales and EUR 1421.22 for PSS for garage boxes. This means that the manufacturer's investment in a PSS is more profitable.

When analysing pure flows without discounting them in a traditional sales option, the customer incurs costs of EUR 1015 already in the first year, rising to EUR 2140 after 10 years. In the case of PSS, this is a charge of EUR 190 in the first year, while after 10 years, it is EUR 1900. After discounting the costs in the traditional option, the NPV after 10 years is EUR −1789.57, and in the PSS, it is EUR −1467.13. This means that in the PSS model, the customer incurs lower costs, i.e., this investment is more profitable for him.

In the next stage of the financial analysis, the Internal Rate of Return (IRR) (Figure 14) and Modified Internal Rate of Return (MIRR) (Figure 15) were used. These indices are economically defined as the rate of return on the expenditure incurred for the project. The analyses were aimed at investigating the profitability of the project from the manufacturer's perspective. IRR assumes continuous reinvestment of the payments received at a rate equal to the value of the calculated IRR. Such an assumption may prove to be oversimplistic and unrealistic under normal market conditions. For this reason, MIRR was used, which allows us to determine the rate of financing all flows (costs) incurred by the company (e.g., interest on credit), as well as the reinvestment rate of all positive flows (e.g., interest on deposits offered by banks). For the purposes of this study, a financing rate of 5% and a reinvestment rate of 1% were assumed.

$$IRR = \sum_{t=1}^{n} \frac{CF_t}{(1+r)^t} - I_0 = 0$$

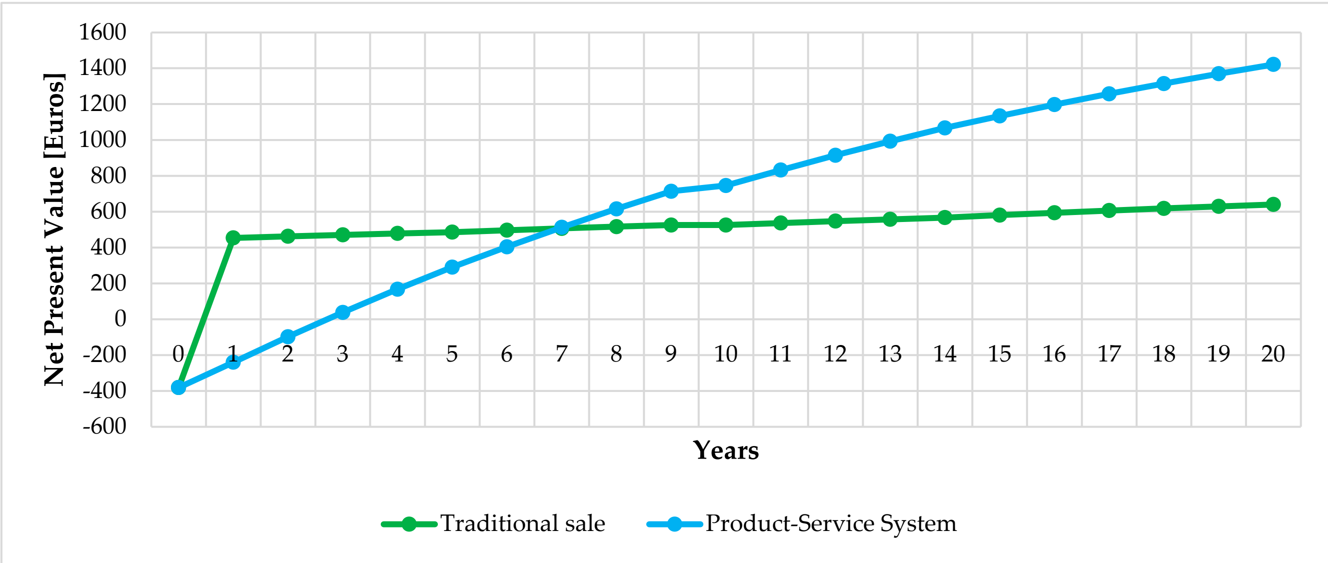


Figure 12. Manufacturer perspective: Net Present Value for traditional sales and a Product-Service System—option for a 20-year garage box life cycle.

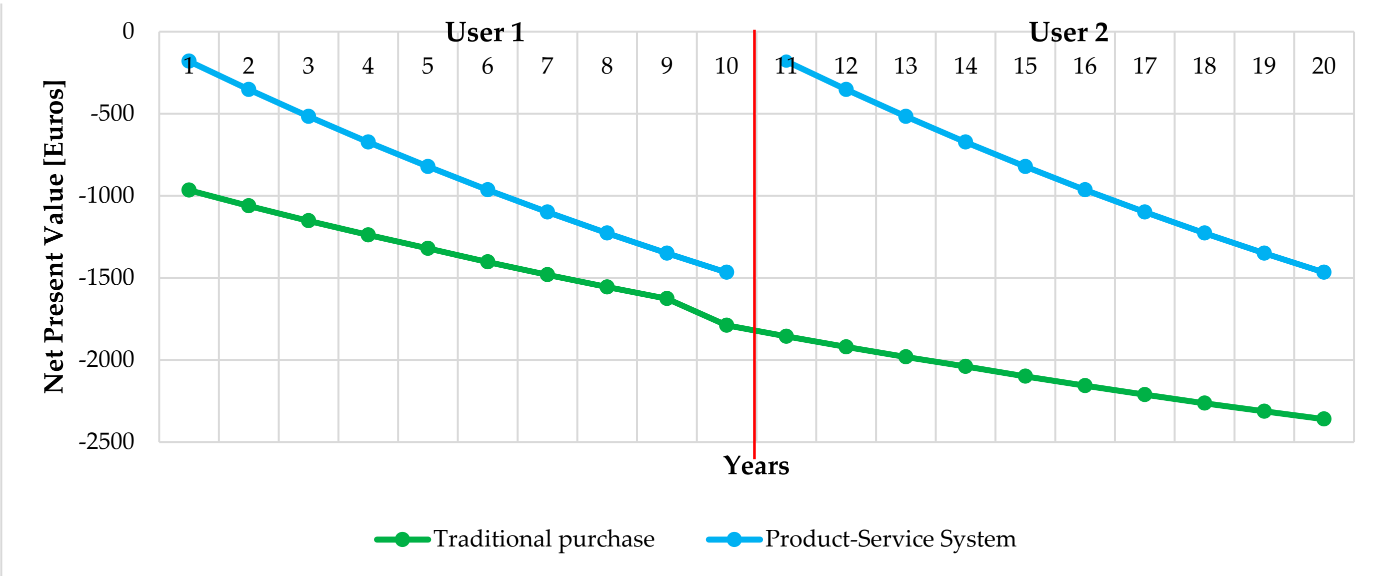


Figure 13. User perspective: Net Present Value for traditional sales and a Product-Service System—option for a 20-year garage box life cycle.

In both examined cases, the IRR increases. In the case of traditional sales, the ratio is already 130% in the first year and changes slightly in the third year to reach 132%. This is due to the fact that the highest flow occurs at the very beginning. In the following years, it remains at the same level. This means that invested money brings profit immediately. In the case of PSS, the IRR in the first year is −61% and changes from year to year. After 10 years, the IRR is 38%, and after 20 years, it is 40%. The money invested does not bring immediate returns in this case. The average annual returns will increase over time.

$$MIRR = \sqrt[n]{\frac{\sum_{t=0}^{n} CIF_t (1+r)^{n-t}}{\sum_{t=0}^{n} \frac{COF_t}{(1+r)^t}}}$$

In the first year, the MIRR for traditional sales was 130%, while for the new model, it was –61%. In the case of PSS, the MIRR increases year on year to reach 17% seven years later, i.e., the level higher than in the traditional sales option, which decreases year on year. According to the MIRR indicator, PSS is a more favourable solution for the manufacturer because its value both after 10 years (15%) and after 20 years (11%) is higher than in the traditional sales option (after 10 years (11%), after 20 years (7%)).

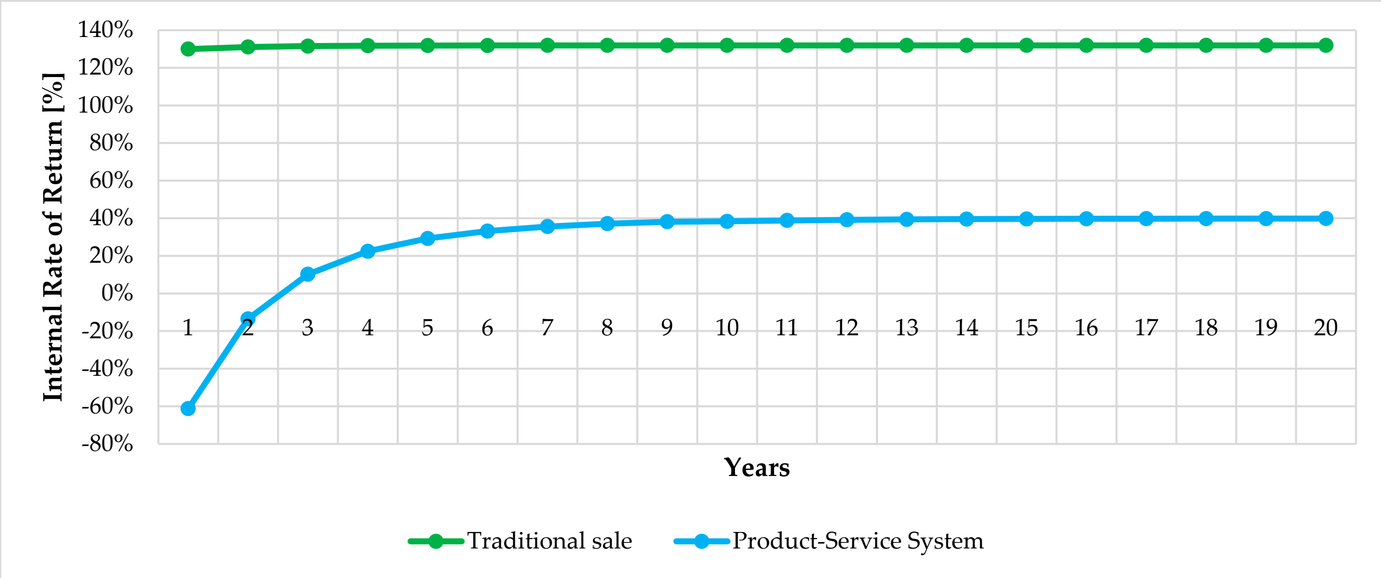


Figure 14. Manufacturer perspective—Internal Rate of Return (IRR) that compares traditional sales and a Product-Service System—an option for 20-year lifetime of a garage box.

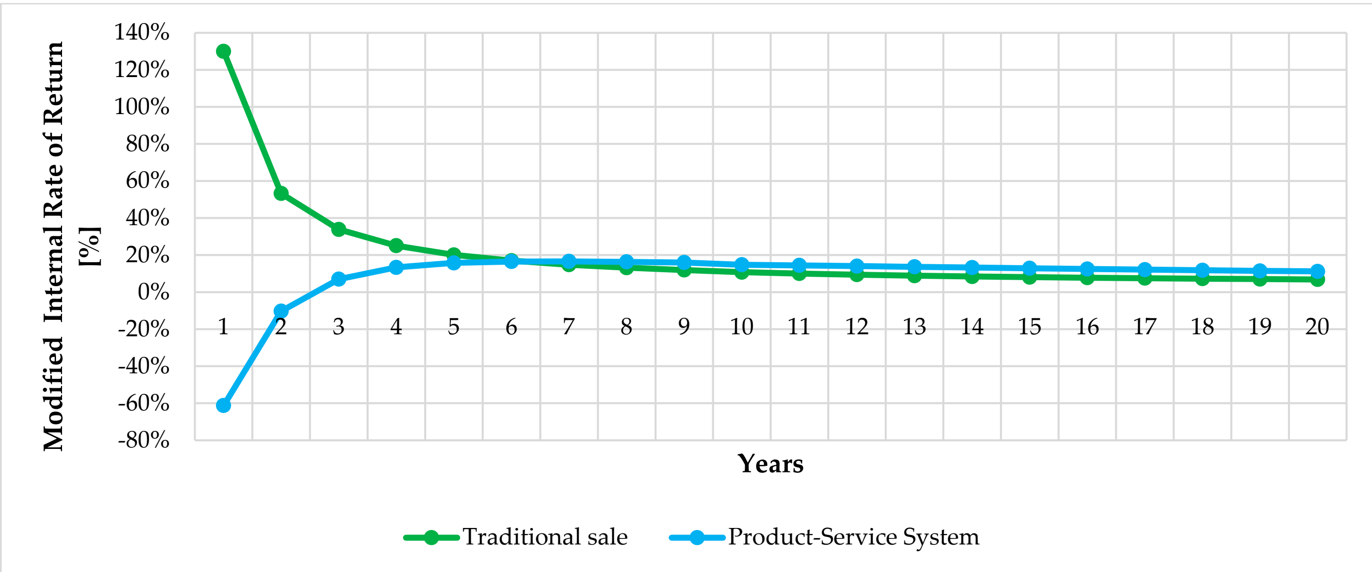


Figure 15. Manufacturer perspective: Modified Internal Rate of Return (MIRR) for traditional sales and a Product-Service System—option for a 20-year garage box life cycle.

8. Discussion and Conclusions

8.1. Discussion

The main problem, which is the focus of this article, is the lack of storage space for residents' belongings, as reported by tenants of blocks of flats. This is due to the absence of storage rooms that would be assigned to each flat and the unwillingness of the local administration to allow garage boxes to be installed in parking spaces. Through research and consultations with residents of the blocks of flats, a PSS business model for garage boxes was developed. It was created to provide storage space for tenants. It meets the specific needs of users and ensures that all the guidelines imposed by homeowner associations and fire regulations are met.

This research contributes to the development of several components of the PSS that integrate two industries, parking furniture and modern housing, to complement each other. All this is aimed at providing a tool to solve the problems of residents of blocks of flats without putting pressure on property developers' budgets. It combines simple parking furniture, services, and new technologies. The system emphasises collaboration between stakeholders as an element through which solutions can be created which require compliance with many requirements. The research also shows the growth potential for the new business model and the interest in it expressed by potential users.

In addition to extra services, features that impact the level of approval for the business model discussed in this article by customers in addition to extra services include security of storage of residents' possessions, equipment options, a mobile application and agreements with homeowner associations. The combination of these elements gives customers access to safe and legal storage space at an affordable price that they can check out anytime, anywhere—encouraging the use of the proposed model.

Several sets of equipment and services respond to the problems and needs of users. The manufacturer of garage boxes gives the customer a wide choice of services and additional equipment. The flexibility of choice has a positive effect on the understanding and acceptance of PSS. The services and equipment proposed in packages are also closely linked to the fulfilment of the installation requirements laid down by homeowner associations.

The PSS business model discussed in this paper can impact the development of the parking and garage furniture industry. A tailor-made solution and building partnerships with homeowner associations can contribute to the spread of this solution. Additionally, it will have an effect on how garage boxes are designed and developed. Moreover, in case of moving or loss of liquidity by the user, he/she can withdraw from the contract, which is a very attractive option. In the case at hand, the model shifts the manufacturer towards being a service provider, operator and supplier of parking furniture systems. This means the manufacturer focuses on many more activities than the production of the garage boxes itself.

This approach is beneficial for the main stakeholders: manufacturers, users, homeowner associations and real estate developers. From a manufacturer's perspective, one might expect that the model could increase the number of customers and, therefore, garage boxes installed in car parks or garages. By increasing the installed product base, creating closed-loop recycling and making garage boxes available again, the manufacturer will reap greater benefits than with traditional sales. This will allow them to work on improving the solution by optimising financing mechanisms and developing a better pricing strategy, applications and return logistics. In this type of business model, a group of residents installing garage boxes come together to form a cooperative, which increases the purchasing power and produces the economies of scale effects and thus offers promising research prospects. From the customer's perspective, it eliminates the need to invest considerable financial resources in the purchase of a garage box. The user also does not bear the costs associated with additional equipment and maintenance. The solution also guarantees the safety of stored items. Thanks to agreements with homeowner associations, it eliminates problems, risks and consequences associated with installation without permission. From the property developer's perspective, the developed model may turn out to be a success. It solves the problem of unsatisfied residents who do not have a storage room assigned to their flats. This solution may turn out to be particularly beneficial for property developers in large cities, where land prices are high, and the heights of constructed residential buildings are regulated by law. From the perspective of the homeowner association, it may support work and decision-making concerning car park management, as well as inform about dangerous situations. This is all thanks to the use of new technologies that provide them with tools to fully monitor car parks or garages.

The development of maintenance and service schedules will extend the life cycle of garage boxes. By creating a closed-loop recycling, the manufacturer will be able to remanufacture and make the garage box available again. The use of recycling, especially of steel components and parts, which are the main materials used in the manufacture of garage boxes, will reduce the amount of waste. This is important because recycling of steel and steel products, compared to production from virgin raw materials, provides energy savings of 74%, reduces atmospheric emissions by up to 86% and saves raw materials such as coke and water. As a result, recycled steel is also cheaper. Recycling also produces by-products, such as minerals, e.g., steelmaking slag and foundry furnace slag. These are used to build roads, enrich soil and produce industrial raw materials.

The business model presented here provides parking furniture manufacturers with the solution that residents of blocks of flats are very much looking for. It focuses on cooperation and information exchange between stakeholders. The implementation of the developed concept, regardless of the equipment option and the set of services, brings benefits for each stakeholder and the environment (Table 9). The ten-year lease cycles of the garage box and the ownership enable the manufacturer to keep it under constant observation, react to damage and repair it or replace parts. By replacing, remanufacturing, and reusing the same parts, the solution helps develop environmentally friendly maintenance activities. In addition, the data obtained is taken into account when designing new products and solutions. The results of all statistical analyzes are presented in Tables 10 and 11.

Table 10. Benefits of the proposed PSS business model.

Customer	Manufacturer	Homeowner Association	Developer	Environment
Garage box with equipment components required by the customer	The first PSS model for garage boxes in the market	Continuous monitoring of a garage/parking space in the building	Elimination of costs involved in building storage rooms for tenants	Waste reduction
A tailor-made solution	Opportunity to attract new clients	Information about hazardous substances	Reduction of construction costs	Reverse Logistics and re-use
No need to invest in traditional purchase of a garage box	Entry into the service market—garage box rental	Automated notification sent to appropriate emergency services	Solution to the problem of lack of additional storage space	Several customers may use one garage box—the exploitation period gets extended
No conflicts over installation with the homeowner association	Operations in the new market of second-hand garage boxes	No additional investment		Using recycled materials
Access to mobile application	Life cycle analysis	Profit from permits issued to users		Garage box remanufacturing
Alarm, protection and monitoring of stored items	Analysing how garage boxes are used by customers	Access to Internet portal		Replacing only necessary parts
Improved efficiency of using storage space and monitoring storage box occupancy	Data showing what equipment components and services related with them customers need			Reduced pollution emission to air
Ability to store items that do not fit in the flat	Identification of the most vulnerable components of garage boxes and making them more resilient			Energy savings compared to manufacturing from virgin raw material
No law infringement problems: responsibility for hazardous situations, losses in assets, damages and all related costs remain on the side of the owner	Much longer period of receiving returns from a single product			Saving other raw materials used in steel production, such as coke or water

The Canvas Business Model was used to identify the manufacturer's activity areas that are subject to change by the PSS business model described in this paper compared to a traditional operating mode (Table 11).

Table 11. Garage box manufacturer business model: changes.

Canvas Business Model Element	Traditional Business Model	PSS Business Model for Garage Boxes
Key Partners	Close cooperation with the suppliers of steel, plastic components and semi-finished products.	Cooperation with a wider group of partners. Relations with property developers, examination of their plans. Establishing close relations with homeowner associations. The need to inform law enforcement authorities in hazardous situations.
Key Activities	Design, manufacturing and installation of garage boxes.	Cover not just the design, manufacturing and installation of garage boxes but also a series of services linked with garage boxes and managing them. Focus is placed on the development of a new offer and delivering solutions expected by the client. Reverse logistics, remanufacturing and making a garage box available again together with the system of waste management make one of core elements. Another important component is the development of an application and smart solutions used by garage boxes. An additional aspect is the collection of data from property developers and working out agreements with homeowner associations.
Value Proposition	A garage box sold to residents of blocks of flats and other interested clients. The ownership is transferred to the user.	Access to storage space—a garage box is made available to the client with different equipment options together with services. Agreements with homeowner associations that allow for lawful installation. A mobile application and access to the garage box anytime from any place. The ownership rests with the manufacturer, the client gets access to the storage space of a garage box.
Customer Relationships	Centred around the sales transaction of a garage box.	Focus on long-term relations with customers. Cooperation and information exchange concerning the development of new garage boxes, improved application operation and the development of garage boxes network. Collaboration in the field of monitoring and product control.
Customer Segments	Residents of blocks of flats and terraced houses who own parking spaces and want to own a garage box.	Primarily residents of blocks of flats and terraced houses who own parking spaces and need storage space. Owning a garage box is not their major concern. They do not want to have problems connected with illegal installation, but they want to store their possessions safely and know what they are storing.
Key resources	Workers, machines and equipment, materials and semi-finished products, utilities.	Critical Key resources remain unchanged. Expanded scope of activities shifts the principal role to workers responsible for the surveillance over all installed garage boxes, maintenance services, as well as employees responsible for: agreements with homeowner associations and partners as well as the maintenance of application and the IT system. Means of transportation and equipment that enables remote control of the product are vital components of the solution.
Channels	Direct contacts with interested clients until a product is sold.	Relations with property developers and homeowner associations. Direct contact with every garage box user throughout the duration of the agreement.
Cost structure	Cost of the purchase of materials and semi-finished products; cost of manufacturing and installation, cost of labour and utilities.	Cost of the purchase of materials and semi-finished products; cost of manufacturing and installation of garage boxes. Cost of servicing, maintenance, repair and remanufacturing. Cost of administration, drafting and maintaining of application together with elements connected with it, cost of labour and utilities.
Revenue streams	Revenue comes mainly from sales of the garage boxes and their additional equipment.	Revenue comes from monthly subscription fee paid for making garage boxes available.

The use of new technologies and the resulting digitisation of the economy are opportunities for such solutions. Cameras and sensors installed on a garage box monitor what is going on in the car park or garage. The data is sent to the homeowner association on an ongoing basis, and in the event of a dangerous situation or threat to people's health and life, all the relevant emergency services are automatically informed. This helps the homeowner association to manage their car parks/garages, making it easier to look after the space. Thanks to intelligent solutions, the garage box not only stores things that do not fit in the flat but also protects and monitors them. The use of these technologies means that the manufacturer, users and apartment communities are fully synchronised, ensuring that no one misses anything and eliminating potential risks. All this and more in terms of ensuring security and meeting the requirements of the homeowner associations.

The future may witness a continued reduction in the number of tenant storage rooms in newly constructed residential buildings. Perhaps storage solutions will emerge that will be dedicated to several blocks of flats. Each resident will have storage space allocated to her/him for which they will pay depending on the volume of items stored. Sharing or pooling storage space for residents of blocks of flats or storing things on demand could also become popular solutions. As it is today, residents will not have access to the allocated storage space, as others will be able to use it when it is free.

8.2. Conclusions

The article focuses on answering the research question: "What opportunities does a PSS offer to parking furniture manufacturers, residents of blocks of flats (customers) and residential property management administration?"

One of the first surveys of residents of blocks of flats (customers) was carried out to develop a PSS business model for garage boxes. Below, we list its major findings:

- Residents of blocks of flats suffer from the lack of storage space for items that do not fit in the flats. This is due to the lack of storage rooms assigned to each flat. An alternative solution to this situation is to install garage boxes in parking spaces.
- At present, the customer cannot count on additional services when purchasing a garage box. Additional equipment is chargeable.
- The problem is that the homeowner associations do not permit residents to install garage boxes. Hence, residents who install them in parking spaces without permission may suffer consequences.
- Residents of blocks of flats are interested in PSS for garage boxes—75% of the respondents expressed interest. With this solution, they receive a secure storage space that they can access at any time.
- Every household will be able to benefit from the developed model as it will not be a big financial burden for them.
- When developing the PSS business model for garage boxes, it is advisable for the manufacturer to create sets of services and equipment items that address different customer profiles. This will ensure flexibility and choice.

The results of the survey, as mentioned above, indicate that the residents of new blocks of flats experience a problem with storing their belongings due to the lack of storage rooms assigned to the flats. The results of the survey allowed us to design a preliminary version of the PSS business model for garage boxes. The initial framework was consulted with representatives of the manufacturer, homeowner associations, retail trade, finance and residents of blocks of flats. The previously developed business model was then improved with their suggestions. It aims to ensure that residents have access to cheap and secure storage space available in their parking spaces. This is a solution to the previously mentioned problems that every resident can afford without having to invest a lot of money. The model developed is a potential alternative to having storage rooms.

The revised business model is the answer to the research question posed in this paper. Its design allowed us to find a set of specific elements that respond to the needs of residents of blocks of flats looking for additional storage space. While identifying these elements, the characteristics differentiating individual customers were taken into account (e.g., gender, age, size of the city in which they live, etc.). This allows the model to be applied to virtually any group of residents. Using such a model, the manufacturer of garage boxes can obtain accurate information on the opportunities, advantages and risks of implementing a PSS.

The directions of the authors' future work will focus on the development of a PSS design method and its validation in industrial settings. Further surveys related to the development of new business models for other industries are also planned.

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Acknowledgments: "Welcome, Queen, in the sunshine; A wreath of twelve stars above Your head. Welcome, Queen, in the sunshine: Stay with us, stay with us, until the end!"

Conflicts of Interest: The authors declare no conflicts of interest.

Appendix A.

Appendix A.1. One-Way ANOVA—Calculations

Table A1. One-Way ANOVA—calculations—sex [A] (female [A1], male [A2]); marital status [B] (in a relationship [B1], single [B2]); having children [C] (yes [C1], no [C2]).

		Sex						Marital Status						Having Children					
	[AS]	Fisher Statistics	*p*-Value	Female [A1]		Male [A2]		Fisher Statistics	*p*-Value	In a Relationship [B1]		Single [B2]		Fisher Statistics	*p*-Value	Have Children [C1]		Have Not Children [C2]	
				Mean [A11]	Standard Deviation [A12]	Mean [A21]	Standard Deviation [A22]			Mean [B11]	Standard Deviation [B12]	Mean [B21]	Standard Deviation [B22]			Mean [C11]	Standard Deviation [C12]	Mean [C21]	Standard Deviation [C22]
[AS]	AS 1	0.19	0.663	0.7	0.459	0.682	0.467	0.08	0.778	0.698	0.46	0.687	0.465	2.561	0.11	0.728	0.446	0.662	0.474
	AS 2	0.571	0.45	0.525	0.5	0.491	0.501	1.723	0.19	0.478	0.501	0.537	0.5	1.709	0.192	0.478	0.501	0.537	0.5
	AS 3	1.839	0.176	0.543	0.802	0.459	0.499	0.362	0.548	0.526	0.852	0.489	0.501	0.118	0.731	0.478	0.501	0.537	0.5
	AS 4	12.87	0	0.239	0.427	0.386	0.488	5.959	0.015	0.358	0.48	0.257	0.438	4.374	0.037	0.351	0.478	0.265	0.442
	AS 5	1.198	0.274	0.318	0.466	0.273	0.446	0.001	0.979	0.297	0.458	0.299	0.458	0.145	0.703	0.289	0.455	0.305	0.461
	AS 6	5.845	0.016	0.243	0.43	0.341	0.475	0.217	0.641	0.276	0.448	0.295	0.457	1.52	0.218	0.259	0.439	0.309	0.463
	AS 7	2.188	0.14	0.286	0.453	0.227	0.42	0.459	0.498	0.246	0.431	0.272	0.446	2.878	0.09	0.224	0.418	0.29	0.455
	AS 8	0.76	0.384	0.243	0.43	0.277	0.449	0.144	0.704	0.25	0.434	0.265	0.442	0.733	0.392	0.276	0.448	0.243	0.429
	AS 9	2.259	0.133	0.257	0.438	0.2	0.401	4.377	0.037	0.19	0.393	0.269	0.444	2.153	0.143	0.202	0.402	0.257	0.438
	AS 10	2.258	0.134	0.211	0.409	0.268	0.444	3.426	0.065	0.198	0.4	0.269	0.444	0.647	0.422	0.219	0.415	0.25	0.434
	AS 11	0.007	0.932	0.229	0.421	0.232	0.423	0.006	0.939	0.228	0.421	0.231	0.422	0.537	0.464	0.215	0.412	0.243	0.429
	AS 12	2.488	0.115	0.175	0.381	0.232	0.423	8.076	0.005	0.254	0.436	0.153	0.361	2.764	0.097	0.232	0.423	0.173	0.379
	AS 13	5.105	0.024	0.164	0.371	0.245	0.431	0.65	0.421	0.216	0.412	0.187	0.39	1.468	0.226	0.224	0.418	0.18	0.385
	AS 14	2.488	0.115	0.175	0.381	0.232	0.423	15.632	0	0.125	0.331	0.265	0.442	6.846	0.009	0.149	0.357	0.243	0.429
	AS 15	0.254	0.614	0.182	0.387	0.2	0.401	0.868	0.352	0.172	0.379	0.205	0.405	0.976	0.324	0.171	0.377	0.206	0.405
	AS 16	0.33	0.566	0.179	0.384	0.159	0.367	0.72	0.396	0.185	0.389	0.157	0.364	0.176	0.675	0.162	0.37	0.176	0.382
	AS 17	0.255	0.614	0.171	0.378	0.155	0.362	0	0.991	0.164	0.371	0.164	0.371	0.151	0.697	0.171	0.377	0.158	0.365
	AS 18	1.257	0.263	0.143	0.351	0.109	0.312	0.381	0.537	0.138	0.346	0.119	0.325	1.674	0.196	0.149	0.357	0.11	0.314
	AS 19	4.462	0.035	0.093	0.291	0.155	0.362	0.759	0.384	0.134	0.341	0.108	0.311	3.375	0.067	0.149	0.357	0.096	0.295
	AS 20	0.419	0.518	0.104	0.305	0.086	0.282	1.286	0.257	0.112	0.316	0.082	0.275	0.413	0.521	0.105	0.308	0.088	0.284
	AS 21	0.044	0.834	0.096	0.296	0.091	0.288	0.308	0.579	0.086	0.281	0.101	0.302	0.558	0.455	0.083	0.277	0.103	0.304
	AS 22	0.3	0.584	0.086	0.28	0.1	0.301	0.011	0.915	0.091	0.288	0.093	0.291	0.394	0.53	0.101	0.302	0.085	0.279
	AS 23	0.024	0.876	0.086	0.28	0.082	0.275	0.238	0.626	0.091	0.288	0.078	0.269	0.357	0.551	0.092	0.29	0.077	0.267
	AS 24	0.634	0.426	0.071	0.258	0.091	0.288	0.021	0.885	0.082	0.275	0.078	0.269	0.833	0.362	0.092	0.29	0.07	0.255
	AS 25	6.426	0.012	0.107	0.31	0.045	0.209	0.649	0.421	0.091	0.288	0.071	0.257	0.063	0.802	0.083	0.277	0.077	0.267
	AS 26	5.139	0.024	0.093	0.291	0.041	0.199	4.837	0.028	0.043	0.204	0.093	0.291	3.053	0.081	0.048	0.215	0.088	0.284
	AS 27	1.12	0.29	0.079	0.27	0.055	0.228	0.399	0.528	0.06	0.239	0.075	0.263	0.796	0.373	0.057	0.232	0.077	0.267
	AS 28	1.126	0.289	0.05	0.218	0.073	0.26	6.895	0.009	0.03	0.171	0.086	0.281	0.402	0.526	0.053	0.224	0.066	0.249
	AS 29	0.009	0.926	0.057	0.233	0.059	0.236	0.886	0.347	0.047	0.213	0.067	0.251	0.22	0.639	0.053	0.224	0.063	0.243
	AS 30	4.958	0.026	0.079	0.27	0.032	0.176	0.043	0.835	0.06	0.239	0.056	0.23	0.22	0.639	0.053	0.224	0.063	0.243
	AS 31	0.123	0.726	0.057	0.233	0.05	0.218	0.366	0.545	0.047	0.213	0.06	0.237	1.73	0.189	0.039	0.195	0.066	0.249
	AS 32	2.252	0.134	0.061	0.239	0.032	0.176	0.61	0.435	0.056	0.23	0.041	0.199	0.744	0.389	0.057	0.232	0.04	0.197

Table A1. *Cont.*

	[AE]	Fisher Statistics	*p*-value	[A1]		[A2]			Fisher Statistics	*p*-value	[B1]		[B2]			Fisher Statistics	*p*-value	[C1]		[C2]	
				[A11]	[A12]	[A21]	[A22]				[B11]	[B12]	[B21]	[B22]				[C11]	[C12]	[C21]	[C22]
[AE]	AE 1	0.529	0.467	0.504	0.501	0.536	0.5		4.526	0.034	0.569	0.496	0.474	0.5		0.039	0.843	0.513	0.501	0.522	0.5
	AE 2	3.126	0.078	0.475	0.5	0.555	0.498		0.056	0.813	0.504	0.501	0.515	0.501		0.589	0.443	0.491	0.501	0.526	0.5
	AE 3	0.968	0.326	0.471	0.5	0.427	0.496		0.854	0.356	0.474	0.5	0.433	0.496		0.505	0.478	0.469	0.5	0.438	0.497
	AE 4	1.556	0.213	0.446	0.498	0.391	0.489		0.119	0.73	0.414	0.494	0.429	0.496		0.002	0.969	0.421	0.495	0.423	0.495
	AE 5	6.027	0.014	0.425	0.495	0.318	0.467		0.003	0.955	0.379	0.486	0.377	0.486		0.92	0.338	0.355	0.48	0.397	0.49
	AE 6	0.585	0.445	0.332	0.472	0.3	0.459		3.162	0.076	0.358	0.48	0.284	0.452		3.361	0.067	0.36	0.481	0.283	0.451
	AE 7	9.779	0.002	0.261	0.44	0.391	0.489		0.002	0.966	0.319	0.467	0.317	0.466		0.232	0.63	0.307	0.462	0.327	0.47
	AE 8	0.778	0.378	0.264	0.442	0.3	0.459		3.21	0.074	0.241	0.429	0.313	0.465		0.935	0.334	0.259	0.439	0.298	0.458
	AE 9	1.85	0.174	0.286	0.453	0.232	0.423		0.594	0.441	0.246	0.431	0.276	0.448		0.933	0.334	0.241	0.429	0.279	0.45
	AE 10	0.741	0.39	0.261	0.44	0.227	0.42		5.211	0.023	0.293	0.456	0.205	0.405		2.725	0.099	0.281	0.45	0.217	0.413
	AE 11	2.668	0.103	0.214	0.411	0.277	0.449		2.061	0.152	0.272	0.446	0.216	0.413		0.349	0.555	0.254	0.436	0.232	0.423
	AE 12	5.098	0.024	0.2	0.401	0.286	0.453		5.619	0.018	0.19	0.393	0.28	0.45		4.709	0.03	0.193	0.396	0.276	0.448
	AE 13	0.082	0.774	0.243	0.43	0.232	0.423		4.264	0.039	0.28	0.45	0.201	0.402		2.664	0.103	0.272	0.446	0.21	0.408
	AE 14	0.105	0.747	0.211	0.409	0.223	0.417		1.133	0.288	0.237	0.426	0.198	0.399		5.543	0.019	0.263	0.441	0.176	0.382
	AE 15	6.924	0.009	0.171	0.378	0.268	0.444		0.536	0.465	0.228	0.421	0.201	0.402		1.099	0.295	0.193	0.396	0.232	0.423
	AE 16	4.683	0.031	0.221	0.416	0.145	0.353		5.722	0.017	0.233	0.424	0.149	0.357		2.693	0.101	0.219	0.415	0.162	0.369
	AE 17	0.535	0.465	0.157	0.365	0.182	0.387		0.055	0.815	0.168	0.374	0.164	0.371		0.028	0.868	0.171	0.377	0.165	0.372
	AE 18	0.185	0.667	0.15	0.358	0.136	0.344		2.683	0.102	0.116	0.321	0.168	0.374		0.002	0.966	0.145	0.353	0.143	0.351
	AE 19	7.853	0.005	0.093	0.291	0.177	0.383		0.05	0.823	0.134	0.341	0.127	0.333		2.048	0.153	0.154	0.361	0.11	0.314
	AE 20	1.079	0.299	0.125	0.331	0.095	0.295		0.317	0.574	0.103	0.305	0.119	0.325		1.667	0.197	0.092	0.29	0.129	0.335
	AE 21	0.719	0.397	0.107	0.31	0.132	0.339		0.435	0.51	0.108	0.311	0.127	0.333		0.001	0.979	0.118	0.324	0.118	0.323
	AE 22	0.241	0.624	0.1	0.301	0.114	0.318		0.03	0.863	0.103	0.305	0.108	0.311		0.059	0.809	0.11	0.313	0.103	0.304
	AE 23	1.047	0.307	0.114	0.319	0.086	0.282		1.177	0.279	0.086	0.281	0.116	0.32		2.434	0.119	0.079	0.27	0.121	0.327
	AE 24	2.646	0.104	0.05	0.218	0.086	0.282		0.941	0.333	0.078	0.268	0.056	0.23		1.138	0.287	0.079	0.27	0.055	0.229
	AE 25	0.228	0.634	0.054	0.226	0.064	0.245		0.35	0.555	0.065	0.246	0.052	0.223		0.464	0.496	0.066	0.248	0.051	0.221
	AE 26	3.639	0.057	0.057	0.233	0.023	0.149		0.606	0.437	0.034	0.183	0.049	0.215		1.328	0.25	0.031	0.173	0.051	0.221
	AE 27	1.656	0.199	0.05	0.218	0.027	0.163		0.108	0.742	0.043	0.204	0.037	0.19		0.942	0.332	0.031	0.173	0.048	0.214
	AE 28	0.067	0.797	0.032	0.177	0.036	0.188		0.192	0.661	0.03	0.171	0.037	0.19		0.015	0.902	0.035	0.184	0.033	0.179

Table A2. One-Way ANOVA—calculations—age [D] (18–25 years old [D1], 26–35 years old [D2], 36–45 years old [D3], 46–55 years old [D4], >55 years old [D5]); Level of education [E] (Ph.D. or higher [E1], higher [E2], secondary [E3], vocational [E4]).

		Age												Level of Education									
				18–25 Years Old [D1]		26–35 Years Old [D2]		36–45 Years Old [D3]		46–55 Years Old [D4]		>55 Years Old [D5]				PhD or Higher [E1]		Higher Education [E3]		Secondary Education [E3]		Vocational Education [E4]	
	[AS]	Fisher Statistics	p-Value	Mean [D11]	Standard Deviation [D12]	Mean [D21]	Standard Deviation [D22]	Mean [D31]	Standard Deviation [D32]	Mean [D41]	Standard Deviation [D42]	Mean [D51]	Standard Deviation [D52]	Fisher Statistics	p-Value	Mean [E11]	Standard Deviation [E12]	Mean [E21]	Standard Deviation [E22]	Mean [E31]	Standard Deviation [E32]	Mean [E41]	Standard Deviation [E42]
[AS]	AS 1	1.78	0.132	0.664	0.474	0.781	0.415	0.638	0.483	0.679	0.471	0.694	0.466	1.824	0.142	0.806	0.402	0.683	0.466	0.645	0.48	0.773	0.422
	AS 2	3.727	0.005	0.517	0.501	0.641	0.482	0.449	0.499	0.415	0.497	0.408	0.497	2.976	0.031	0.645	0.486	0.546	0.499	0.475	0.501	0.379	0.489
	AS 3	1.056	0.378	0.441	0.498	0.508	0.502	0.488	0.502	0.642	1.533	0.592	0.497	0.232	0.874	0.516	0.508	0.527	0.82	0.489	0.502	0.455	0.502
	AS 4	6.588	0	0.168	0.375	0.305	0.462	0.331	0.472	0.491	0.505	0.429	0.5	2.386	0.068	0.419	0.502	0.302	0.46	0.241	0.429	0.394	0.492
	AS 5	0.361	0.836	0.336	0.474	0.289	0.455	0.283	0.452	0.283	0.455	0.265	0.446	0.818	0.484	0.387	0.495	0.286	0.453	0.277	0.449	0.348	0.48
	AS 6	0.175	0.951	0.287	0.454	0.266	0.443	0.307	0.463	0.302	0.463	0.265	0.446	1.575	0.195	0.355	0.486	0.29	0.455	0.312	0.465	0.182	0.389
	AS 7	1.058	0.377	0.28	0.45	0.313	0.465	0.213	0.411	0.226	0.423	0.224	0.422	1.46	0.225	0.387	0.495	0.229	0.421	0.284	0.452	0.273	0.449
	AS 8	0.347	0.846	0.266	0.443	0.242	0.43	0.283	0.452	0.264	0.445	0.204	0.407	2.718	0.044	0.161	0.374	0.302	0.46	0.248	0.434	0.152	0.361
	AS 9	5.829	0	0.357	0.481	0.219	0.415	0.189	0.393	0.189	0.395	0.061	0.242	0.683	0.563	0.161	0.374	0.233	0.423	0.262	0.442	0.197	0.401
	AS 10	1.953	0.101	0.231	0.423	0.25	0.435	0.299	0.46	0.17	0.379	0.122	0.331	0.79	0.5	0.29	0.461	0.256	0.437	0.199	0.4	0.212	0.412
	AS 11	0.279	0.891	0.217	0.414	0.211	0.41	0.236	0.426	0.264	0.445	0.265	0.446	1.047	0.371	0.226	0.425	0.233	0.423	0.262	0.442	0.152	0.361
	AS 12	3.115	0.015	0.119	0.325	0.195	0.398	0.244	0.431	0.321	0.471	0.204	0.407	2.46	0.062	0.355	0.486	0.206	0.405	0.191	0.395	0.121	0.329
	AS 13	2.041	0.088	0.161	0.369	0.164	0.372	0.213	0.411	0.245	0.434	0.327	0.474	2.272	0.079	0.355	0.486	0.168	0.375	0.213	0.411	0.227	0.422
	AS 14	4.439	0.002	0.273	0.447	0.25	0.435	0.087	0.282	0.17	0.379	0.184	0.391	0.724	0.538	0.129	0.341	0.195	0.397	0.234	0.425	0.182	0.389
	AS 15	1.191	0.314	0.182	0.387	0.219	0.415	0.134	0.342	0.226	0.423	0.245	0.434	2.083	0.101	0.323	0.475	0.16	0.368	0.191	0.395	0.242	0.432
	AS 16	0.78	0.538	0.182	0.387	0.203	0.404	0.142	0.35	0.113	0.32	0.184	0.391	0.903	0.44	0.065	0.25	0.176	0.381	0.184	0.389	0.167	0.376
	AS 17	1.531	0.192	0.175	0.381	0.164	0.372	0.118	0.324	0.264	0.445	0.143	0.354	2.112	0.098	0.065	0.25	0.187	0.391	0.121	0.327	0.212	0.412
	AS 18	2.467	0.044	0.112	0.316	0.195	0.398	0.094	0.294	0.057	0.233	0.163	0.373	3.42	0.017	0.29	0.461	0.137	0.345	0.085	0.28	0.106	0.31
	AS 19	1.385	0.238	0.112	0.316	0.102	0.303	0.094	0.294	0.189	0.395	0.184	0.391	4.029	0.008	0.032	0.18	0.126	0.332	0.078	0.269	0.227	0.422
	AS 20	2.704	0.03	0.07	0.256	0.063	0.243	0.11	0.314	0.208	0.409	0.102	0.306	1.483	0.218	0.032	0.18	0.115	0.319	0.099	0.3	0.045	0.21
	AS 21	0.657	0.622	0.091	0.288	0.063	0.243	0.118	0.324	0.113	0.32	0.102	0.306	1.408	0.24	0.161	0.374	0.088	0.284	0.113	0.318	0.045	0.21
	AS 22	1.364	0.245	0.07	0.256	0.086	0.281	0.102	0.304	0.17	0.379	0.061	0.242	0.601	0.615	0.097	0.301	0.107	0.31	0.078	0.269	0.061	0.24
	AS 23	1.541	0.189	0.063	0.244	0.086	0.281	0.063	0.244	0.113	0.32	0.163	0.373	0.554	0.646	0.032	0.18	0.08	0.272	0.092	0.29	0.106	0.31
	AS 24	1.838	0.12	0.056	0.231	0.125	0.332	0.094	0.294	0.038	0.192	0.041	0.2	2.77	0.041	0.161	0.374	0.099	0.3	0.035	0.186	0.061	0.24
	AS 25	2.18	0.07	0.042	0.201	0.055	0.228	0.11	0.314	0.132	0.342	0.122	0.331	0.306	0.821	0.065	0.25	0.088	0.284	0.064	0.245	0.091	0.29
	AS 26	1.495	0.203	0.07	0.256	0.063	0.243	0.11	0.314	0.038	0.192	0.02	0.143	0.841	0.472	0	0	0.076	0.266	0.071	0.258	0.076	0.267
	AS 27	1.098	0.357	0.098	0.298	0.047	0.212	0.047	0.213	0.094	0.295	0.061	0.242	2.16	0.092	0	0	0.084	0.278	0.078	0.269	0.015	0.123
	AS 28	1.364	0.245	0.07	0.256	0.094	0.293	0.031	0.175	0.038	0.192	0.041	0.2	0.186	0.906	0.032	0.18	0.065	0.247	0.057	0.232	0.061	0.24
	AS 29	0.665	0.616	0.049	0.217	0.047	0.212	0.087	0.282	0.057	0.233	0.041	0.2	3.888	0.009	0.032	0.18	0.092	0.289	0.021	0.145	0.015	0.123
	AS 30	1.478	0.208	0.063	0.244	0.023	0.152	0.055	0.229	0.094	0.295	0.102	0.306	1.098	0.35	0.032	0.18	0.053	0.225	0.085	0.28	0.03	0.173
	AS 31	0.279	0.892	0.07	0.256	0.047	0.212	0.047	0.213	0.057	0.233	0.041	0.2	0.151	0.929	0.032	0.18	0.057	0.233	0.057	0.232	0.045	0.21
	AS 32	0.752	0.557	0.028	0.165	0.047	0.212	0.071	0.258	0.038	0.192	0.061	0.242	0.93	0.426	0.032	0.18	0.061	0.24	0.043	0.203	0.015	0.123

Table A2. *Cont.*

				[D1]		[D2]		[D3]		[D4]		[D5]					[E1]		[E2]		[E3]		[E4]	
	[AE]	Fisher Statistics	*p*-value	[D11]	[D11]	[D21]	[D21]	[D31]	[D31]	[D41]	[D41]	[D51]	[D52]		Fisher Statistics	*p*-value	[E11]	[E12]	[E21]	[E22]	[E31]	[E32]	[E41]	[E42]
	AE 1	1.367	0.244	0.524	0.501	0.477	0.501	0.504	0.502	0.66	0.478	0.49	0.505		1.595	0.19	0.613	0.495	0.55	0.498	0.454	0.5	0.485	0.504
	AE 2	2.805	0.025	0.448	0.499	0.609	0.49	0.465	0.501	0.453	0.503	0.612	0.492		1.711	0.164	0.677	0.475	0.519	0.501	0.489	0.502	0.439	0.5
	AE 3	0.565	0.688	0.455	0.5	0.406	0.493	0.496	0.502	0.472	0.504	0.429	0.5		1.015	0.386	0.581	0.502	0.458	0.499	0.411	0.494	0.455	0.502
	AE 4	1.806	0.126	0.497	0.502	0.438	0.498	0.386	0.489	0.377	0.489	0.306	0.466		2.157	0.092	0.258	0.445	0.424	0.495	0.482	0.501	0.364	0.485
	AE 5	1.191	0.314	0.343	0.476	0.336	0.474	0.449	0.499	0.415	0.497	0.367	0.487		3.125	0.026	0.484	0.508	0.408	0.492	0.277	0.449	0.424	0.498
	AE 6	3.268	0.012	0.301	0.46	0.266	0.443	0.315	0.466	0.528	0.504	0.286	0.456		0.834	0.476	0.387	0.495	0.294	0.456	0.319	0.468	0.379	0.489
	AE 7	1.619	0.168	0.28	0.45	0.398	0.492	0.268	0.445	0.321	0.471	0.347	0.481		1.118	0.341	0.452	0.506	0.324	0.469	0.291	0.456	0.288	0.456
	AE 8	0.464	0.762	0.301	0.46	0.242	0.43	0.299	0.46	0.302	0.463	0.245	0.434		2.213	0.086	0.258	0.445	0.298	0.458	0.312	0.465	0.152	0.361
	AE 9	0.646	0.63	0.266	0.443	0.258	0.439	0.228	0.421	0.264	0.445	0.347	0.481		0.543	0.653	0.29	0.461	0.267	0.443	0.227	0.42	0.303	0.463
	AE 10	0.42	0.794	0.217	0.414	0.242	0.43	0.26	0.44	0.302	0.463	0.245	0.434		0.158	0.925	0.29	0.461	0.248	0.433	0.241	0.429	0.227	0.422
	AE 11	0.74	0.565	0.231	0.423	0.211	0.41	0.26	0.44	0.226	0.423	0.327	0.474		1.815	0.143	0.355	0.486	0.214	0.411	0.234	0.425	0.318	0.469
	AE 12	1.769	0.134	0.273	0.447	0.289	0.455	0.213	0.411	0.17	0.379	0.143	0.354		1.352	0.257	0.194	0.402	0.221	0.416	0.234	0.425	0.333	0.475
	AE 13	1.655	0.159	0.217	0.414	0.211	0.41	0.236	0.426	0.377	0.489	0.224	0.422		0.178	0.911	0.194	0.402	0.248	0.433	0.234	0.425	0.227	0.422
	AE 14	2.78	0.026	0.196	0.398	0.172	0.379	0.268	0.445	0.132	0.342	0.347	0.481		8.656	0	0.484	0.508	0.248	0.433	0.106	0.309	0.197	0.401
[AE]	AE 15	1.966	0.099	0.168	0.375	0.195	0.398	0.213	0.411	0.34	0.478	0.265	0.446		2.324	0.074	0.161	0.374	0.206	0.405	0.184	0.389	0.333	0.475
	AE 16	1.473	0.209	0.189	0.393	0.164	0.372	0.15	0.358	0.283	0.455	0.245	0.434		0.787	0.501	0.097	0.301	0.187	0.391	0.191	0.395	0.227	0.422
	AE 17	2.97	0.019	0.189	0.393	0.18	0.385	0.079	0.27	0.208	0.409	0.265	0.446		0.464	0.707	0.129	0.341	0.168	0.375	0.156	0.364	0.212	0.412
	AE 18	1.296	0.271	0.175	0.381	0.094	0.293	0.165	0.373	0.17	0.379	0.102	0.306		0.624	0.6	0.129	0.341	0.141	0.349	0.128	0.335	0.197	0.401
	AE 19	1.59	0.176	0.098	0.298	0.094	0.293	0.173	0.38	0.151	0.361	0.184	0.391		2.886	0.035	0.129	0.341	0.141	0.349	0.071	0.258	0.212	0.412
	AE 20	2.602	0.035	0.112	0.316	0.18	0.385	0.079	0.27	0.038	0.192	0.102	0.306		1.159	0.325	0.129	0.341	0.118	0.324	0.128	0.335	0.045	0.21
	AE 21	3.17	0.014	0.063	0.244	0.141	0.349	0.189	0.393	0.075	0.267	0.082	0.277		1.066	0.363	0.065	0.25	0.141	0.349	0.092	0.29	0.106	0.31
	AE 22	2.766	0.027	0.084	0.278	0.172	0.379	0.11	0.314	0.075	0.267	0.02	0.143		1.617	0.184	0.032	0.18	0.13	0.337	0.099	0.3	0.061	0.24
	AE 23	2.061	0.085	0.126	0.333	0.102	0.303	0.071	0.258	0.038	0.192	0.184	0.391		1.814	0.144	0.161	0.374	0.103	0.305	0.121	0.327	0.03	0.173
	AE 24	3.824	0.005	0.056	0.231	0.016	0.125	0.071	0.258	0.113	0.32	0.163	0.373		6.135	0	0	0	0.05	0.218	0.057	0.232	0.182	0.389
	AE 25	1.58	0.178	0.077	0.267	0.023	0.152	0.087	0.282	0.038	0.192	0.041	0.2		5.293	0.001	0.032	0.18	0.023	0.15	0.099	0.3	0.121	0.329
	AE 26	0.518	0.723	0.028	0.165	0.063	0.243	0.039	0.195	0.038	0.192	0.041	0.2		2.476	0.061	0.032	0.18	0.065	0.247	0.014	0.119	0.015	0.123
	AE 27	0.355	0.84	0.035	0.184	0.055	0.228	0.039	0.195	0.019	0.137	0.041	0.2		4.048	0.007	0	0	0.069	0.253	0.007	0.084	0.015	0.123
	AE 28	0.453	0.771	0.042	0.201	0.031	0.175	0.024	0.152	0.057	0.233	0.02	0.143		1.115	0.342	0.065	0.25	0.038	0.192	0.035	0.186	0	0

Table A3. One-Way ANOVA—calculations—level of monthly earnings [F] (≤250 Euro [F1], 251–500 Euro [F2], 501–1000 Euro [F3], 1001–1500 Euro [F4], 1501–2000 Euro [F5], 2001–2500 Euro [F6], 2501–4000 Euro [F7], >4000 Euro [F8]); place of residence [G] (city of ≤20,000 inhabitants [G1], city of 20.001–100,000 inhabitants [G2], city of 100,001–500,000 inhabitants [G3], city > 500,000 inhabitants [G4]).

		Level of monthly earnings																	
	[AS]	Fisher Statistics	*p*-Value	≤250 € [F1]		251–500 € [F2]		501–1000 € [F3]		1001–1500 € [F4]		1501–2000 € [F5]		2001–2500 € [F6]		2501–4000 € [F7]		>4000 € [F8]	
				Mean [F11]	Standard Deviation [F12]	Mean [F21]	Standard Deviation [F22]	Mean [F31]	Standard Deviation [F32]	Mean [F41]	Standard Deviation [F42]	Mean [F51]	Standard Deviation [F52]	Mean [F61]	Standard Deviation [F62]	Mean [F71]	Standard Deviation [F72]	Mean [F81]	Standard Deviation [F82]
	AS 1	0.742	0.636	0.721	0.454	0.683	0.469	0.67	0.471	0.657	0.477	0.839	0.374	0.756	0.435	0.667	0.485	0.688	0.479
	AS 2	1.484	0.17	0.395	0.495	0.556	0.501	0.5	0.501	0.463	0.501	0.484	0.508	0.667	0.477	0.5	0.514	0.688	0.479
	AS 3	0.662	0.704	0.512	0.506	0.492	0.504	0.489	0.501	0.481	0.502	0.452	0.506	0.711	1.646	0.5	0.514	0.438	0.512
	AS 4	1.246	0.276	0.302	0.465	0.381	0.49	0.301	0.46	0.213	0.411	0.387	0.495	0.311	0.468	0.444	0.511	0.313	0.479
	AS 5	0.503	0.833	0.349	0.482	0.302	0.463	0.318	0.467	0.241	0.43	0.258	0.445	0.289	0.458	0.389	0.502	0.313	0.479
	AS 6	2.261	0.028	0.256	0.441	0.302	0.463	0.256	0.437	0.333	0.474	0.29	0.461	0.133	0.344	0.444	0.511	0.563	0.512
	AS 7	2.305	0.026	0.395	0.495	0.302	0.463	0.284	0.452	0.167	0.374	0.226	0.425	0.333	0.477	0.056	0.236	0.188	0.403
	AS 8	2.188	0.034	0.186	0.394	0.397	0.493	0.188	0.391	0.315	0.467	0.258	0.445	0.311	0.468	0.222	0.428	0.188	0.403
	AS 9	1.62	0.127	0.326	0.474	0.27	0.447	0.227	0.42	0.176	0.383	0.387	0.495	0.222	0.42	0.111	0.323	0.125	0.342
	AS 10	1.061	0.387	0.302	0.465	0.286	0.455	0.199	0.4	0.185	0.39	0.29	0.461	0.267	0.447	0.278	0.461	0.375	0.5
	AS 11	0.997	0.433	0.186	0.394	0.302	0.463	0.193	0.396	0.231	0.424	0.258	0.445	0.2	0.405	0.333	0.485	0.375	0.5
	AS 12	2.623	0.011	0.093	0.294	0.238	0.429	0.153	0.361	0.194	0.398	0.258	0.445	0.289	0.458	0.222	0.428	0.5	0.516
	AS 13	1.097	0.363	0.116	0.324	0.19	0.396	0.193	0.396	0.278	0.45	0.194	0.402	0.133	0.344	0.167	0.383	0.25	0.447
	AS 14	1.381	0.211	0.209	0.412	0.254	0.439	0.25	0.434	0.148	0.357	0.161	0.374	0.133	0.344	0.056	0.236	0.188	0.403
	AS 15	2.48	0.016	0.186	0.394	0.349	0.481	0.21	0.409	0.111	0.316	0.129	0.341	0.156	0.367	0.111	0.323	0.188	0.403
	AS 16	0.781	0.604	0.093	0.294	0.159	0.368	0.182	0.387	0.176	0.383	0.258	0.445	0.178	0.387	0.056	0.236	0.188	0.403
[AS]	AS 17	0.86	0.538	0.093	0.294	0.206	0.408	0.199	0.4	0.148	0.357	0.161	0.374	0.133	0.344	0.111	0.323	0.063	0.25
	AS 18	0.844	0.551	0.233	0.427	0.127	0.336	0.108	0.311	0.13	0.337	0.161	0.374	0.111	0.318	0.111	0.323	0.063	0.25
	AS 19	1.288	0.254	0.047	0.213	0.159	0.368	0.119	0.325	0.111	0.316	0.097	0.301	0.222	0.42	0.056	0.236	0.063	0.25
	AS 20	0.967	0.455	0.023	0.152	0.095	0.296	0.074	0.262	0.12	0.327	0.129	0.341	0.156	0.367	0.111	0.323	0.125	0.342
	AS 21	0.839	0.555	0.116	0.324	0.048	0.215	0.085	0.28	0.083	0.278	0.129	0.341	0.156	0.367	0.167	0.383	0.063	0.25
	AS 22	1.193	0.305	0.07	0.258	0.127	0.336	0.074	0.262	0.083	0.278	0.129	0.341	0.178	0.387	0.056	0.236	0	0
	AS 23	0.562	0.787	0.07	0.258	0.095	0.296	0.097	0.296	0.083	0.278	0.129	0.341	0.044	0.208	0.056	0.236	0	0
	AS 24	0.599	0.757	0.093	0.294	0.079	0.272	0.063	0.243	0.102	0.304	0.129	0.341	0.089	0.288	0	0	0.063	0.25
	AS 25	0.507	0.829	0.07	0.258	0.079	0.272	0.08	0.271	0.074	0.263	0.065	0.25	0.133	0.344	0.111	0.323	0	0
	AS 26	0.804	0.584	0.023	0.152	0.079	0.272	0.091	0.288	0.056	0.23	0.032	0.18	0.089	0.288	0.111	0.323	0	0
	AS 27	2.657	0.01	0.093	0.294	0.016	0.126	0.091	0.288	0.056	0.23	0.097	0.301	0	0	0	0	0.25	0.447
	AS 28	0.73	0.647	0.07	0.258	0.111	0.317	0.063	0.243	0.046	0.211	0.032	0.18	0.044	0.208	0	0	0.063	0.25
	AS 29	1.517	0.159	0.023	0.152	0.063	0.246	0.074	0.262	0.046	0.211	0.097	0.301	0	0	0.167	0.383	0	0
	AS 30	1.335	0.231	0.093	0.294	0.016	0.126	0.085	0.28	0.056	0.23	0.065	0.25	0	0	0	0	0.063	0.25
	AS 31	0.658	0.707	0.093	0.294	0.048	0.215	0.068	0.253	0.037	0.19	0.065	0.25	0.022	0.149	0	0	0.063	0.25
	AS 32	0.983	0.443	0.023	0.152	0.016	0.126	0.051	0.221	0.074	0.263	0.097	0.301	0.022	0.149	0	0	0.063	0.25

		Place of residence									
	[AS]	Fisher Statistics	*p*-Value	City ≤ 20,000 Inhabitants [G1]		City 20.001–100,000 Inhabitants [G2]		City 100,001–500,000 Inhabitants [G3]		City > 500,000 Inhabitants [G4]	
				Mean [G11]	Standard Deviation [G12]	Mean [G21]	Standard Deviation [G22]	Mean [G31]	Standard Deviation [G32]	Mean [G41]	Standard Deviation [G42]
	AS 1	0.591	0.621	0.678	0.469	0.68	0.469	0.75	0.435	0.679	0.468
	AS 2	3.117	0.026	0.6	0.492	0.402	0.493	0.467	0.502	0.531	0.5
	AS 3	1.165	0.323	0.426	0.497	0.485	0.502	0.489	0.503	0.571	0.9
	AS 4	0.683	0.562	0.339	0.475	0.32	0.469	0.25	0.435	0.301	0.46
	AS 5	2.571	0.054	0.304	0.462	0.402	0.493	0.228	0.422	0.276	0.448
	AS 6	1.34	0.261	0.296	0.458	0.206	0.407	0.326	0.471	0.301	0.46
	AS 7	2.158	0.092	0.252	0.436	0.165	0.373	0.293	0.458	0.296	0.458
	AS 8	0.348	0.791	0.278	0.45	0.237	0.428	0.228	0.422	0.27	0.445
	AS 9	1.48	0.219	0.296	0.458	0.175	0.382	0.228	0.422	0.224	0.418
	AS 10	3.488	0.016	0.322	0.469	0.134	0.342	0.228	0.422	0.24	0.428
	AS 11	3.962	0.008	0.148	0.356	0.165	0.373	0.272	0.447	0.291	0.455
	AS 12	1.14	0.332	0.174	0.381	0.155	0.363	0.25	0.435	0.214	0.411
	AS 13	2.621	0.05	0.122	0.328	0.268	0.445	0.185	0.39	0.219	0.415
	AS 14	2.989	0.031	0.296	0.458	0.186	0.391	0.152	0.361	0.173	0.38
	AS 15	0.942	0.42	0.235	0.426	0.144	0.353	0.185	0.39	0.189	0.392
	AS 16	1.441	0.23	0.13	0.338	0.227	0.421	0.196	0.399	0.153	0.361
[AS]	AS 17	0.56	0.642	0.191	0.395	0.186	0.391	0.152	0.361	0.143	0.351
	AS 18	0.548	0.65	0.122	0.328	0.093	0.292	0.141	0.35	0.143	0.351
	AS 19	0.973	0.405	0.139	0.348	0.103	0.306	0.076	0.267	0.138	0.346
	AS 20	1.286	0.278	0.096	0.295	0.113	0.319	0.043	0.205	0.112	0.316
	AS 21	3.455	0.016	0.139	0.348	0.031	0.174	0.054	0.228	0.117	0.323
	AS 22	2.733	0.043	0.148	0.356	0.041	0.2	0.065	0.248	0.097	0.297
	AS 23	0.647	0.585	0.104	0.307	0.103	0.306	0.076	0.267	0.066	0.249
	AS 24	0.383	0.765	0.087	0.283	0.062	0.242	0.065	0.248	0.092	0.29
	AS 25	0.166	0.92	0.078	0.27	0.093	0.292	0.065	0.248	0.082	0.275
	AS 26	0.369	0.775	0.061	0.24	0.052	0.222	0.076	0.267	0.082	0.275
	AS 27	2.072	0.103	0.087	0.283	0.113	0.319	0.043	0.205	0.046	0.21
	AS 28	1.215	0.304	0.07	0.256	0.052	0.222	0.022	0.147	0.077	0.267
	AS 29	0.261	0.854	0.061	0.24	0.041	0.2	0.054	0.228	0.066	0.249
	AS 30	2.525	0.057	0.078	0.27	0.103	0.306	0.043	0.205	0.031	0.173
	AS 31	1.668	0.173	0.061	0.24	0.021	0.143	0.033	0.179	0.077	0.267
	AS 32	2.821	0.038	0.096	0.295	0.052	0.222	0.033	0.179	0.026	0.158

Table A3. *Cont.*

	[AE]	Fisher Statistics	*p*-value	[F1]		[F2]		[F3]		[F4]		[F5]		[F6]		[F7]		[F8]			Fisher Statistics	*p*-value	[G1]		[G2]		[G3]		[G4]	
				[F11]	[F12]	[F21]	F22]	[F31]	[F32]	[F41]	[F42]	[F51]	[F52]	[F61]	[F62]	[F71]	[F72]	[F81]	[F82]				[G11]	[G12]	[G21]	[G22]	[G31]	[G32]	[G41]	[42]
	AE 1	1.614	0.129	0.465	0.505	0.492	0.504	0.46	0.5	0.546	0.5	0.484	0.508	0.644	0.484	0.722	0.461	0.688	0.479		2.507	0.058	0.409	0.494	0.546	0.5	0.576	0.497	0.541	0.5
	AE 2	1.711	0.104	0.395	0.495	0.476	0.503	0.494	0.501	0.556	0.499	0.387	0.495	0.556	0.503	0.778	0.428	0.625	0.5		0.72	0.54	0.496	0.502	0.505	0.503	0.457	0.501	0.546	0.499
	AE 3	1.598	0.134	0.395	0.495	0.476	0.503	0.494	0.501	0.556	0.499	0.387	0.495	0.556	0.503	0.778	0.428	0.625	0.5		1.355	0.256	0.452	0.5	0.381	0.488	0.424	0.497	0.5	0.501
	AE 4	1.364	0.218	0.372	0.489	0.476	0.503	0.42	0.495	0.481	0.502	0.452	0.506	0.4	0.495	0.222	0.428	0.188	0.403		1.605	0.187	0.426	0.497	0.464	0.501	0.489	0.503	0.367	0.483
	AE 5	2.193	0.034	0.326	0.474	0.286	0.455	0.33	0.471	0.407	0.494	0.419	0.502	0.578	0.499	0.389	0.502	0.563	0.512		0.804	0.492	0.348	0.478	0.34	0.476	0.37	0.485	0.418	0.495
	AE 6	0.939	0.476	0.209	0.412	0.381	0.49	0.352	0.479	0.259	0.44	0.29	0.461	0.356	0.484	0.333	0.485	0.313	0.479		3.986	0.008	0.339	0.475	0.34	0.476	0.435	0.498	0.24	0.428
	AE 7	3.802	0	0.279	0.454	0.413	0.496	0.273	0.447	0.278	0.45	0.613	0.495	0.178	0.387	0.444	0.511	0.5	0.516		1.577	0.194	0.357	0.481	0.237	0.428	0.293	0.458	0.347	0.477
	AE 8	0.979	0.446	0.349	0.482	0.365	0.485	0.267	0.444	0.259	0.44	0.161	0.374	0.267	0.447	0.222	0.428	0.375	0.5		1.291	0.277	0.243	0.431	0.227	0.421	0.293	0.458	0.321	0.468
	AE 9	1.275	0.261	0.279	0.454	0.254	0.439	0.261	0.441	0.287	0.454	0.194	0.402	0.2	0.405	0.5	0.514	0.125	0.342		2.473	0.061	0.243	0.431	0.32	0.469	0.163	0.371	0.291	0.455
	AE 10	0.592	0.763	0.233	0.427	0.206	0.408	0.244	0.431	0.269	0.445	0.323	0.475	0.289	0.458	0.167	0.383	0.125	0.342		2.393	0.068	0.278	0.45	0.216	0.414	0.152	0.361	0.286	0.453
	AE 11	1.489	0.169	0.186	0.394	0.175	0.383	0.261	0.441	0.25	0.435	0.226	0.425	0.178	0.387	0.333	0.485	0.5	0.516		0.14	0.936	0.252	0.436	0.227	0.421	0.261	0.442	0.235	0.425
	AE 12	1.22	0.29	0.163	0.374	0.302	0.463	0.227	0.42	0.259	0.44	0.258	0.445	0.2	0.405	0.389	0.502	0.063	0.25		1.215	0.304	0.287	0.454	0.186	0.391	0.207	0.407	0.25	0.434
	AE 13	0.966	0.456	0.163	0.374	0.238	0.429	0.239	0.427	0.222	0.418	0.387	0.495	0.267	0.447	0.278	0.461	0.125	0.342		1.695	0.167	0.209	0.408	0.32	0.469	0.25	0.435	0.209	0.408
	AE 14	1.257	0.27	0.256	0.441	0.159	0.368	0.193	0.396	0.222	0.418	0.161	0.374	0.289	0.458	0.222	0.428	0.438	0.512		3.134	0.025	0.139	0.348	0.196	0.399	0.196	0.399	0.281	0.45
[AE]	AE 15	2.704	0.009	0.256	0.441	0.333	0.475	0.233	0.424	0.13	0.337	0.194	0.402	0.178	0.387	0	0	0.375	0.5		2.118	0.097	0.2	0.402	0.299	0.46	0.228	0.422	0.173	0.38
	AE 16	0.969	0.453	0.186	0.394	0.159	0.368	0.239	0.427	0.185	0.39	0.097	0.301	0.111	0.318	0.167	0.383	0.188	0.403		1.532	0.205	0.183	0.388	0.237	0.428	0.228	0.422	0.148	0.356
	AE 17	0.509	0.828	0.186	0.394	0.19	0.396	0.193	0.396	0.157	0.366	0.097	0.301	0.111	0.318	0.167	0.383	0.125	0.342		1.703	0.165	0.226	0.42	0.186	0.391	0.12	0.326	0.148	0.356
	AE 18	2.917	0.005	0.116	0.324	0.27	0.447	0.125	0.332	0.093	0.291	0.097	0.301	0.244	0.435	0	0	0.25	0.447		0.884	0.449	0.122	0.328	0.144	0.353	0.196	0.399	0.133	0.34
	AE 19	1.023	0.414	0.093	0.294	0.111	0.317	0.108	0.311	0.12	0.327	0.161	0.374	0.2	0.405	0.222	0.428	0.25	0.447		2.787	0.04	0.157	0.365	0.134	0.342	0.196	0.399	0.082	0.275
	AE 20	0.684	0.686	0.093	0.294	0.159	0.368	0.114	0.318	0.13	0.337	0.032	0.18	0.067	0.252	0.111	0.323	0.125	0.342		1.839	0.139	0.157	0.365	0.103	0.306	0.054	0.228	0.117	0.323
	AE 21	1.597	0.134	0.07	0.258	0.127	0.336	0.091	0.288	0.111	0.316	0.258	0.445	0.178	0.387	0.056	0.236	0.188	0.403		1.212	0.305	0.148	0.356	0.113	0.319	0.065	0.248	0.128	0.334
	AE 22	0.773	0.61	0.07	0.258	0.127	0.336	0.097	0.296	0.102	0.304	0.097	0.301	0.133	0.344	0.056	0.236	0.25	0.447		4.067	0.007	0.096	0.295	0.031	0.174	0.185	0.39	0.112	0.316
	AE 23	0.741	0.637	0.14	0.351	0.048	0.215	0.114	0.318	0.111	0.316	0.129	0.341	0.089	0.288	0.111	0.323	0	0		1.32	0.267	0.087	0.283	0.124	0.331	0.054	0.228	0.122	0.329
	AE 24	0.894	0.511	0.093	0.294	0.095	0.296	0.04	0.196	0.083	0.278	0.032	0.18	0.044	0.208	0.111	0.323	0.125	0.342		2.816	0.039	0.043	0.205	0.093	0.292	0.12	0.326	0.041	0.198
	AE 25	2.05	0.047	0.163	0.374	0.048	0.215	0.068	0.253	0.056	0.23	0	0	0.022	0.149	0	0	0	0		1.291	0.277	0.087	0.283	0.041	0.2	0.076	0.267	0.041	0.198
	AE 26	1.549	0.149	0.023	0.152	0.079	0.272	0.011	0.106	0.074	0.263	0.065	0.25	0.044	0.208	0	0	0.063	0.25		0.164	0.921	0.052	0.223	0.041	0.2	0.043	0.205	0.036	0.186
	AE 27	1.236	0.281	0.023	0.152	0.079	0.272	0.017	0.13	0.065	0.247	0.065	0.25	0.022	0.149	0.056	0.236	0	0		0.395	0.756	0.043	0.205	0.021	0.143	0.043	0.205	0.046	0.21
	AE 28	1.975	0.057	0.07	0.258	0.016	0.126	0.028	0.167	0.009	0.096	0.032	0.18	0.111	0.318	0	0	0.063	0.25		2.443	0.063	0.017	0.131	0.041	0.2	0	0	0.056	0.231

Appendix A.2. ANOVA Post Hoc—Test Least Significant Difference Fishera—Calculations

Table A4. ANOVA Post hoc—Test Least Significant Difference Fishera—calculations.

Test factor	[AS]	Pair: Feature 1	Mean	Feature 2	Mean	*p*-Value	[AE]	Pair: Feature 1	Mean	Feature 2	Mean	*p*-Value
[A]	AS 4	[A1]	0.239	[A2]	0.386	0	AE 5	[A1]	0.425	[A2]	0.318	0.014
	AS 6	[A1]	0.243	[A2]	0.341	0.016	AE 7	[A1]	0.261	[A2]	0.391	0.002
	AS 13	[A1]	0.164	[A2]	0.245	0.024	AE 12	[A1]	0.2	[A2]	0.286	0.024
	AS 19	[A1]	0.093	[A2]	0.155	0.035	AE 15	[A1]	0.171	[A2]	0.268	0.009
	AS 25	[A1]	0.107	[A2]	0.045	0.012	AE 16	[A1]	0.221	[A2]	0.145	0.031
	AS 26	[A1]	0.093	[A2]	0.041	0.024	AE 19	[A1]	0.093	[A2]	0.177	0.005
	AS 30	[A1]	0.079	[A2]	0.032	0.026						
[B]	AS 4	[B1]	0.358	[B2]	0.257	0.015	AE 1	[B1]	0.569	[B2]	0.474	0.034
	AS 9	[B1]	0.19	[B2]	0.269	0.037	AE 10	[B1]	0.281	[B2]	0.217	0.099
	AS 12	[B1]	0.254	[B2]	0.153	0.005	AE 12	[B1]	0.193	[B2]	0.276	0.03
	AS 14	[B1]	0.125	[B2]	0.265	0	AE 13	[B1]	0.272	[B2]	0.21	0.103
	AS 26	[B1]	0.043	[B2]	0.093	0.028	AE 16	[B1]	0.219	[B2]	0.162	0.101
	AS 28	[B1]	0.03	[B2]	0.086	0.009						
[C]	AS 4	[C1]	0.351	[C2]	0.265	0.037	AE 12	[C1]	0.193	[C2]	0.276	0.03
	AS 14	[C1]	0.149	[C2]	0.243	0.009	AE 14	[C1]	0.263	[C2]	0.176	0.019
[D]	AS 2	[D1]	0.517	[D2]	0.641	0.041	AE 2	[D1]	0.448	[D2]	0.609	0.008
		[D1]	0.517	[D3]	0.449	0.256		[D1]	0.448	[D3]	0.465	0.779
		[D1]	0.517	[D4]	0.415	0.199		[D1]	0.448	[D4]	0.453	0.947
		[D1]	0.517	[D5]	0.408	0.183		[D1]	0.448	[D5]	0.612	0.046
		[D2]	0.641	[D3]	0.449	0.002		[D2]	0.609	[D3]	0.465	0.02
		[D2]	0.641	[D4]	0.415	0.005		[D2]	0.609	[D4]	0.453	0.054
		[D2]	0.641	[D5]	0.408	0.005		[D2]	0.609	[D5]	0.612	0.973
		[D3]	0.449	[D4]	0.415	0.677		[D3]	0.465	[D4]	0.453	0.885
		[D3]	0.449	[D5]	0.408	0.944		[D3]	0.465	[D5]	0.612	0.078

Test Factor	[AS]	Pair: Feature 1	Mean	Feature 2	Mean	*p*-Value	[AE]	Pair: Feature 1	Mean	Feature 2	Mean	*p*-Value
[F]	AS 7	[F1]	0.395	[F2]	0.302	0.277	AE 7	[F1]	0.279	[F2]	0.413	0.14
		[F1]	0.395	[F3]	0.284	0.133		[F1]	0.279	[F3]	0.273	0.935
		[F1]	0.395	[F4]	0.167	0.004		[F1]	0.279	[F4]	0.278	0.988
		[F1]	0.395	[F5]	0.226	0.099		[F1]	0.279	[F5]	0.613	0.002
		[F1]	0.395	[F6]	0.333	0.504		[F1]	0.279	[F6]	0.178	0.299
		[F1]	0.395	[F7]	0.056	0.006		[F1]	0.279	[F7]	0.444	0.198
		[F1]	0.395	[F8]	0.188	0.103		[F1]	0.279	[F8]	0.5	0.1
		[F2]	0.302	[F3]	0.284	0.784		[F2]	0.413	[F3]	0.273	0.038
		[F2]	0.302	[F4]	0.167	0.051		[F2]	0.413	[F4]	0.278	0.063
		[F2]	0.302	[F5]	0.226	0.428		[F2]	0.413	[F5]	0.613	0.047
		[F2]	0.302	[F6]	0.333	0.709		[F2]	0.413	[F6]	0.178	0.009
		[F2]	0.302	[F7]	0.056	0.035		[F2]	0.413	[F7]	0.444	0.795
		[F2]	0.302	[F8]	0.188	0.349		[F2]	0.413	[F8]	0.5	0.496
		[F3]	0.284	[F4]	0.167	0.028		[F3]	0.273	[F4]	0.278	0.928
		[F3]	0.284	[F5]	0.226	0.492		[F3]	0.273	[F5]	0.613	0
		[F3]	0.284	[F6]	0.333	0.498		[F3]	0.273	[F6]	0.178	0.214
		[F3]	0.284	[F7]	0.056	0.034		[F3]	0.273	[F7]	0.444	0.13
		[F3]	0.284	[F8]	0.188	0.396		[F3]	0.273	[F8]	0.5	0.058
		[F4]	0.167	[F5]	0.226	0.505		[F4]	0.278	[F5]	0.613	0
		[F4]	0.167	[F6]	0.333	0.031		[F4]	0.278	[F6]	0.178	0.218
		[F4]	0.167	[F7]	0.056	0.316		[F4]	0.278	[F7]	0.444	0.153
		[F4]	0.167	[F8]	0.188	0.858		[F4]	0.278	[F8]	0.5	0.07
		[F5]	0.226	[F6]	0.333	0.29		[F5]	0.613	[F6]	0.178	0
		[F5]	0.226	[F7]	0.056	0.187		[F5]	0.613	[F7]	0.444	0.214

Table A4. *Cont.*

Test factor	[AS]	Pair					[AE]	Pair					Test Factor	[AS]	Pair					[AE]	Pair				
		Feature 1	Mean	Feature 2	Mean	*p*-Value		Feature 1	Mean	Feature 2	Mean	*p*-Value			Feature 1	Mean	Feature 2	Mean	*p*-Value		Feature 1	Mean	Feature 2	Mean	*p*-Value
		[D4]	0.415	[D5]	0.408	0.626		[D4]	0.453	[D5]	0.612	0.106			[F5]	0.226	[F8]	0.188	0.775		[F5]	0.613	[F8]	0.5	0.423
	AS 4	[D1]	0.168	[D2]	0.305	0.013	AE 6	[D1]	0.301	[D2]	0.266	0.533			[F6]	0.333	[F7]	0.056	0.022		[F6]	0.178	[F7]	0.444	0.037
		[D1]	0.168	[D3]	0.331	0.003		[D1]	0.301	[D3]	0.315	0.8			[F6]	0.333	[F8]	0.188	0.25		[F6]	0.178	[F8]	0.5	0.016
		[D1]	0.168	[D4]	0.491	0		[D1]	0.301	[D4]	0.528	0.002			[F7]	0.056	[F8]	0.188	0.378		[F7]	0.444	[F8]	0.5	0.724
		[D1]	0.168	[D5]	0.429	0.001		[D1]	0.301	[D5]	0.286	0.845		AS 8	[F1]	0.186	[F2]	0.397	0.015	AE 15	[F1]	0.256	[F2]	0.333	0.335
		[D2]	0.305	[D3]	0.331	0.645		[D2]	0.266	[D3]	0.315	0.394			[F1]	0.186	[F3]	0.188	0.984		[F1]	0.256	[F3]	0.233	0.741
		[D2]	0.305	[D4]	0.491	0.012		[D2]	0.266	[D4]	0.528	0.001			[F1]	0.186	[F4]	0.315	0.101		[F1]	0.256	[F4]	0.13	0.085
		[D2]	0.305	[D5]	0.429	0.102		[D2]	0.266	[D5]	0.286	0.796			[F1]	0.186	[F5]	0.258	0.482		[F1]	0.256	[F5]	0.233	0.515
		[D3]	0.331	[D4]	0.491	0.03		[D3]	0.315	[D4]	0.528	0.005			[F1]	0.186	[F6]	0.311	0.178		[F1]	0.256	[F6]	0.178	0.368
		[D3]	0.331	[D5]	0.429	0.197		[D3]	0.315	[D5]	0.286	0.707			[F1]	0.186	[F7]	0.222	0.767		[F1]	0.256	[F7]	0	0.025
		[D4]	0.491	[D5]	0.429	0.488		[D4]	0.528	[D5]	0.286	0.008			[F1]	0.186	[F8]	0.188	0.991		[F1]	0.256	[F8]	0.375	0.316
	AS 9	[D1]	0.357	[D2]	0.219	0.006	AE 14	[D1]	0.196	[D2]	0.172	0.631			[F2]	0.397	[F3]	0.188	0.001		[F2]	0.333	[F3]	0.233	0.093
		[D1]	0.357	[D3]	0.189	0.001		[D1]	0.196	[D3]	0.268	0.15			[F2]	0.397	[F4]	0.315	0.234		[F2]	0.333	[F4]	0.13	0.002
		[D1]	0.357	[D4]	18868	0.012		[D1]	0.196	[D4]	0.132	0.333			[F2]	0.397	[F5]	0.258	0.146		[F2]	0.333	[F5]	0.233	0.117
		[D1]	0.357	[D5]	0.061	0		[D1]	0.196	[D5]	0.347	0.026			[F2]	0.397	[F6]	0.311	0.313		[F2]	0.333	[F6]	0.178	0.05
		[D2]	0.219	[D3]	0.189	0.567		[D2]	0.172	[D3]	0.268	0.062			[F2]	0.397	[F7]	0.222	0.133		[F2]	0.333	[F7]	0	0.002
		[D2]	0.219	[D4]	18868	0.657		[D2]	0.172	[D4]	0.132	0.552			[F2]	0.397	[F8]	0.188	0.086		[F2]	0.333	[F8]	0.375	0.714
		[D2]	0.219	[D5]	0.061	0.024		[D2]	0.172	[D5]	0.347	0.011			[F3]	0.188	[F4]	0.315	0.017		[F3]	0.233	[F4]	0.13	0.038
		[D3]	0.189	[D4]	18868	0.997		[D3]	0.268	[D4]	0.132	0.043			[F3]	0.188	[F5]	0.258	0.405		[F3]	0.233	[F5]	0.233	0.618
		[D3]	0.189	[D5]	0.061	0.068		[D3]	0.268	[D5]	0.347	0.25			[F3]	0.188	[F6]	0.311	0.089		[F3]	0.233	[F6]	0.178	0.416
		[D4]	0.189	[D5]	0.061	0.121		[D4]	0.132	[D5]	0.347	0.008			[F3]	0.188	[F7]	0.222	0.747		[F3]	0.233	[F7]	0	0.021
	AS 12	[D1]	0.119	[D2]	0.195	0.114	AE 17	[D1]	0.189	[D2]	0.18	0.84			[F3]	0.188	[F8]	0.188	1		[F3]	0.233	[F8]	0.375	0.181
		[D1]	0.119	[D3]	0.244	0.01		[D1]	0.189	[D3]	0.079	0.015			[F4]	0.315	[F5]	0.258	0.522		[F4]	0.13	[F5]	0.233	0.44
		[D1]	0.119	[D4]	0.321	0.002		[D1]	0.189	[D4]	0.208	0.754			[F4]	0.315	[F6]	0.311	0.962		[F4]	0.13	[F6]	0.178	0.504
		[D1]	0.119	[D5]	0.204	0.195		[D1]	0.189	[D5]	0.265	0.214			[F4]	0.315	[F7]	0.222	0.403		[F4]	0.13	[F7]	0	0.21
		[D2]	0.195	[D3]	0.244	0.327		[D2]	0.18	[D3]	0.079	0.03			[F4]	0.315	[F8]	0.188	0.274		[F4]	0.13	[F8]	0.375	0.024
		[D2]	0.195	[D4]	0.321	0.054		[D2]	0.18	[D4]	0.208	0.646			[F5]	0.258	[F6]	0.311	0.601		[F5]	0.233	[F6]	0.178	0.868
		[D2]	0.195	[D5]	0.204	0.895		[D2]	0.18	[D5]	0.265	0.171			[F5]	0.258	[F7]	0.222	0.781		[F5]	0.233	[F7]	0	0.108
		[D3]	0.244	[D4]	0.321	0.238		[D3]	0.079	[D4]	0.208	0.034			[F5]	0.258	[F8]	0.188	0.598		[F5]	0.233	[F8]	0.375	0.147
		[D3]	0.244	[D5]	0.204	0.549		[D3]	0.079	[D5]	0.265	0.003			[F6]	0.311	[F7]	0.222	0.463		[F6]	0.178	[F7]	0	0.117
		[D4]	0.321	[D5]	0.204	0.139		[D4]	0.208	[D5]	0.265	0.433			[F6]	0.311	[F8]	0.188	0.329		[F6]	0.178	[F8]	0.375	0.096

Table A4. *Cont.*

Test factor	[AS]	Feature 1	Mean	Feature 2	Mean	*p*-Value	[AE]	Feature 1	Mean	Feature 2	Mean	*p*-Value
	AS 14	[D1]	0.273	[D2]	0.25	0.637	AE 20	[D1]	0.112	[D2]	0.18	0.076
		[D1]	0.273	[D3]	0.087	0		[D1]	0.112	[D3]	0.079	0.387
		[D1]	0.273	[D4]	0.17	0.106		[D1]	0.112	[D4]	0.038	0.142
		[D1]	0.273	[D5]	0.184	0.174		[D1]	0.112	[D5]	0.102	0.85
		[D2]	0.25	[D3]	0.087	0.001		[D2]	0.18	[D3]	0.079	0.01
		[D2]	0.25	[D4]	0.17	0.215		[D2]	0.18	[D4]	0.038	0.006
		[D2]	0.25	[D5]	0.184	0.318		[D2]	0.18	[D5]	0.102	0.141
		[D3]	0.087	[D4]	0.17	0.198		[D3]	0.079	[D4]	0.038	0.424
		[D3]	0.087	[D5]	0.184	0.145		[D3]	0.079	[D5]	0.102	0.659
		[D4]	0.17	[D5]	0.184	0.86		[D4]	0.038	[D5]	0.102	0.301
	AS 18	[D1]	0.112	[D2]	0.195	0.04	AE 21	[D1]	0.063	[D2]	0.141	0.047
		[D1]	0.112	[D3]	0.094	0.668		[D1]	0.063	[D3]	0.189	0.001
		[D1]	0.112	[D4]	0.057	0.302		[D1]	0.063	[D4]	0.075	0.808
		[D1]	0.112	[D5]	0.163	0.351		[D1]	0.063	[D5]	0.082	0.724
		[D2]	0.195	[D3]	0.094	0.016		[D2]	0.141	[D3]	0.189	0.228
		[D2]	0.195	[D4]	0.057	0.011		[D2]	0.141	[D4]	0.075	0.213
		[D2]	0.195	[D5]	0.163	0.566		[D2]	0.141	[D5]	0.082	0.273
		[D3]	0.094	[D4]	0.057	0.486		[D3]	0.189	[D4]	0.075	0.031
		[D3]	0.094	[D5]	0.163	0.219		[D3]	0.189	[D5]	0.082	0.047
		[D4]	0.057	[D5]	0.163	0.106		[D4]	0.075	[D5]	0.082	0.923
	AS 20	[D1]	0.07	[D2]	0.063	0.835	AE 22	[D1]	0.084	[D2]	0.172	0.019
		[D1]	0.07	[D3]	0.11	0.26		[D1]	0.084	[D3]	0.11	0.481
		[D1]	0.07	[D4]	0.208	0.004		[D1]	0.084	[D4]	0.075	0.864
		[D1]	0.07	[D5]	0.102	0.508		[D1]	0.084	[D5]	0.02	0.21
		[D2]	0.063	[D3]	0.11	0.194		[D2]	0.172	[D3]	0.11	0.108
		[D2]	0.063	[D4]	0.208	0.003		[D2]	0.172	[D4]	0.075	0.054
		[D2]	0.063	[D5]	0.102	0.422		[D2]	0.172	[D5]	0.02	0.003
		[D3]	0.11	[D4]	0.208	0.043		[D3]	0.11	[D4]	0.075	0.488

Test Factor	[AS]	Feature 1	Mean	Feature 2	Mean	*p*-Value	[AE]	Feature 1	Mean	Feature 2	Mean	*p*-Value
		[F7]	0.222	[F8]	0.188	0.816		[F7]	0	[F8]	0.375	0.007
	AS 12	[F1]	0.093	[F2]	0.238	0.065	AE 18	[F1]	0.116	[F2]	0.27	0.026
		[F1]	0.093	[F3]	0.153	0.37		[F1]	0.116	[F3]	0.125	0.883
		[F1]	0.093	[F4]	0.194	0.156		[F1]	0.116	[F4]	0.093	0.705
		[F1]	0.093	[F5]	0.258	0.077		[F1]	0.116	[F5]	0.097	0.811
		[F1]	0.093	[F6]	0.289	0.021		[F1]	0.116	[F6]	0.244	0.084
		[F1]	0.093	[F7]	0.222	0.246		[F1]	0.116	[F7]	0	0.233
		[F1]	0.093	[F8]	0.5	0		[F1]	0.116	[F8]	0.25	0.189
		[F2]	0.238	[F3]	0.153	0.146		[F2]	0.27	[F3]	0.125	0.005
		[F2]	0.238	[F4]	0.194	0.487		[F2]	0.27	[F4]	0.093	0.001
		[F2]	0.238	[F5]	0.258	0.818		[F2]	0.27	[F5]	0.097	0.023
		[F2]	0.238	[F6]	0.289	0.511		[F2]	0.27	[F6]	0.244	0.708
		[F2]	0.238	[F7]	0.222	0.881		[F2]	0.27	[F7]	0	0.004
		[F2]	0.238	[F8]	0.5	0.019		[F2]	0.27	[F8]	0.25	0.838
		[F3]	0.153	[F4]	0.194	0.397		[F3]	0.125	[F4]	0.093	0.445
		[F3]	0.153	[F5]	0.258	0.175		[F3]	0.125	[F5]	0.097	0.676
		[F3]	0.153	[F6]	0.289	0.041		[F3]	0.125	[F6]	0.244	0.04
		[F3]	0.153	[F7]	0.222	0.483		[F3]	0.125	[F7]	0	0.146
		[F3]	0.153	[F8]	0.5	0.001		[F3]	0.125	[F8]	0.25	0.168
		[F4]	0.194	[F5]	0.258	0.431		[F4]	0.093	[F5]	0.097	0.953
		[F4]	0.194	[F6]	0.289	0.179		[F4]	0.093	[F6]	0.244	0.014
		[F4]	0.194	[F7]	0.222	782984		[F4]	0.093	[F7]	0	0.295
		[F4]	0.194	[F8]	0.5	0.004		[F4]	0.093	[F8]	0.25	0.091
		[F5]	0.258	[F6]	0.289	0.739		[F5]	0.097	[F6]	0.244	0.069
		[F5]	0.258	[F7]	0.222	0.76		[F5]	0.097	[F7]	0	0.347
		[F5]	0.258	[F8]	0.5	0.048		[F5]	0.097	[F8]	0.25	0.152
		[F6]	0.289	[F7]	0.222	0.546		[F6]	0.244	[F7]	0	0.012
		[F6]	0.289	[F8]	0.5	0.068		[F6]	0.244	[F8]	0.25	0.956

Table A4. *Cont.*

Test factor	[AS]	Pair: Feature 1	Mean	Feature 2	Mean	*p*-Value	[AE]	Pair: Feature 1	Mean	Feature 2	Mean	*p*-Value
		[D3]	0.11	[D5]	0.102	0.868		[D3]	0.11	[D5]	0.02	0.081
		[D4]	0.208	[D5]	0.102	0.07		[D4]	0.075	[D5]	0.02	0.364
							AE 24	[D1]	0.056	[D2]	0.016	0.178
								[D1]	0.056	[D3]	0.071	0.619
								[D1]	0.056	[D4]	0.113	0.148
								[D1]	0.056	[D5]	0.163	0.009
								[D2]	0.016	[D3]	0.071	0.073
								[D2]	0.016	[D4]	0.113	0.015
								[D2]	0.016	[D5]	0.163	0
								[D3]	0.071	[D4]	0.113	0.293
								[D3]	0.071	[D5]	0.163	0.026
								[D4]	0.113	[D5]	0.163	0.305
[E]	AS 2	[E1]	0.645	[E2]	0.546	0.293	AE 5	[E1]	0.484	[E2]	0.408	0.41
		[E1]	0.645	[E3]	0.475	0.086		[E1]	0.484	[E3]	0.277	0.031
		[E1]	0.645	[E4]	0.379	0.014		[E1]	0.484	[E4]	0.424	0.57
		[E2]	0.546	[E3]	0.475	0.175		[E2]	0.408	[E3]	0.277	0.009
		[E2]	0.546	[E4]	0.379	0.015		[E2]	0.408	[E4]	0.424	0.812
		[E3]	0.475	[E4]	0.379	0.194		[E3]	0.277	[E4]	0.424	0.041
	AS 8	[E1]	0.161	[E2]	0.302	0.091	AE 14	[E1]	0.161	[E2]	0.302	0.091
		[E1]	0.161	[E3]	0.248	0.315		[E1]	0.161	[E3]	0.248	0.315
		[E1]	0.161	[E4]	0.152	0.918		[E1]	0.161	[E4]	0.152	0.918
		[E2]	0.302	[E3]	0.248	0.242		[E2]	0.302	[E3]	0.248	0.242
		[E2]	0.302	[E4]	0.152	0.013		[E2]	0.302	[E4]	0.152	0.013
		[E3]	0.248	[E4]	0.152	0.137		[E3]	0.248	[E4]	0.152	0.137
	AS 18	[E1]	0.29	[E2]	0.137	0.016	AE 19	[E1]	0.129	[E2]	0.141	0.848
		[E1]	0.29	[E3]	0.085	0.002		[E1]	0.129	[E3]	0.071	0.382
		[E1]	0.29	[E4]	0.106	0.011		[E1]	0.129	[E4]	0.212	0.255
		[E2]	0.137	[E3]	0.085	0.132		[E2]	0.141	[E3]	0.071	0.045

Test Factor	[AS]	Pair: Feature 1	Mean	Feature 2	Mean	*p*-Value	[AE]	Pair: Feature 1	Mean	Feature 2	Mean	*p*-Value
		[F7]	0.222	[F8]	0.5	0.042		[F7]	0	[F8]	0.25	0.036
	AS 15	[F1]	0.186	[F2]	0.349	0.034	AE 25	[F1]	0.163	[F2]	0.048	0.013
		[F1]	0.186	[F3]	0.21	0.715		[F1]	0.163	[F3]	0.068	0.017
		[F1]	0.186	[F4]	0.111	0.286		[F1]	0.163	[F4]	0.056	0.011
		[F1]	0.186	[F5]	0.129	0.534		[F1]	0.163	[F5]	0	0.003
		[F1]	0.186	[F6]	0.156	0.713		[F1]	0.163	[F6]	0.022	0.005
		[F1]	0.186	[F7]	0.111	0.493		[F1]	0.163	[F7]	0	0.013
		[F1]	0.186	[F8]	0.188	0.99		[F1]	0.163	[F8]	0	0.017
		[F2]	0.349	[F3]	0.21	0.015		[F2]	0.048	[F3]	0.068	0.547
		[F2]	0.349	[F4]	0.111	0		[F2]	0.048	[F4]	0.056	0.829
		[F2]	0.349	[F5]	0.129	0.01		[F2]	0.048	[F5]	0	0.351
		[F2]	0.349	[F6]	0.156	0.011		[F2]	0.048	[F6]	0.022	0.576
		[F2]	0.349	[F7]	0.111	0.022		[F2]	0.048	[F7]	0	0.443
		[F2]	0.349	[F8]	0.188	0.138		[F2]	0.048	[F8]	0	0.464
		[F3]	0.21	[F4]	0.111	0.037		[F3]	0.068	[F4]	0.056	0.657
		[F3]	0.21	[F5]	0.129	0.284		[F3]	0.068	[F5]	0	0.132
		[F3]	0.21	[F6]	0.156	0.4		[F3]	0.068	[F6]	0.022	0.237
		[F3]	0.21	[F7]	0.111	0.303		[F3]	0.068	[F7]	0	0.236
		[F3]	0.21	[F8]	0.188	0.823		[F3]	0.068	[F8]	0	0.261
		[F4]	0.111	[F5]	0.129	0.821		[F4]	0.056	[F5]	0	0.241
		[F4]	0.111	[F6]	0.156	0.52		[F4]	0.056	[F6]	0.022	0.419
		[F4]	0.111	[F7]	0.111	1		[F4]	0.056	[F7]	0	0.348
		[F4]	0.111	[F8]	0.188	0.464		[F4]	0.056	[F8]	0	0.372
		[F5]	0.129	[F6]	0.156	0.77		[F5]	0	[F6]	0.022	0.682
		[F5]	0.129	[F7]	0.111	0.876		[F5]	0	[F7]	0	1
		[F5]	0.129	[F8]	0.188	0.625		[F5]	0	[F8]	0	1
		[F6]	0.156	[F7]	0.111	0.682		[F6]	0.022	[F7]	0	0.732
		[F6]	0.156	[F8]	0.188	0.778		[F6]	0.022	[F8]	0	0.743

Table A4. *Cont.*

Test factor	[AS]	Pair: Feature 1	Mean	Feature 2	Mean	p-Value	[AE]	Pair: Feature 1	Mean	Feature 2	Mean	p-Value
		[E2]	0.137	[E4]	0.106	0.493		[E2]	0.141	[E4]	0.212	0.125
		[E3]	0.085	[E4]	0.106	0.672		[E3]	0.071	[E4]	0.212	0.005
	AS 19	[E1]	0.032	[E2]	0.126	0.127	AE 24	[E1]	0	[E2]	0.05	0.286
		[E1]	0.032	[E3]	0.078	0.475		[E1]	0	[E3]	0.057	0.243
		[E1]	0.032	[E4]	0.227	0.006		[E1]	0	[E4]	0.182	0.001
		[E2]	0.126	[E3]	0.078	0.155		[E2]	0.05	[E3]	0.057	0.781
		[E2]	0.126	[E4]	0.227	0.023		[E2]	0.05	[E4]	0.182	0
		[E3]	0.078	[E4]	0.227	0.002		[E3]	0.057	[E4]	0.182	0.001
	AS 24	[E1]	0.161	[E2]	0.099	0.227	AE 25	[E1]	0.032	[E2]	0.023	0.831
		[E1]	0.161	[E3]	0.035	0.019		[E1]	0.032	[E3]	0.099	0.144
		[E1]	0.161	[E4]	0.061	0.088		[E1]	0.032	[E4]	0.121	0.078
		[E2]	0.099	[E3]	0.035	0.024		[E2]	0.023	[E3]	0.099	0.002
		[E2]	0.099	[E4]	0.061	0.3		[E2]	0.023	[E4]	0.121	0.002
		[E3]	0.035	[E4]	0.061	0.533		[E3]	0.099	[E4]	0.121	0.525
	AS 29	[E1]	0.032	[E2]	0.092	0.179	AE 27	[E1]	0	[E2]	0.069	0.063
		[E1]	0.032	[E3]	0.021	0.811		[E1]	0	[E3]	0.007	0.854
		[E1]	0.032	[E4]	0.015	0.735		[E1]	0	[E4]	0.015	0.721
		[E2]	0.092	[E3]	0.021	0.004		[E2]	0.069	[E3]	0.007	0.003
		[E2]	0.092	[E4]	0.015	0.017		[E2]	0.069	[E4]	0.015	0.046
		[E3]	0.021	[E4]	0.015	0.86		[E3]	0.007	[E4]	0.015	0.781
[F]	AS 6	[F1]	0.256	[F2]	0.302	0.606	AE 5	[F1]	0.326	[F2]	0.286	0.676
		[F1]	0.256	[F3]	0.256	0.999		[F1]	0.326	[F3]	0.33	0.961
		[F1]	0.256	[F4]	0.333	0.338		[F1]	0.326	[F4]	0.407	0.346
		[F1]	0.256	[F5]	0.29	0.744		[F1]	0.326	[F5]	0.419	0.409
		[F1]	0.256	[F6]	0.133	0.201		[F1]	0.326	[F6]	0.578	0.014
		[F1]	0.256	[F7]	0.444	0.135		[F1]	0.326	[F7]	0.389	0.64
		[F1]	0.256	[F8]	0.563	0.02		[F1]	0.326	[F8]	0.563	0.093
		[F2]	0.302	[F3]	0.256	0.486		[F2]	0.286	[F3]	0.33	0.535

Test Factor	[AS]	Pair: Feature 1	Mean	Feature 2	Mean	p-Value	[AE]	Pair: Feature 1	Mean	Feature 2	Mean	p-Value
		[F7]	0.111	[F8]	0.188	0.568		[F7]	0	[F8]	0	1
	AS 27	[F1]	0.093	[F2]	0.016	0.118						
		[F1]	0.093	[F3]	0.091	0.96						
		[F1]	0.093	[F4]	0.056	0.405						
		[F1]	0.093	[F5]	0.097	0.949						
		[F1]	0.093	[F6]	0	0.081						
		[F1]	0.093	[F7]	0	0.184						
		[F1]	0.093	[F8]	0.25	0.032						
		[F2]	0.016	[F3]	0.091	0.041						
		[F2]	0.016	[F4]	0.056	0.315						
		[F2]	0.016	[F5]	0.097	0.139						
		[F2]	0.016	[F6]	0	0.744						
		[F2]	0.016	[F7]	0	0.812						
		[F2]	0.016	[F8]	0.25	0.001						
		[F3]	0.091	[F4]	0.056	0.246						
		[F3]	0.091	[F5]	0.097	0.904						
		[F3]	0.091	[F6]	0	0.029						
		[F3]	0.091	[F7]	0	0.141						
		[F3]	0.091	[F8]	0.25	0.015						
		[F4]	0.056	[F5]	0.097	0.417						
		[F4]	0.056	[F6]	0	0.209						
		[F4]	0.056	[F7]	0	0.381						
		[F4]	0.056	[F8]	0.25	0.004						
		[F5]	0.097	[F6]	0	0.097						
		[F5]	0.097	[F7]	0	0.191						
		[F5]	0.097	[F8]	0.25	0.046						
		[F6]	0	[F7]	0	1						
		[F6]	0	[F8]	0.25	0.001						

Table A4. *Cont.*

Test factor	[AS]	Pair: Feature 1	Mean	Feature 2	Mean	*p*-Value	[AE]	Pair: Feature 1	Mean	Feature 2	Mean	*p*-Value
		[F2]	0.302	[F4]	0.333	0.655		[F2]	0.286	[F4]	0.407	0.111
		[F2]	0.302	[F5]	0.29	0.909		[F2]	0.286	[F5]	0.419	0.206
		[F2]	0.302	[F6]	0.133	0.055		[F2]	0.286	[F6]	0.578	0.002
		[F2]	0.302	[F7]	0.444	0.234		[F2]	0.286	[F7]	0.389	0.423
		[F2]	0.302	[F8]	0.563	0.038		[F2]	0.286	[F8]	0.563	0.041
		[F3]	0.256	[F4]	0.333	0.157		[F3]	0.33	[F4]	0.407	0.186
		[F3]	0.256	[F5]	0.29	0.692		[F3]	0.33	[F5]	0.419	0.339
		[F3]	0.256	[F6]	0.133	0.103		[F3]	0.33	[F6]	0.578	0.002
		[F3]	0.256	[F7]	0.444	0.09		[F3]	0.33	[F7]	0.389	0.619
		[F3]	0.256	[F8]	0.563	0.009		[F3]	0.33	[F8]	0.563	0.064
		[F4]	0.333	[F5]	0.29	0.638		[F4]	0.407	[F5]	0.419	0.903
		[F4]	0.333	[F6]	0.133	0.012		[F4]	0.407	[F6]	0.578	0.047
		[F4]	0.333	[F7]	0.444	0.331		[F4]	0.407	[F7]	0.389	0.88
		[F4]	0.333	[F8]	0.563	0.057		[F4]	0.407	[F8]	0.563	0.23
		[F5]	0.29	[F6]	0.133	0.134		[F5]	0.419	[F6]	0.578	0.159
		[F5]	0.29	[F7]	0.444	0.247		[F5]	0.419	[F7]	0.389	0.831
		[F5]	0.29	[F8]	0.563	0.049		[F5]	0.419	[F8]	0.563	0.334
		[F6]	0.133	[F7]	0.444	0.013		[F6]	0.578	[F7]	0.389	0.16
		[F6]	0.133	[F8]	0.563	0.001		[F6]	0.578	[F8]	0.563	0.294
		[F7]	0.444	[F8]	0.563	0.444		[F7]	0.389	[F8]	0.563	0.913

Test Factor	[AS]	Pair: Feature 1	Mean	Feature 2	Mean	*p*-Value	[AE]	Pair: Feature 1	Mean	Feature 2	Mean	*p*-Value
		[F7]	0	[F8]	0.25	0.004						
[G]	AS 2	[G1]	0.6	[G2]	0.402	0.004	AE 6	[G1]	0.339	[G2]	0.34	0.987
		[G1]	0.6	[G3]	0.467	0.057		[G1]	0.339	[G3]	0.435	0.139
		[G1]	0.6	[G4]	0.531	0.235		[G1]	0.339	[G4]	0.24	0.068
		[G2]	0.402	[G3]	0.467	0.367		[G2]	0.34	[G3]	0.435	0.16
		[G2]	0.402	[G4]	0.531	0.038		[G2]	0.34	[G4]	0.24	0.081
		[G3]	0.467	[G4]	0.531	0.315		[G3]	0.435	[G4]	0.24	0.001
	AS 10	[G1]	0.322	[G2]	0.134	0.001	AE 14	[G1]	0.139	[G2]	0.196	0.315
		[G1]	0.322	[G3]	0.228	0.114		[G1]	0.139	[G3]	0.196	0.324
		[G1]	0.322	[G4]	0.24	0.099		[G1]	0.139	[G4]	0.281	0.003
		[G2]	0.134	[G3]	0.228	0.125		[G2]	0.196	[G3]	0.196	0.997
		[G2]	0.134	[G4]	0.24	0.044		[G2]	0.196	[G4]	0.281	0.096
		[G3]	0.228	[G4]	0.24	0.829		[G3]	0.196	[G4]	0.281	0.101
	AS 11	[G1]	0.148	[G2]	0.165	0.766	AE 19	[G1]	0.157	[G2]	0.134	0.626
		[G1]	0.148	[G3]	0.272	0.034		[G1]	0.157	[G3]	0.196	0.404
		[G1]	0.148	[G4]	0.291	0.004		[G1]	0.157	[G4]	0.082	0.057
		[G2]	0.165	[G3]	0.272	0.079		[G2]	0.134	[G3]	0.196	0.207
		[G2]	0.165	[G4]	0.291	0.016		[G2]	0.134	[G4]	0.082	0.208
		[G3]	0.272	[G4]	0.291	0.718		[G3]	0.196	[G4]	0.082	0.007
	AS 14	[G1]	0.296	[G2]	0.186	0.045	AE 22	[G1]	0.096	[G2]	0.031	0.125
		[G1]	0.296	[G3]	0.152	0.01		[G1]	0.096	[G3]	0.185	0.037
		[G1]	0.296	[G4]	0.173	0.009		[G1]	0.096	[G4]	0.112	0.644
		[G2]	0.186	[G3]	0.152	0.672		[G2]	0.031	[G3]	0.185	0.001
		[G2]	0.186	[G4]	0.173	0.807		[G2]	0.031	[G4]	0.112	0.032
		[G3]	0.152	[G4]	0.173	0.672		[G3]	0.185	[G4]	0.112	0.061
	AS 21	[G1]	0.139	[G2]	0.031	0.007	AE 22	[G1]	0.096	[G2]	0.031	0.125
		[G1]	0.139	[G3]	0.054	0.037		[G1]	0.096	[G3]	0.185	0.037
		[G1]	0.139	[G4]	0.117	0.523		[G1]	0.096	[G4]	0.112	0.644
		[G2]	0.031	[G3]	0.054	0.579		[G2]	0.031	[G3]	0.185	0.001
		[G2]	0.031	[G4]	0.117	0.017		[G2]	0.031	[G4]	0.112	0.032

Table A4. *Cont.*

		Pair							Pair							Pair							Pair				
Test factor	**[AS]**	**Feature 1**	**Mean**	**Feature 2**	**Mean**	***p*-Value**		**[AE]**	**Feature 1**	**Mean**	**Feature 2**	**Mean**	***p*-Value**	**Test Factor**	**[AS]**	**Feature 1**	**Mean**	**Feature 2**	**Mean**	***p*-Value**		**[AE]**	**Feature 1**	**Mean**	**Feature 2**	**Mean**	***p*-Value**
																[G3]	0.054	[G4]	0.117	0.086			[G3]	0.185	[G4]	0.112	0.061
															AS 22	[G1]	0.148	[G2]	0.041	0.007		AE 24	[G1]	0.043	[G2]	0.093	0.149
																[G1]	0.148	[G3]	0.065	0.041			[G1]	0.043	[G3]	0.12	0.028
																[G1]	0.148	[G4]	0.097	0.133			[G1]	0.043	[G4]	0.041	0.927
																[G2]	0.041	[G3]	0.065	0.567			[G2]	0.093	[G3]	0.12	0.457
																[G2]	0.041	[G4]	0.097	0.12			[G2]	0.093	[G4]	0.041	0.091
																[G3]	0.065	[G4]	0.097	0.384			[G3]	0.12	[G4]	0.041	0.012
															AS 32	[G1]	0.096	[G2]	0.052	0.133							
																[G1]	0.096	[G3]	0.033	0.035							
																[G1]	0.096	[G4]	0.026	0.005							
																[G2]	0.052	[G3]	0.033	0.541							
																[G2]	0.052	[G4]	0.026	0.325							
																[G3]	0.033	[G4]	0.026	0.792							

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